The Final Exam Textbook

General Psychology, Psychological Assessment, Social Psychology, Personality, Physiological Psychology, Developmental Psychology, Clinical Psychology, Psychiatric Disorders and Research Methodology, in one book

Written for Sihan, for the JSS AHER MA Clinical Psychology entrance, 27 October 2026. Built from every paper and mock on the site. This printing has chapter 1 General Psychology; the rest go in as they are finished, with the same page layout.

BLACK AND WHITE EDITION · A4, DOUBLE SIDED · 22 OCTOBER 2026

How this book works

This is the whole JSS syllabus in one book: one chapter for each of the nine subjects JSS names, in the order JSS lists them, cut so that all nine fit in under 1000 A4 pages. It is built from every past paper, question bank and practice mock on the site. Every question was sorted into the section that should teach it, and every option in it, wrong ones included, is explained somewhere in the book, because next year's paper can make a wrong option the right answer. Chapter 1 is the full-length General Psychology book (821 pages) cut down to its exam core; the full version stays in the Library for looking things up.

How to read it. Each chapter opens with a real case, lists its sections with a learning goal for each, and then teaches them in order, the way Pinel does: the question, the idea, the study behind it and who did it, then what it means. Read the High yield sections first if time is short (the tags are explained below), and do every Scan Your Brain before moving on. Each chapter ends with the case revisited, a summary of every section, a Who's who, the key terms, real past-paper questions with the answers explained, and a one-page High-yield sheet for your last read before the exam.

Key terms are in bold colour where they are defined, and the index at the back points to those pages. People are in bold where they are introduced, with their full name and years.

The boxes

! EXAM TRAP

The confusable pair, the bait option, the famous misattribution. Read every one.

KEY STUDY

A classic experiment told as a story: the question, the method, the result, the criticism.

CLINICAL LINK

Where the idea matters to a clinical psychologist, and to an interview panel.

REMEMBER

The one idea in the section you must not leave without.

INDIAN CONTEXT

Indian research, Indian psychology and Indian applications.

CASE

A person whose story makes the concept concrete.

SCAN YOUR BRAIN

Fill-in-the-blank checks after every two or three sections, answers underneath. Cover the answers first.

The colour code: how often each topic is asked

Every numbered section carries a tag, and every subsection that has been asked carries dots. Both are counted from the past papers and practice mocks on the site: each question was matched to the section that teaches it, a real past-paper question scores 2 points and a mock or question-bank item 1 point.

HIGH YIELD · 23 QS ●●●	MEDIUM YIELD · 8 QS ●●	LOWER YIELD · 1 Q ○
The most-asked quarter of the book's sections (at least 153 points). Read these first and know every trap.	Asked regularly (at least 85 points). Read them fully.	Asked rarely or not yet. Still in the syllabus, so still taught in full.

"23 Qs" is the number of questions on the site that this section answers. The same tags sit next to each section in the Contents, and the page after this one shows where the marks are across the book.

Printing

Print A4, double sided, flip on the long edge. Every chapter opens on a right-hand page, the inner margin is wider for binding, and page numbers sit on the outer corner. The colour and the black and white editions have identical pages, page numbers and index, so either can be printed.

Where the marks are

815 questions from the past papers, question banks and practice mocks on the site are taught in this book. This is how they spread across the chapters.

Chapter	Questions	From real papers	Share of the book	High-yield sections
1 General Psychology	815	402	100.0%	3 of 11

The 12 most-asked sections

143 QS **1.6** Learning: Conditioning, Cognition and Observation 47

112 QS **1.2** Schools and Systems, the Cognitive Revolution and Indian Psychology 16

111 QS **1.7** Memory: Encoding, Storage, Retrieval and Forgetting 58

74 QS **1.11** Emotion and Stress 92

74 QS **1.10** Motivation 84

63 QS **1.9** Intelligence: Theories, Measurement, Heredity and Environment 76

61 QS **1.5** Attention and Consciousness 38

53 QS **1.8** Thinking, Problem Solving, Reasoning, Decision Making and Language 69

46 QS **1.1** Psychology as a Science: Roots, Founders and the First Schools 9

40 QS **1.3** Sensation and Psychophysics 25

38 QS **1.4** Perception: Organising and Interpreting the World 31

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1.2 Schools and Systems, the Cognitive Revolution and Indian Psychology — HIGH · 112 QS — 16

1.3 Sensation and Psychophysics — LOWER · 40 QS 25

1.4 Perception: Organising and Interpreting the World — LOWER · 38 QS 31

1.5 Attention and Consciousness — MEDIUM · 61 QS 38

1.6 Learning: Conditioning, Cognition and Observation — HIGH · 143 QS 47

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CHAPTER 1

General Psychology

IN THIS CHAPTER

1.1 ○ Psychology as a Science: Roots, Founders and the First Schools: Explain how psychology grew out of philosophy and physiology into a laboratory science, and name the founders, methods and central ideas of Wundt's voluntarism, Titchener's structuralism and the functionalism of James and the American pioneers.

1.2 ●●● Schools and Systems, the Cognitive Revolution and Indian Psychology: Trace psychology's rival schools (behaviourism, Gestalt, psychoanalysis, humanistic and existential psychology) through the cognitive revolution to today's perspectives, and explain the core ideas of Indian psychology: its pioneers, Samkhya, the three gunas, Patanjali's yoga and the Vedantic self.

1.3 ○ Sensation and Psychophysics: Explain how physical energy becomes sensation, how thresholds, the psychophysical laws and signal detection theory measure it, and how vision, hearing, the chemical senses, the skin senses and the body senses work.

1.4 ○ Perception: Organising and Interpreting the World: Explain how the brain organises and interprets sensory input, from the bottom-up and top-down theories, Gestalt laws, depth cues and constancies to illusions, motion, pattern recognition, perceptual set and perception without awareness.

1.5 ●● Attention and Consciousness: Explain how attention selects, divides and sustains processing (the filter and attenuation models, automaticity, the Stroop effect and vigilance), and describe consciousness and its states: biological rhythms, sleep stages and disorders, dreams, hypnosis, meditation and psychoactive drugs.

1.6 ●●● Learning: Conditioning, Cognition and Observation: Define learning and explain habituation, classical and operant conditioning, the conditioning therapies, schedules of reinforcement, and cognitive and observational learning, naming who showed what and when.

1.7 ●●● Memory: Encoding, Storage, Retrieval and Forgetting: Explain how memories are encoded, stored, organised, retrieved, distorted and forgotten, using the multi-store and working memory models, the systems of long-term memory, the theories of forgetting and the classic amnesic cases.

1.8 ○ Thinking, Problem Solving, Reasoning, Decision Making and Language: Explain how people form concepts, solve problems, reason, judge, decide and create, and how language is structured, acquired and related to thought, naming the classic studies and their traps.

1.9 ●● Intelligence: Theories, Measurement, Heredity and Environment: Define intelligence, calculate ratio and deviation IQs, describe the Wechsler, Stanford-Binet, culture-fair and Indian tests, compare the factor, multiple, triarchic and emotional intelligence theories, interpret heritability correctly and classify intellectual disability and giftedness.

1.10 ●● Motivation: Explain why people and animals act, from instinct, drive, arousal and incentive theories through Maslow's hierarchy, achievement motivation and self-determination theory, and apply these ideas to conflict, hunger, sex and work.

1.11 ●● Emotion and Stress: Explain what an emotion is, compare the classic theories and brain systems of emotion, and describe how stress arises, how the body responds to it, and how coping, support and personality shape its effects on health.

Weightage. 815 of the 815 questions mapped in this book (100.0%) test this chapter, 402 of them from real past papers. 3 of its 11 sections are High yield.

Leipzig, in the winter of 1879. In a small room of an old university building known as the Konvikt, a doctoral student named Max Friedrich sits in front of a Hipp chronoscope, a clockwork timer that measures intervals in thousandths of a second. A stimulus is presented; Friedrich presses a telegraph key. Then the rule changes: now he must press not the instant he senses the stimulus, but only once he has clearly grasped what it is. The second reaction takes longer, and the extra time, Friedrich and his supervisor reasoned, is the time the mind needs to *apperceive*, to bring an impression into the clear focus of attention. Trial after trial, the numbers go into a notebook.

His supervisor was a 47-year-old professor of philosophy, Wilhelm Wundt. When Wundt arrived in Leipzig four years earlier, the university had given him this room to store the apparatus he used in lecture demonstrations. By the winter of 1879 it had become something new: a place where students carried out original experiments on the mind and published them as research. Friedrich's study of the duration of apperception became the first dissertation to come out of that room, and historians who later combed the Leipzig archives dated the start of the work to the last weeks of 1879. That is why textbooks, and nearly every entrance paper in India, treat Leipzig in 1879 as the birth of psychology as a science.

Look closely and that one small act holds this whole chapter. The stimulus had to be sensed and then perceived as something; attention had to seize it; practice changed the times; the instruction had to be remembered; a decision released the press; and the readiness and strain of waiting were motivation and feeling in miniature. Remember Max Friedrich pressing his key. Each section that follows takes one step of that act and asks how it works.

1.1 Psychology as a Science: Roots, Founders and the First Schools

LOWER YIELD · 46 QS

"Psychology has a long past, but only a short history," wrote Hermann Ebbinghaus in 1908. The long past is philosophy and physiology; the short history, set out in Figure 1.1, begins in Wilhelm Wundt's Leipzig laboratory in 1879.

What psychology is ○

The word joins the Greek *psyche* (soul) and *logos* (study); **Aristotle** (384 to 322 BCE) wrote the first treatise on the soul, *De Anima*. Later schools redefined the subject: mind for the philosophers, conscious experience for Wundt ("the Science of Mental Life" for William James in 1890), then behaviour for **William McDougall** (1905, 1908) and John B. Watson (1913). Woodworth's definition was the individual's activities in relation to the environment; the one Indian textbooks quote, "the scientific study of behaviour", is **Norman L. Munn**'s. Today psychology is *the scientific study of behaviour and mental processes*: behaviour is overt (speaking, blushing), **mental processes** internal and covert (perceiving, remembering).

Its goals are to describe, explain, predict and control (or influence); basic research seeks understanding for its own sake, applied research solves practical problems. Among its fields, **clinical psychology** assesses and treats psychological disorders, counselling psychology

helps people in the normal range with adjustment problems, and abnormal psychology studies the causes of disorders. Animal behaviour across species is the field of **comparative psychology**; **evolutionary psychology** explains behaviour as inherited adaptations shaped by natural selection, and differential psychology studies individual differences. Health, developmental, social, industrial-organisational, educational, forensic, experimental and neuropsychology study illness, the lifespan, people among others, work, schooling, the law, basic processes and the brain.

> **! EXAM TRAP**
>
> **Mental processes** are internal and covert, not outward behaviour, overt actions or animal behaviour. The bank credits **Munn** with "the scientific study of behaviour" ("J. B. William" is no known psychologist). Animal behaviour is **comparative** psychology, not social, abnormal or differential. Clinical psychology is the **assessment and treatment of mental illness**, not industry or organisations. "Behaviour in relation to genetic inheritance" is keyed **evolutionary psychology**.



Figure 1.1 The short history of psychology at a glance. Notice how the roots (left) cluster in German physiology and British biology, and how quickly the institutions (right) followed once Wundt's laboratory opened in 1879.

Philosophical roots ○

Philosophy asked where knowledge comes from: some is inborn, says **nativism**; all comes through the senses, says **empiricism**; reason is the chief route, says **rationalism**. **Plato** (about 427 to 347 BCE) saw learning as the recollection of inborn knowledge. Aristotle, an empiricist, gave the **laws of association**, similarity, contrast and contiguity: "Sachin" calls up "Tendulkar" by contiguity.

Rene Descartes (1596 to 1650) saw the body as a machine run by the reflex, yet gave humans a separate thinking mind that interacts with the body through the pineal gland: **dualism**. He also held **innate ideas** (God, self, infinity). His reflex led to behaviourism, and his nativism, through Kant, to Gestalt psychology.

John Locke (1632 to 1704) answered in 1690 that the mind at birth is "white paper", a **tabula rasa**, filled by sensation and reflection. Simple ideas combine into complex ones; primary qualities (size, shape, motion) lie in objects, secondary qualities (colour, taste, warmth) in the perceiver. Building all mental life from such links became the doctrine of **associationism**, founded by **David Hartley** (1749). **James Mill** (1773 to 1836) held a complex idea to be the sum of its parts, mental mechanics; his son **John Stuart Mill** (1806 to 1873) held that parts fuse into something new, as hydrogen and oxygen make water: **mental chemistry**. Before Locke, **Thomas Hobbes** (1588 to 1679) was a materialist; after him, **George Berkeley** (1685 to 1753) held that to be is to be perceived, **David Hume** (1711 to 1776) gave the laws of resemblance, contiguity and cause and effect, and **Alexander Bain** founded the journal *Mind* (1876).

German rationalists made the mind active. **Gottfried Wilhelm Leibniz** (1646 to 1716), who built reality from active units called monads, held that unnoticed *petites perceptions* (each wave in the sea's roar) add up to clear awareness, which he called apperception. **Johann Friedrich Herbart** (1776 to 1841) added a threshold of consciousness and the **apperceptive mass**, the existing ideas that must absorb a new idea for it to be understood. **Immanuel Kant** (1724 to 1804) gave the mind innate categories and denied that psychology could be a science.

On mind and body, interactionism (Descartes) lets each act on the other; **psychophysical parallelism** (Leibniz, Wundt) has them run in step without causation; epiphenomenalism (**Thomas Henry Huxley**, 1825 to 1895) makes mind a by-product of the brain; and monism admits one substance, or, in the double-aspect theory of **Baruch Spinoza** (1632 to 1677), one reality with two faces. Ontology asks what exists; epistemology, how we know.

> **! EXAM TRAP**
>
> Tabula rasa is **Locke**; Plato, Descartes and Kant were nativists. Empiricists draw knowledge from **sensory experience**; logical reasoning and contemplation are rationalist routes, and theories are products of knowledge. A practice key says Descartes' dualism influenced **behaviourism and Gestalt psychology**, not empiricism and rationalism. Post-positivism treats knowledge as fallible; scientific realism holds unobservables real. The apperceptive mass (Herbart's term, credited to Wundt by IHBAS 2025) is **pure perception and pre-existing ideas**, not associated elements (associationism), "sensation and pure perception" or Helmholtz's unconscious inference.

Physiological and biological roots ○

The **Bell-Magendie law** (**Charles Bell**, 1811; **Francois Magendie**, 1822) split the spinal nerve roots into sensory (dorsal) and motor (ventral). By the doctrine of specific nerve energies of **Johannes Muller** (1801 to 1858; *Handbook of Human Physiology*, 1833 to 1840), a sensation's quality depends on the nerve excited: press on the eye and you see light. **Hermann von Helmholtz** (1821 to 1894) timed the nerve impulse at about 27 metres per second (1850), called perception unconscious inference, and built the 1802 idea of **Thomas Young** into the trichromatic theory of colour vision. Localisation ran from the phrenology of **Franz Joseph Gall** and **Johann Gaspar Spurzheim** (character read from skull bumps, about 1800) through the

ablation studies of **Pierre Flourens** (1820s) to **Paul Broca**'s speech area (1861), **Carl Wernicke**'s comprehension area (1874) and the motor cortex of **Gustav Fritsch** and **Eduard Hitzig** (1870).

Charles Darwin (1809 to 1882; *On the Origin of Species*, 1859; *The Expression of the Emotions*, 1872) gave psychology continuity between species, the question of function and an interest in variation. **Francis Galton** (1822 to 1911) measured that variation, founding the study of individual differences: *Hereditary Genius* (1869), twin studies, an anthropometric laboratory (1884), correlation, the questionnaire, word association and eugenics. **George John Romanes** founded comparative psychology on anecdotes (1882), and **Conwy Lloyd Morgan**'s canon (1894) demanded parsimony: never a higher faculty where a lower one will do.

> ! EXAM TRAP
>
> Helmholtz is keyed for the **trichromatic theory**; the JND is Weber's, the law of effect Thorndike's, the unconscious mind Freud's.

Measuring the mind ○

Kant was answered at Leipzig. **Ernst Heinrich Weber** (1795 to 1878) found that the just noticeable difference is a constant fraction of the standard stimulus. **Gustav Theodor Fechner** (1801 to 1887), after an insight on 22 October 1850, made sensation grow with the logarithm of intensity ($S = k \log I$) and founded **psychophysics** in *Elements of Psychophysics* (1860). (Seeing a contour needs a minimum intensity and exposure time, not a minimum reaction time; Section 1.3.) In 1868 **Franciscus Cornelis Donders** (1818 to 1889) subtracted the time of a simple reaction from that of one with an added stage, such as identifying the stimulus, founding **mental chronometry**. **Hermann Ebbinghaus** (1850 to 1909), working alone, measured memory with nonsense syllables (1885; Section 1.7).

> ! EXAM TRAP
>
> The founder of psychophysics is **Fechner**, not Weber, Wundt or Ebbinghaus. Ebbinghaus, Galton and Freud each worked independently as pioneers and each devised new methods (nonsense syllables; the questionnaire and word association; free association), but Freud founded no laboratory. The Panwar bank keys (d), "(a) and (c)"; RML 2020 keyed (a), "pioneers", alone. Both statements are true, so prefer (d).

Wundt and the Leipzig laboratory ●●

Wilhelm Maximilian Wundt (1832 to 1920), a physician who had assisted Helmholtz, announced a new science in *Principles of Physiological Psychology* (1873 to 1874), "physiological" meaning borrowed experimental method. At Leipzig his students began original research in 1879, the conventional birth of psychology, and the journal *Philosophische Studien* followed in 1881. Weber, Fechner and Helmholtz were precursors; Wundt founded the discipline by giving it a laboratory, a journal and students.

He studied **immediate experience**, sensations and simple feelings that the mind actively organises through attention and will; hence **voluntarism**. Contents in the broad field of consciousness are perceived, those in its focus apperceived: the ceiling fan is perceived, the question paper apperceived. By **creative synthesis** a whole gains new properties, as a melody ex-

ceeds its notes. His **tridimensional theory of feeling** (1896) has three axes: pleasure-displeasure, tension-relaxation and excitement-depression. His method was **introspection** under experimental control (*innere Wahrnehmung*), short judgements of signalled stimuli, not armchair self-observation (*Selbstbeobachtung*). Language, myth and custom he studied in the ten-volume **Volkerpsychologie** (folk psychology, 1900 to 1920). His students included Hall, Cattell, Titchener, Kulpe, Witmer, Munsterberg, **Emil Kraepelin** (1856 to 1926; psychiatric classification), **Charles Spearman** (1863 to 1945; factor analysis) and **Vladimir Bekhterev** (1857 to 1927; reflexology).

> ! EXAM TRAP
>
> The first laboratory is **Wundt's**, at **Leipzig**, not Vienna, Paris or Cambridge; Galton's (1884) was anthropometric, Weber founded none, Jung used word association clinically, and James's 1875 Harvard room was for demonstration. *The Principles of Psychology* (1890) is **James's**. Introspection belongs to **Wundt** (Watson attacked it; Wertheimer founded Gestalt) and to **structuralism** (functionalism added tests, behaviourism rejected it, psychoanalysis used free association). Spearman is to factor analysis as Wundt is to **introspection** (humanism: Maslow, Rogers). Folk psychology (RML 2024) is **Wundt's**; Pavlov found conditioning, Roentgen X-rays.

Structuralism and its rivals ○

Edward Bradford Titchener (1867 to 1927) took a stricter version to Cornell and named it **structuralism** (1898): consciousness analysed into elements, aiming at a "periodic table" of the mind. Its elements were sensations, images and affections, varying in quality, intensity, duration, clearness and extensity; of Wundt's three feeling dimensions he kept only pleasantness-unpleasantness. Introspection was "a highly specialised form of self-observation"; its cardinal sin, the **stimulus error**, was naming the object ("an apple") instead of describing the experience (a glossy red patch). By his **context theory of meaning**, perception is a complex of sensory and imaginal processes. His society, the Experimentalists (1904), barred women.

At Wurzburg **Oswald Kulpe** (1862 to 1915) found **imageless thought**, thought without images, and showed that the task (*Aufgabe*) steers thought through what Narziss Ach called a determining tendency, creating a set (*Einstellung*): "Mysuru" brings "palace" if you want a building, "Karnataka" if a state. The act psychology of **Franz Brentano** (1874) studied mental acts and their intentionality, their pointing at objects; **Carl Stumpf** (1848 to 1936) taught the Gestalt founders; **Georg Elias Muller** (1850 to 1934) extended memory research at Gottingen. Structuralism died with Titchener in 1927: introspection could not settle disputes, and the school excluded animals, children and applied work. Its topics returned in cognitive psychology.

> ! EXAM TRAP
>
> Introspection is **self-observation**, not self-perception (a later social-psychology theory), self-thinking or attention. "Sensory and imaginal" perception is **Titchener**; Bruner and Schafer were "New Look" perception researchers, Kimble a learning theorist. Structuralism aimed at a **periodic table of elements**; "understanding consciousness" is too general and "behaviour, not mind" is behaviourism. Its heir is **cognitive psychology**, not neuroscience, behaviourism or functionalism.

Functionalism and the American pioneers ●●●

Americans asked what the mind is *for*. Their answer, **functionalism**, studies the mind as it functions in adapting the organism to its environment; it drew on Darwin, on pragmatism (ideas judged by their consequences) and on a demand for useful science.

William James (1842 to 1910), father of American psychology and inspiration of functionalism, wrote *The Principles of Psychology* (1890). It described the **stream of consciousness** ("It is nothing jointed; it flows"), personal, changing, continuous and selective; habit, "the enormous fly-wheel of society"; the self, split into the material, social and spiritual "Me" and the knowing "I" or pure ego; and, with **Carl Lange** (1885), the James-Lange theory of emotion (Section 1.11).

At Chicago, **John Dewey** (1859 to 1952) attacked the **reflex arc** in 1896: stimulus and response are not separate links, since a child's burn changes how the next flame is seen. **James Rowland Angell** (1869 to 1949) defined functional psychology (1906) and taught Watson; **Harvey A. Carr** (1873 to 1954) took the adaptive act as its unit. At Columbia, **James McKeen Cattell** (1860 to 1944) measured reaction time and individual differences and coined **mental test** (1890), **Robert Sessions Woodworth** (1869 to 1962) wrote *Dynamic Psychology* (1918) and added the organism to stimulus and response (S-O-R), and **Edward Lee Thorndike** (1874 to 1949) studied animal learning (Section 1.6).

Granville Stanley Hall (1844 to 1924) earned the first American doctorate in psychology (1878), opened the first American research laboratory (Johns Hopkins, 1883), founded the *American Journal of Psychology* (1887) and was first president of the American Psychological Association (APA, 1892); he called adolescence "storm and stress", explained by recapitulation (each child relives human evolution). **Lightner Witmer** (1867 to 1956) opened the first **psychological clinic** (Pennsylvania, 1896) and first printed "clinical psychology" (1907). **Hugo Munsterberg** (1863 to 1916) founded forensic and industrial psychology. **Mary Whiton Calkins** (1863 to 1930), refused a Harvard doctorate as a woman, became the first woman APA president (1905); **Margaret Floy Washburn** (1871 to 1939) was the first woman PhD in psychology (1894) and wrote *The Animal Mind*; **Leta Stetter Hollingworth** (1886 to 1939) disproved claims of female inferiority.

KEY STUDY: THE FIRST CASE AT THE FIRST CLINIC (1896)

A teacher brought Witmer a fourteen-year-old "chronic bad speller". Testing his vision, memory and reading rather than labelling him, Witmer found poor eyesight behind part of the problem; corrected vision and individual teaching helped. Assess first; seek the simplest cause.

Functionalism dissolved into the mainstream: Angell's student Watson dropped consciousness to found behaviourism (Section 1.2), and the functional question passed to evolutionary psychology.

! EXAM TRAP

Functionalism ("how people work, play and adapt") is **William James**, not Titchener, Watson, Hall or "William Wundt"; Darwin inspired it but led no school, and Dewey or Angell is right only for the formal Chicago school. It studies the **mind** adapting the **organism**, grew from **Darwin's evolution** (not Freud, Pavlov or Gestalt laws) and gave rise to the **evolutionary perspective**. Dewey's arc is the **reflex arc**; "efferent" is only the outgoing half; "association" and "dynamic" arcs are invented. "Clinical psychology" in print: **Witmer**, not Freud, Erikson or Cattell. APA: **1892**, not 1896. First woman APA president: **Calkins**, not Washburn, Hollingworth or the psychoanalyst Karen Horney.

Zeitgeist, great persons and "fathers"

Why credit one man and one year? The **great-person theory** (**Thomas Carlyle**, 1841) credits exceptional individuals; the **Zeitgeist** ("spirit of the times") view, championed by **Edwin Garrigues Boring** (1886 to 1968), credits ripe times, pointing to multiple discovery, as when Darwin and **Alfred Russel Wallace** reached natural selection independently (1858). "Father" may mean founder, namer, chief influence or first in a country, so read the qualifier (Table 1.1). Psychology's arrival in India, through N. N. Sengupta at Calcutta (1915, 1916), is told in Section 1.2.

Table 1.1 Who is the "father" of what.

Father of	Usual answer	Common bait
Scientific, experimental psychology	Wundt	Aristotle; Fechner
Psychophysics	Fechner	Weber
Associationism	Hartley	Aristotle
Structuralism	Titchener	Wundt (in Indian texts)
American psychology; functionalism	James	Hall; Titchener (named it)
Behaviourism	Watson	Pavlov; Skinner
Clinical psychology	Witmer	Freud; Kraepelin
Individual differences	Galton	Cattell
Intelligence testing	**Alfred Binet** (1905)	Galton; Cattell
Modern psychology in India	N. N. Sengupta	G. Bose; M. V. Gopalaswamy

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Thinking and remembering are internal, covert _____ processes.
2. Locke called the mind at birth a tabula _____.
3. Herbart's store of existing ideas is the _____ mass.
4. Wundt's system, stressing will and attention, is _____.
5. Naming "an apple" instead of describing the experience is the _____ error.
6. Dewey's 1896 paper attacked the _____ concept.
7. The first psychological clinic (1896) was opened by Lightner _____.

Answers: (1) mental, (2) rasa, (3) apperceptive, (4) voluntarism, (5) stimulus, (6) reflex arc, (7) Witmer.

1.2 Schools and Systems, the Cognitive Revolution and Indian Psychology

HIGH YIELD · 112 QS

Section 1.1 ended with structuralism and functionalism. This section meets the rival schools that followed (about 1900 to 1960), the mind's return in the cognitive revolution, and the Indian traditions, which asked the same questions two thousand years earlier.

Schools, paradigms and the four forces ○

A school of thought is a group of psychologists sharing assumptions about subject matter, method and explanation. **Thomas Samuel Kuhn** (1922 to 1996), in *The Structure of Scientific Revolutions* (1962), argued that a mature science works within one paradigm, a shared framework of problems and methods, until anomalies force a paradigm shift; a field of competing schools, as psychology was, is a **preparadigmatic science**. The humanists later numbered the schools as "forces": psychoanalysis first, behaviourism second, humanistic psychology third, transpersonal psychology fourth.

The schools divided on enduring issues, which the historian **Robert Irving Watson** (1909 to 1980) called **prescriptions** (1967): mind-body, nature-nurture, free will-determinism, mechanism-vitalism, molar-molecular, nomothetic-idiographic (general laws or the unique individual), objectivism-subjectivism and empiricism-rationalism. Rene Descartes fed two schools: his reflexes led to behaviourism, his innate ideas to Gestalt nativism.

! EXAM TRAP

The **fourth force** is **transpersonal** psychology; "experiential" is not a force. **Two Watsons**: John B. founded behaviourism (1913); Robert I. proposed the prescriptions (1967).

Behaviourism: Watson, the neobehaviourists and Skinner ●●

Behaviourism threw consciousness out of psychology altogether. Animal psychology (Thorndike's puzzle-box cats, 1898) and Russian objective psychology had prepared the ground: Pavlov's conditioned reflexes and the reflexology of **Vladimir Mikhailovich Bekhterev** (1857 to 1927), who conditioned movements.

KEY STUDY: WATSON'S MANIFESTO (1913)

John Broadus Watson (1878 to 1958) launched **behaviourism** with "Psychology as the Behaviorist Views It": psychology is "a purely objective experimental branch of natural science", its subject matter behaviour, not consciousness, and "introspection forms no essential part of its methods". Its goal is prediction and control: given the stimulus, to predict the response, and given the response, the stimulus. The same laws hold for humans and animals.

Watson treated thinking as subvocal speech, tiny movements of the larynx; this **peripheralism** puts mental life in the muscles rather than the brain. Such implicit (covert) responses cannot be seen but can be measured with instruments, so covert behaviour is "measurable but not observable", and the "mental processes" in today's definition of psychology are internal, covert processes. Infants, he held, have only three unlearned emotions, fear, rage and love; the rest are conditioned, as he and **Rosalie Rayner** tried to show with Little Albert (1920). His boast "give me a dozen healthy infants" (1925) is environmentalism at its purest.

The neobehaviourists (about 1930 to 1960) drew on logical positivism (only statements that observation can verify are meaningful) and on **operationism**, proposed by the physicist **Percy Williams Bridgman** (1882 to 1961) in 1927: a concept means the operations used to measure it, hence the operational definition. **Edward Chace Tolman** (1886 to 1959) studied molar, purposive acts and inferred intervening variables between stimulus and response, above all the **cognitive map**, a mental representation of the environment. His evidence was **latent learning**: rats that explored a maze unrewarded for ten days matched rewarded rats the day after food appeared (Tolman and Honzik, 1930). **Clark Leonard Hull** (1884 to 1952) built a hypothetico-deductive system of drive reduction in which reaction potential is habit strength multiplied by drive; **John Dollard** and **Neal Elgar Miller** (1950) recast Freud in his terms. **Edwin Ray Guthrie** (1886 to 1959) needed only contiguity and one-trial learning. **Burrhus Frederic Skinner** (1904 to 1990; *Walden Two*, 1948; *Verbal Behavior*, 1957; *Beyond Freedom and Dignity*, 1971), who studied operant behaviour, was a radical ("thoroughgoing") behaviourist: thoughts are real behaviour but not causes, whereas methodological behaviourism simply sets them aside. **Zing-Yang Kuo** (1898 to 1970) rejected instinct outright.

From the late 1950s Chomsky's attack (1959, below), instinctive drift (1961) and Kohler's insight weakened behaviourism; it left operational definitions and behaviour therapy, such as **Joseph Wolpe**'s systematic desensitisation by reciprocal inhibition (1958).

! EXAM TRAP

Watson founded behaviourism (not rationalism) and defined psychology as the science of **behaviour** (not soul, mind or consciousness). One Panwar item keys **N. L. Munn**, whose textbooks popularised "the scientific study of behaviour": prefer Watson if offered, else Munn (not Woodworth, of the S-O-R formula, or Wundt). **Cognitive map and latent learning** are Tolman's, not Kohler's (insight), Harlow's (learning sets), Skinner's, Gould's (he only tested the idea in honeybees, 1986) or Bandura's. **Dollard and Miller** = early social learning; **Bandura and Walters** (1963) = later.

Gestalt psychology and Kurt Lewin ●●●

A second rebellion, in Germany, kept consciousness but denied that experience is built from elements.

KEY STUDY: WERTHEIMER'S PHI PHENOMENON (1912)

At Frankfurt, **Max Wertheimer** (1880 to 1943) flashed two lines in quick succession. About 30 milliseconds apart they looked simultaneous, about 200 apart successive, but at about 60 a single line seemed to move: the **phi phenomenon**. The motion lay in neither stimulus, so no sum of sensations could produce it. The paper founded Gestalt psychology.

Gestalt means an organised whole, and the school's slogan is that the whole is different from (popularly, greater than) the sum of its parts: a square has closedness and an inside that none of its four lines has. Field theory holds that perception, thought and behaviour are set by the whole field of forces present at the moment. Against the elements of structuralism and the reflexes of behaviourism, Gestalt studied wholes and relations and leaned nativist; its laws of organisation (Pragnanz, figure and ground) are in Section 1.4.

Wolfgang Kohler (1887 to 1967), stranded on Tenerife from 1913 to 1920, saw the chimpanzee Sultan join two sticks to reach a banana: **insight**, a sudden reorganisation of the perceptual field (German 1917; *The Mentality of Apes*, 1925). His hens learned the relation "darker" rather than a shade (**transposition**), and his isomorphism matched the structure of experience to brain fields. **Kurt Koffka** (1886 to 1941; *Growth of the Mind*, 1921; *Principles of Gestalt Psychology*, 1935) separated the geographical environment from the behavioural one, the world as perceived. Wertheimer's *Productive Thinking* (1945) set understanding against rote. Wertheimer, Kohler and Lewin fled the Nazis to America, where Koffka had settled in 1927.

Kurt Lewin (1890 to 1947) carried field theory into motivation and social life. In B = f(P, E), behaviour is a function of the person and the environment, together the **life space**: the totality of facts determining behaviour at a moment, the environment being psychological, as it exists for that person. Topological psychology maps its regions and boundaries, hodological space its paths; valences attract or repel, and vectors push. His students found that interrupted tasks are better recalled (**Bluma Zeigarnik**, 1927) and spontaneously resumed (**Maria Ovsiankina**, 1928). He coined group dynamics, compared autocratic, democratic and laissez-faire leaders (Social chapter), and founded **action research**, a cycle of planning, acting, observing and reflecting with the people affected; he is the father of modern social psychology.

Gestalt therapy is different: an experiential, here-and-now therapy of **Frederick Salomon Perls** (Fritz Perls, 1893 to 1970), Laura Perls and Paul Goodman, famous for the empty chair, in which the client addresses an imagined person or a disowned part of the self.

! EXAM TRAP

Founders: Wertheimer, Kohler, Koffka (not Watson, James, Pavlov or Lewin); the school most important for perception. **Empty chair** = Gestalt therapy, not psychoanalysis, cognitive-behavioural or dialectical behaviour therapy, or person-centred therapy. Loose keys stretch "Gestalt" to cover the therapy: "here-and-now" is keyed to Gestalt psychology, and "Perls was a proponent of the Gestalt perspective" is accepted because "Gestalt focuses on wholeness in the present moment". A management stem adding that the parts relate to each other and to the whole means **systems theory**. **Life space** = person plus psychological environment, not groups. Lewin founded **action research**, not psychoanalysis, psychophysics or functionalism.

Psychoanalysis: the first force ○

Sigmund Freud (1856 to 1939) began as a neurologist treating hysteria (paralysis or blindness without nerve damage). In Paris in 1885 **Jean-Martin Charcot** (1825 to 1893) showed him hysterical symptoms appearing and vanishing under hypnosis; **Pierre Janet** (1859 to 1947) explained hysteria as dissociation, and **Hippolyte Bernheim** (1840 to 1919) of Nancy saw hypnosis as suggestion. **Josef Breuer** (1842 to 1925) found that his patient Anna O. lost symptoms by talking about their origins, the "talking cure" or cathartic method, and their *Studies on Hysteria* (1895) began psychoanalysis. Freud turned to free association, called dreams "the royal road" to the unconscious (*The Interpretation of Dreams*, 1900) and lectured at Clark University in 1909.

His topographic model divides the mind into conscious, preconscious and unconscious; his **structural model** (1923) into the id (impulsive, irrational, seeking immediate gratification), the ego (reality) and the superego (conscience), the ego guarding against anxiety with defence mechanisms such as repression. (Freud's neurotic anxiety is the fear that id impulses will break through.) Dissenters and neo-Freudians (Personality chapter) stressed relationships and culture: Adler (individual psychology), Jung (analytical psychology, the collective unconscious), Karen Horney (**basic anxiety**, a pervasive loneliness and helplessness), Fromm (humanistic psychoanalysis), Sullivan (interpersonal psychiatry), Erikson (psychosocial stages), Anna Freud (ego defences) and Melanie Klein (object relations). From psychoanalysis, the first force, descends the psychodynamic perspective, which explains behaviour, and many disorders, by unconscious urges and the hidden struggle over their expression.

> **! EXAM TRAP**
>
> Freud began as a **neurologist**. **Father of psychoanalysis**: Freud, not Bernheim, Adler or Jung. Psychoanalytic theory is Freud's, **psychosocial** theory Erikson's. Gratification, irrational, impulsive = **id**. Horney's **basic anxiety** is not Freud's neurotic anxiety, existential anxiety or "innate anxiety" (no such technical term).

Humanistic, existential and positive psychology ●●●

Against the determinism of psychoanalysis (instinct) and behaviourism (reward), **Abraham Harold Maslow** (1908 to 1970) launched a third force and studied self-actualisation, the fulfilment of one's potential (Section 1.10); **Carl Ransom Rogers** (1902 to 1987) described an innate actualising tendency that thrives on unconditional positive regard. **Charlotte Buhler** (1893 to 1974) added a lifespan view and **Rollo May** (1909 to 1994) existentialism. The Association for Humanistic Psychology formed in 1961 or 1962 (sources differ), and the Old Saybrook conference (1964) marked the movement's formal birth. Humanistic psychology emphasises free will, the whole person, an innate tendency to grow and reaching maximum potential, understood from inside the person's experience: **phenomenology**, after **Edmund Husserl** (1859 to 1938), describes experience by bracketing assumptions about its causes. Disorder means blocked growth; healing comes from acceptance, respect and belief in the person's potential, not from reinforcement or uncovering unconscious conflict. Critics call humanistic and phenomenological approaches vague and blind to the wider social and cultural contexts of experience.

Existential psychology starts from the givens no one escapes, which **Irvin David Yalom** (1980) named the four ultimate concerns: death, freedom, isolation and meaninglessness.

Soren Kierkegaard (1813 to 1855) called anxiety the dizziness of freedom; **Martin Heidegger** (1889 to 1976) named human existence **Dasein**, being-in-the-world (1927); **Jean-Paul Sartre** (1905 to 1980) held that existence precedes essence, so each person must keep recreating himself and meaning is never fixed. **Ludwig Binswanger** (1881 to 1966) and **Medard Boss** (1903 to 1990) built Daseinsanalysis on Heidegger. **Viktor Emil Frankl** (1905 to 1997) founded **logotherapy**: the will to meaning is the primary human motive, and its frustration brings an existential vacuum or a **noogenic neurosis**. May (*The Meaning of Anxiety*, 1950) separated normal, usable anxiety, existential anxiety included, from neurotic anxiety.

Maslow later foresaw a fourth force (1968), **transpersonal psychology**, the study of peak experiences, meditation and self-transcendence. Positive psychology, the science of positive experiences, traits and institutions, was launched by **Martin Elias Peter Seligman** (born 1942) as President of the American Psychological Association in 1998, with **Mihaly Csikszentmihalyi** (1934 to 2021), known for flow. Its themes include the PERMA model of well-being (Seligman, 2011; positive emotion, engagement, relationships, meaning and accomplishment), C. R. Snyder's hope (the will and the ways to reach a goal), which its interventions promote, and **savouring**, deliberately enhancing the enjoyment of positive experiences.

> ! EXAM TRAP
>
> **Third force** = humanism (Maslow, Rogers), not cognitivism or the biopsychosocial model. "Uncovering unconscious fears" is psychoanalytic, not existential. **Dasein** and **noogenic anxiety** belong to the existential school. **Will to meaning** = logotherapy, not Ellis's rational emotive behaviour therapy, acceptance and commitment therapy or the broader existential-humanistic therapy. Positive psychology was founded by **Seligman**, not Gardner (multiple intelligences), Pinker (language and evolutionary psychology) or Bandura; its interventions promote **hope**; savouring is not avoidance, relaxation or trauma work.

The cognitive revolution and today's perspectives ●●

Other critics charged behaviourism with leaving out the mind. Tolman's maps, Kohler's insight and **Karl Spencer Lashley**'s (1890 to 1958) argument of 1948 that rapid sequences such as speech need central plans were anomalies, and the computer made inner states respectable. The symbolic birth year of the **cognitive revolution** is 1956: **Allen Newell** and **Herbert Alexander Simon**'s Logic Theorist modelled problem solving, **Noam Chomsky** (born 1928) presented formal models of language, **George Armitage Miller** (1920 to 2012) gave "the magical number seven, plus or minus two", and the Dartmouth project launched artificial intelligence. **Donald Eric Broadbent** (1926 to 1993) produced the first major information-processing theory (1958); Chomsky's 1959 review of Skinner's *Verbal Behavior* argued that children produce sentences never reinforced; and **Ulric Neisser** (1928 to 2012), the father of cognitive psychology, named the field in *Cognitive Psychology* (1967).

Cognitive psychology studies internal mental processes such as attention, memory and problem solving through the **information-processing approach**, which treats the mind as a computer passing information through stages of limited capacity. Its forerunners were **Frederic Charles Bartlett** (schemas, 1932) and **Jean Piaget** (1896 to 1980), who studied children's thinking from the 1920s. Cognitive neuroscience asks how the brain does it, and **connectionism**, or parallel distributed processing (David Rumelhart and James McClelland,

1986), stores knowledge as weights across many units working at once. **Albert Bandura**'s (1925 to 2021) self-efficacy (1977), belief in one's capacity to act, and the cognitive therapies of Albert Ellis and Aaron Beck, which target maladaptive thoughts and cognitive distortions, fed cognitive-behavioural therapy (CBT; Clinical chapter).

By the 1970s schools had given way to coexisting perspectives (Table 1.2, applied to Ravi, a Mysuru student anxious before every examination). Some lists add a developmental perspective, the study of change as people mature. Cross-cultural psychology compares processes across cultures, cultural psychology studies how culture and mind make each other up, indigenous psychology builds theory from within one culture, and **eclecticism** selects concepts and techniques from several schools to fit the problem.

Table 1.2 The contemporary perspectives, applied to one case of examination anxiety.

Perspective	Explains behaviour through	Ravi's anxiety comes from
Biological (neurobiological)	Brain, nervous system, genes	An over-reactive threat system
Evolutionary	Natural selection	An ancient alarm misfiring
Psychodynamic	Unconscious conflict, early childhood	Fear of parental disapproval
Behavioural	Stimulus-response conditioning	Failure paired with humiliation
Cognitive	Internal mental processes	Catastrophic thoughts
Humanistic	Free will and growth	Worth conditional on marks
Sociocultural	Culture and social context	Competitive family pressure
Biopsychosocial (George Engel, 1977)	All three levels together	All of these, interacting

> ! EXAM TRAP
>
> **First major information-processing theory**: Broadbent, 1958 (not Treisman, about 1960; Chomsky, 1957; or Watson, 1920). **Pioneer of cognitive psychology** against Kohler, Chomsky and Kohlberg (moral development): **Piaget**. "Behaviour is governed by the brain and nervous system" = **neurobiological**, not psychophysical (Fechner's stimulus-sensation measurement, Section 1.3) or neo-behaviouristic. Cultural differences are studied by **cross-cultural psychologists** (a Panwar key says social psychologists, the vaguer answer).

Indian psychology: the idea and its pioneers ●●●

Psychology in India is the discipline as practised here, long mostly Western psychology on Indian soil. **Indian psychology** is an approach that integrates the classical Indian traditions with psychological concepts. It is consciousness-centred; holistic, treating body, mind, spirit, family and community as one system; transformative; open to disciplined first-person methods such as meditation; and value-laden. In mental health it combines traditional practices such as yoga with modern therapy.

Modern psychology entered India at Calcutta. **Narendra Nath Sengupta** (1889 to 1944) set up the first laboratory there in 1915 and the first Department of Psychology in 1916, and founded the Indian Psychological Association (1925), whose *Indian Journal of Psychology* began in 1926; he is the father of modern psychology in India. **Girindrasekhar Bose** (1887 to 1953) founded the Indian Psychoanalytical Society (1922) and corresponded with Freud; the father of psychoanalysis in India, he proposed the **theory of opposite wishes**, in which every wish is

paired with its opposite, one expressed and one repressed. **M. V. Gopalaswamy** began psychology at Mysore in 1924; Madras (1943) and Pune (1950) followed.

Durganand Sinha (1922 to 1998) urged **indigenisation**, making psychology's concepts and methods fit Indian culture, either from without, by adapting Western tools (as in **Uma Chowdhury**'s Indian Thematic Apperception Test, about 1960), or from within, from the culture's own traditions. **Jai B. P. Sinha** proposed the **nurturant-task leader** (1980), whose warm care depends on hard work; **Udai Pareek** (1925 to 2010) studied role stress; and **Sudhir Kakar** (born 1938) applied psychoanalysis to Indian childhood and society. The Indian Psychology movement dates from the Pondicherry Manifesto (2002). **Matthijs Cornelissen** founded the Indian Psychology Institute; **Koneru Ramakrishna Rao**, **Anand C. Paranjpe** and **Ajit K. Dalal** edited the *Handbook of Indian Psychology* (2008); and Cornelissen, **Girishwar Misra** and **Suneet Varma** edited *Foundations of Indian Psychology* (2011).

> **! EXAM TRAP**
>
> Indian psychology is not Western theory applied exclusively, folklore, or meditation without psychological context. Keys naming Bose (IHBAS) or Uma Chaudhuri (Panwar) for the **first laboratory** are wrong: it was Sengupta's; without Sengupta, choose Bose (Mangal and A. K. Singh are textbook authors; neither they nor Rathi founded a laboratory). **Sequence**: Calcutta (1916), Mysore (1924), journal (1926), Madras, Pune; a key putting the journal before Mysore is wrong. Bose's theory is **opposite wishes** (not repression, dharma-sankata or "guilt theory"). "Amitosh Mukherjee" garbles Asutosh Mookerjee, a Calcutta vice-chancellor. Kakar is keyed for Horney's basic anxiety only as the sole psychoanalyst offered ("Kamala Chaudhuri" there probably garbles Uma Chowdhury). "G. R. S. Rao" as father of Indian psychology (RML 2025) probably means K. Ramakrishna Rao.

Samkhya, the three gunas and the inner instrument ●●●

Indian philosophies (darshanas) are **astika** if they accept the Vedas (Nyaya, logic; Vaisheshika; Samkhya; Yoga; Mimamsa, ritual; Vedanta) and nastika if not (Buddhism, Jainism, the materialist Charvaka); Sufism, Islamic mysticism, lies outside the scheme.

Samkhya, attributed to the sage **Kapila** and set out in Ishvarakrishna's *Samkhya Karika*, is a dualism of **purusha**, pure consciousness, the passive, unchanging witness, and **prakriti**, primordial matter, unconscious, active and ever-changing, from which all forms of life emerge. In purusha's presence prakriti evolves, down to the five great elements (pancha mahabhutas), as Figure 1.2 shows. We suffer by mistaking prakriti's activity for the self ("I am angry"); discriminating the seer from the seen brings liberation, kaivalya.

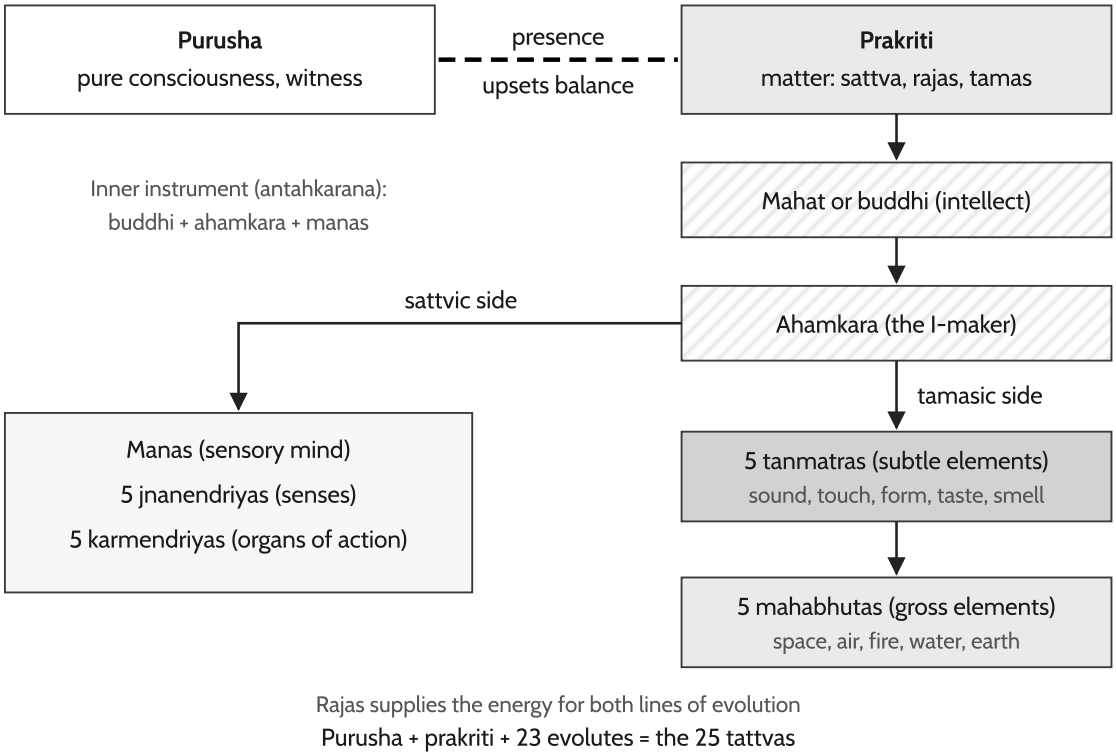


Figure 1.2 Samkhya's map of mind and matter. In the presence of purusha, prakriti evolves into intellect, the I-maker, the sensory mind, the ten organs and the elements. Notice that everything below the top row, including thinking, feeling and the sense of "I", is a product of prakriti; only purusha is conscious.

Prakriti is woven of three strands, the **gunas**, always mixed in varying proportions: sattva (clarity, calm, harmony, balance), rajas (activity, passion, desire, restlessness) and tamas (inertia, lethargy, darkness, dullness, ignorance). The *Bhagavad Gita* (chapters 14, 17 and 18) made them a psychology of personality. Before the same examination, Ananya keeps her plan and sleeps on time (sattva), Rohan flits between six guidebooks and lies awake racing (rajas), and Karthik sleeps till noon, telling himself there is no point (tamas). Nobody is pure; diet, routine and practice shift the balance.

The mind's working part is the **antahkarana**, the inner instrument: manas, the sensory mind that attends and doubts; ahamkara, the I-maker or ego; buddhi, the intellect that decides; and, in later Vedanta, chitta, the store of memory impressions.

> **! EXAM TRAP**
>
> Purusha is **not** dynamic or ever-changing; prakriti is not pure consciousness, unchanging or the intelligent self. The gunas belong to **prakriti** (not purusha, atman, Brahman, samadhi or samskara) and come from **Samkhya** with the *Gita* (not Nyaya, Charvaka or Mimamsa). Bavisha, who has stopped eating and cannot act, shows **tamas alone** (yama is a yoga limb, not a state). Not gunas: vata (a dosha), ahimsa, prakriti, ahamkara, or "karma, dharma, moksha". The ego is **ahamkara**; the *Gita*'s three elements of mind are manas, buddhi and ahamkara, not "prakriti, purusha, kshetra" (kshetra, the field, is the body that the self knows) or "dharma, artha, kama".

Yoga, the self and other Indian paths ●●●

Samkhya says what the mind is made of; yoga says how to change it. **Patanjali**'s *Yoga Sutras* adopt Samkhya's map, and sutra 1.2 defines yoga as **chitta-vritti-nirodha**, the stilling of the mind's fluctuations; the goal is liberation (kaivalya). The five fluctuations (vrittis) are valid cognition, error, verbal imagination, sleep and memory. They are stilled by practice (abhyasa) and non-attachment (vairagya), and driven by five afflictions (kleshas): ignorance (avidya), egoism, attachment, aversion and clinging to life. The method is the eight limbs, in fixed order (Table 1.3).

Table 1.3 The eight limbs of Patanjali's yoga, in order.

Limb	Meaning
1. Yama	Ethical restraints, such as non-violence (ahimsa)
2. Niyama	Personal observances, such as purity
3. Asana	Steady, comfortable posture
4. Pranayama	Breath regulation
5. Pratyahara	Withdrawal of the senses inward
6. Dharana	Concentration on one point
7. Dhyana	Meditation: an unbroken flow of attention
8. Samadhi	Absorption in the object

A student meditating on a candle flame turns her senses away from the hostel's noise and smells (pratyahara), fixes on the flame and brings attention back whenever it drifts (dharana), finds it flowing unbroken (dhyana), and for a moment loses even "I am meditating" (samadhi).

The *Gita*'s principle of disinterestedness is **nishkama karma**, acting without attachment to the fruits. It drives karma yoga, beside the paths of knowledge (jnana) and devotion (bhakti), and answers Arjuna's dharma-sankata, a dilemma in which every action breaks some duty. The Upanishads call the true self **atman**, eternal consciousness distinct from the body and from passing states of mind; **Adi Shankaracharya**'s Advaita Vedanta (eighth century CE) makes it one with Brahman, ultimate reality, and calls the world of many things maya, appearance. The *Mandukya Upanishad*'s four states are waking (jagrat), dreaming (svapna), deep sleep (sushupti) and **turiya**, "the fourth", the consciousness that witnesses the other three. The *Taittiriya Upanishad*'s five sheaths (**panchakosha**), outer to inner, are annamaya (the food body), pranamaya (breath), manomaya (mind and senses, the sheath of perception), vijnanamaya (intellect) and anandamaya (bliss); the middle three form the subtle body (linga sharira).

Buddhism (**Siddhartha Gautama**, the Buddha, traditionally 563 to 483 BCE) teaches four Noble Truths: suffering, its cause in craving, its cessation and the Noble Eightfold Path. Its anatta denies a permanent self, its Abhidhamma catalogues states of mind, and vipassana (insight meditation) trains sati, mindfulness. Sufi psychology refines the nafs, the lower self, from the commanding ammara to the tranquil mutma'inna and the ego's passing away (fana). **Sri Aurobindo** (1872 to 1950) put the **psychic being**, the evolving soul, at the centre of integral yoga, behind an outer being of body, vital (life-energy, desire and emotion) and

mind. Ayurveda's three doshas (vata, pitta, kapha) govern the body. Karma's impressions (samskaras) work like habits, and dharma, the order of duties, heads the four aims of life (purusharthas): dharma, artha (wealth), kama (pleasure) and moksha (liberation).

CLINICAL LINK

Buddhist mindfulness underlies third-wave CBT, such as **Marsha Linehan**'s dialectical behaviour therapy, which blends CBT with Zen mindfulness and dialectical philosophy (Clinical chapter). Psychoanalysis is no wave of CBT.

! EXAM TRAP

Goal of yoga: control of the mind and kaivalya, not fitness, cognitive enhancement or emotion regulation only. **Last four limbs**: pratyahara, dharana, dhyana, samadhi. One point = **dharana**; senses turned inward = **pratyahara** (not pranayama or niyama); karma and jnana are not limbs. Nishkama karma is the *principle*, karma yoga the path. **Atman** is not the body, a passing state or Jung's collective unconscious; turiya is not deep sleep. **Annamaya** is outermost, **manomaya** perceptual; the atman is never a sheath, so a key adding it to the subtle body is wrong.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Kuhn called a field of competing schools a _____ science.
2. Tolman's unrewarded rats showed _____ learning.
3. Lewin's life space is the person plus the _____ environment.
4. Frankl called neurosis from frustrated meaning _____ neurosis.
5. India's first psychology laboratory (1915) was set up by _____.
6. The guna of inertia and ignorance is _____.
7. The yoga limb that withdraws the senses inward is _____.

Answers: (1) preparadigmatic, (2) latent, (3) psychological, (4) noogenic, (5) Narendra Nath Sengupta, (6) tamas, (7) pratyahara.

1.3 Sensation and Psychophysics LOWER YIELD · 40 QS

Everything you know about the world arrives as physical energy. This section follows that energy into experience, shows how sensation is measured, and then takes each sense in turn.

From energy to experience ●●

In **sensation** the sense organs detect physical energy and convert it into neural signals; in **perception** the brain organises and interprets those signals. Optical illusions therefore distort perception, not sensation: in the Muller-Lyer figure the receptors register the lines correctly (Section 1.4). The structuralist **Edward Bradford Titchener** (1867 to 1927) defined a perception as a complex of sensory and imaginal elements.

A receptor is a cell specialised for one form of energy, and its conversion of that energy into neural signals is **transduction**. Because all impulses look alike, **Johannes Peter Muller** (1801 to 1858) proposed the **doctrine of specific nerve energies** (today, labelled-line coding): the quality of a sensation depends on which nerve is stimulated, not on what stimulates it, so pressing a closed eyelid makes you see light. Intensity is coded differently. By the **all-or-none law** an action potential fires at full size once threshold is reached; a stronger stimulus never makes a bigger impulse, and a sub-threshold one makes none, not a miniature or graded one. Intensity is therefore carried by rate coding (faster firing) and by recruiting more neurons. Every sense except smell relays to the cortex through the thalamus (vision via its lateral geniculate nucleus, hearing via the medial). Finally, **sensory adaptation** is the fading of response to an unchanging stimulus, which is why you soon stop feeling your wristwatch.

> **! EXAM TRAP**
>
> Illusions distort *perception*; "translation" and "adaptation" are bait. "Sensory and imaginal" is Tit*chener*, not Bruner (Section 1.4). Specific nerve energies is *Muller*, not Helmholtz. The relay for every sense but smell is the *thalamus*, not the hypothalamus (drives), medulla (breathing), corpus callosum (joins the hemispheres), basal ganglia (movement) or hippocampus (memory).

Thresholds and the psychophysical laws ○

Gustav Theodor Fechner (1801 to 1887) founded **psychophysics**, the measurement of the relation between stimuli and sensations, with his *Elements of Psychophysics* (1860). The **absolute threshold** (or limen) is the minimum intensity a person can detect. It is not a switch: as a faint tone grows stronger, "yes" answers climb gradually along an S-shaped psychometric function, since neurons fire spontaneously. The threshold is therefore defined as the intensity detected on *50 percent* of trials; weaker stimuli are subliminal. **Eugene Galanter** (1962) gave the classic examples: a candle flame 48 kilometres away on a dark night, a watch ticking at 6 metres. The difference threshold, or **just noticeable difference** (JND), is the smallest change detected 50 percent of the time.

Ernst Heinrich Weber (1795 to 1878), comparing lifted weights in 1834, found the JND to be a constant fraction of the standard: **Weber's law**, $\Delta I/I = k$. If k for weights is 0.02, 500 grams needs 10 grams more to feel heavier, but 5 kilograms needs 100. The fraction is finest for pitch and among the coarsest for taste and smell. Fechner then assumed that every JND is an equal unit of sensation; counting JNDs gives **Fechner's law**, $S = k \log I$, so each doubling of light adds about one equal step of brightness. **Stanley Smith Stevens** (1906 to 1973) instead had people give each stimulus a number proportional to its strength (**magnitude estimation**) and found **Stevens' power law** (1957), $S = kI^n$: exponents below 1 compress (brightness, about 0.33), exponents above 1 expand (electric shock, about 3.5). Cross-modality matching (squeezing a hand grip as hard as a sound is loud) confirmed them (Table 1.4).

Table 1.4 The three psychophysical laws compared.

Law	Formula	Main limitation
Weber (1834)	$\Delta I/I = k$	Fails near threshold
Fechner (1860)	$S = k \log I$	Assumes equal JNDs; no expansion

Law	Formula	Main limitation
Stevens (1957)	$S = kI^n$	Swayed by range and number habits

> **! EXAM TRAP**
>
> Minimum detectable stimulus: *absolute threshold*; smallest detectable change: *difference threshold*. "Sensory adaptation point", "perceptual boundary" and "signal detection of response" are invented; response bias is a signal detection term, and the phi phenomenon is apparent movement (Section 1.4). Weber's JND is a *constant fraction*, not a fixed amount. The equal-JND assumption is *Fechner's*, not Weber's. The father of psychophysics is Fechner, not Ernst Weber (whose JND he built on), Wilhelm Wundt (experimental psychology), Hermann Ebbinghaus (memory, inspired by Fechner) or Edward Tolman (learning).

Measuring thresholds and signal detection ○

Fechner's **method of limits** changes the stimulus in steps, in alternating ascending and descending series. Its errors of *habituation* (keeping an answer too long) and *anticipation* (switching too early) are cancelled by counterbalancing. The **method of constant stimuli** presents fixed values many times in random order: most accurate, but slow. In the method of adjustment (average error) the observer sets the stimulus: fastest, but least precise. Modern staircases weaken the stimulus after each "yes" and strengthen it after each "no".

Yet a threshold mixes what an observer can sense with how readily the observer says "yes", as wartime radar operators showed; **signal detection theory** (SDT), set out by **David M. Green** and **John A. Swets** in 1966, separates the two. Noise alone and signal plus noise produce overlapping distributions of internal evidence, and the observer says "yes" above a cut-off, the criterion. "Yes" to a signal is a *hit*, "no" a *miss*; "yes" to noise alone is a *false alarm*, "no" a *correct rejection*. Sensitivity, or **d prime** (d′), is the distance between the distributions, *z*(hit rate) minus *z*(false alarm rate): zero when hits equal false alarms, larger as hits outnumber them. The criterion (β or *c*), or **response bias**, follows expectations and payoffs. A *liberal* observer (a smoke alarm) gives many hits and false alarms; a *conservative* one (a court) few false alarms but more misses. Plotting hit rate against false alarm rate as the criterion moves gives the **receiver operating characteristic** (ROC) curve. It is bowed, not straight as a fixed threshold predicts, which is why SDT replaced the threshold. Lie detection and claims of subliminal perception are judged the same way.

> **! EXAM TRAP**
>
> Telling a signal from background noise is *sensitivity* (*d′*); willingness to say "yes" is the *criterion* (β); the ROC curve plots both; the JND and Weber's constant are classical psychophysics. SDT added the *response criterion*. "Receiver-operant characteristics in communication studies" (IHBAS 2025) means *audience factors influencing persuasion* (Social Psychology chapter), not the ROC curve.

Vision: light, the eye, rods and cones ●●●

The eye responds to wavelengths of about 380 to 760 *nanometres*; shorter waves are ultraviolet, longer ones infrared. Since speed equals frequency times wavelength, frequency and wavelength are *inversely* related, for light and sound alike. Wavelength is seen as hue, inten-

sity as brightness and purity as saturation. Light passes through the cornea, the pupil (sized by the iris) and the lens, which changes shape to focus (**accommodation**), to the retina. In the retina's macula lies the **fovea**, packed only with cones and giving the sharpest vision. The optic disc, where the optic nerve leaves, has no receptors and forms the **blind spot**.

About 120 million rods, with one pigment, rhodopsin, serve **scotopic vision**, vision in dim light: a rod responds to a single quantum of light, but many rods converge on each ganglion cell, so rod vision is blurred (and, with one pigment, colourless). Rods dominate the periphery and are absent from the fovea. About 6 to 7 million cones, with three pigments, serve photopic vision: colour and fine detail in moderate to bright light. Walk into a dark cinema and **dark adaptation** slowly raises your sensitivity to dim light, first in the cones and then, after the rod-cone break, in the rods as bleached rhodopsin regenerates (Figure 1.3). Light adaptation takes seconds to a minute; both are visual adaptation, adjustment to different light intensities. At dusk, as peak sensitivity shifts from cones (about 555 nanometres) to rods (500), reds darken before blues: the **Purkinje shift**, described by **Jan Evangelista Purkyne** (Purkinje) in 1825.

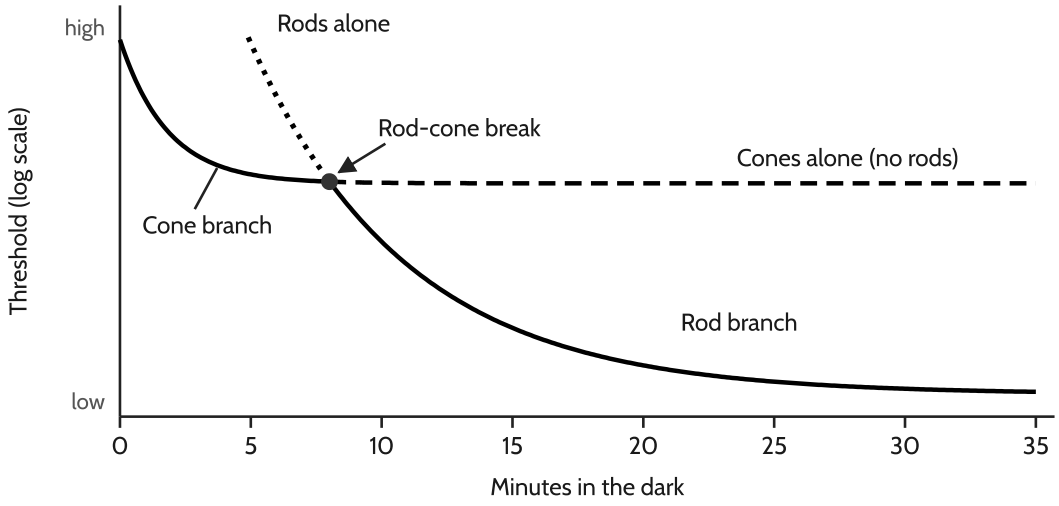


Figure 1.3 The dark adaptation curve. The threshold for detecting light falls rapidly as the cones adapt and levels off after a few minutes (cone branch). At the rod-cone break, about 7 to 10 minutes into darkness, the more sensitive rods take over and the threshold continues to fall for 20 minutes or more (rod branch). A person with no functioning rods would follow the dashed line; one with no functioning cones would show only a rod curve.

Visual acuity, the ability to see fine detail, peaks at the fovea. Rapid jumps between fixations, **saccades**, and tiny microsaccades keep the image moving on the retina; a stabilised image fades within seconds, so these movements keep perception constant. Illusory light and dark stripes at a light-dark edge, **Mach bands** (**Ernst Mach**, 1865), come from **lateral inhibition**: retinal cells suppress their neighbours, so a cell on the light side of the edge, having dark neighbours, is less inhibited and signals "brighter", exaggerating contrast.

! EXAM TRAP

Scotopic vision is *rods*; the retina and iris are structures, not receptors. Cones peak in the *fovea centralis*, not the optic disc, the periphery, "the macula but not the fovea" (the fovea lies inside the macula) or the ora serrata (its front edge). "Cones give only monochromatic information" is false: rods do. Dark adaptation raises sensitivity to *low* illumination, not bright or coloured light; adjusting to light levels is *visual adaptation*, not accommodation or Piaget's assimilation. *Saccades* ensure constant perception; fixations are pauses. Mach bands come from *lateral inhibition*, not the opponent process (colour afterimages), the rod-cone dichotomy (day versus night vision) or temporal summation (summing weak signals over time).

Colour vision ○

Lights mix additively (red, green and blue light make white, as on a screen); paints mix subtractively (each absorbs wavelengths, so blue and yellow leave green). **Thomas Young** (1802) and **Hermann von Helmholtz** (1821 to 1894, in 1852) proposed the **trichromatic theory**: three receptor types whose ratio of activity codes colour, confirmed in 1964 as short-, medium- and long-wavelength cones. **Ewald Hering** (1834 to 1918) proposed the **opponent-process theory** (1878): opposed pairs of red versus green, blue versus yellow and black versus white. It explains why nobody sees a reddish green and why staring at red leaves a green negative afterimage. The dual-process theory unites them: trichromatic in the cones, opponent in later cells. In colour vision deficiency, anomalous trichromacy (one cone type shifted) is commonest; in dichromacy one cone type is missing (protanopia, long, and deuteranopia, medium, confuse red with green; tritanopia, short, blue with yellow); in monochromacy at most one works. Red-green defects are X-linked, so far commoner in men; the plates of **Shinobu Ishihara** (1917) screen for them.

! EXAM TRAP

Helmholtz gave the trichromatic theory (with Young), the place theory of hearing and unconscious inference, not the JND (Weber and Fechner), the unconscious (Sigmund Freud) or the law of effect (Edward Thorndike).

Hearing ○

A sound's frequency, in hertz (Hz), is heard as pitch (humans hear about 20 to 20,000 Hz); its amplitude as loudness, in decibels (dB), a logarithmic scale on which each 10 dB is a tenfold rise in intensity; its complexity as timbre. Sound passes from the pinna to the tympanic membrane, across the ossicles (malleus, incus, stapes) to the oval window and into the cochlea, where waves along the basilar membrane bend the hair cells of the organ of Corti. The bending opens channels mechanically, so potassium flows in and the cell depolarises.

Helmholtz's **place theory** (1863), confirmed by the travelling wave of **Georg von Bekesy** (Nobel Prize 1961), holds that each frequency peaks at its own place: high near the narrow base, low near the floppy apex. It best explains high pitches. In the **frequency theory** of **William Rutherford** (1886) the nerve fires in step with the sound, which suits low pitches, but no neuron fires much faster than 1,000 times a second. **Ernest Glen Wever** and **Charles William Bray** (1930) answered with the **volley principle**: fibres fire in staggered turns, like soldiers in rotation, following up to about 4,000 to 5,000 Hz. Today, timing codes low pitches and place

high ones. Sounds are located by interaural differences in time (low frequencies) and intensity (high). Outer or middle ear damage (wax, otosclerosis) causes conductive hearing loss; damage to hair cells or the auditory nerve (noise, ageing) causes permanent sensorineural loss.

> **! EXAM TRAP**
>
> Frequencies flexing different parts of the basilar membrane is *place coding*; frequency coding is firing in time with the wave, rate coding signals loudness, and "pitch coding" is not a term. Place theory does not explain low pitches, localisation (comparing the ears) or loudness (firing rate). Hair cells transduce by *mechanical bending*; closing sodium channels happens in photoreceptors, and GABA and chemical signalling are wrong. The frequency theory is William, not Ernest, Rutherford's.

Taste and smell ○

Taste receptors sit in taste buds inside the **papillae** of the tongue (fungiform, foliate and circumvallate; filiform papillae have none and sense texture). The five basic tastes are sweet, salty, sour, bitter and **umami**, the savoury taste of glutamate named by **Kikunae Ikeda** (1908); the "tongue map" of taste zones is a myth. Smell receptors lie in the olfactory epithelium high in the nose. Pheromones, chemical signals between members of a species, are clear in many animals but doubtful in humans. Flavour is taste plus smell, which is why food tastes flat with a cold. **Hans Henning** (1916) arranged six primary odours on a smell prism and four tastes on a taste tetrahedron, but the hundreds of smell receptor types found since 1991 rule out a small set of primary odours.

> **! EXAM TRAP**
>
> Papillae serve *taste*. The olfactory nerve (I) is the only sensory nerve whose pathway bypasses the thalamus; the optic (II), trigeminal (V) and vestibulocochlear (VIII) relay there.

Skin, pain and the body senses ○

The skin senses are touch, pressure, temperature and pain. Pain is detected by free nerve endings, nociceptors, and ascends in the anterolateral system; fine touch, vibration and limb position take the dorsal columns. In the **gate control theory** of **Ronald Melzack** and **Patrick David Wall** (1965), a gate in the spinal cord's substantia gelatinosa is opened by small pain fibres, closed by large touch fibres (so rubbing a bump helps) and set from above by attention and emotion; endorphins, the body's own opiates, help to close it.

> **KEY STUDY: RAMACHANDRAN FINDS A HAND ON THE FACE (1990S)**
>
> Most amputees still feel the missing limb, a **phantom limb** (the term of **Silas Weir Mitchell**, 1871). **Vilayanur Subramanian Ramachandran** (born 1951) touched an arm amputee's face with a cotton bud, and the man felt it on specific fingers of the phantom hand. The face area of the somatosensory cortex borders the hand area, so phantoms reflect *cortical reorganisation*: neighbours invade the lost limb's territory.

Kinesthesis is the sense of limb position and movement, from proprioceptors in muscles, tendons and joints. The vestibular sense (balance) comes from the inner ear. The three **semicir-**

cular canals detect rotational acceleration, so after spinning the still-moving fluid makes you feel you are turning; the otolith organs (utricle and saccule), whose calcium carbonate crystals are pulled by gravity, detect tilt and linear acceleration.

> **! EXAM TRAP**
>
> The anterolateral system carries *nociceptive* signals, not proprioception, vibration or touch. Phantoms are *cortical remapping*, not nerve regrowth, denial or cerebellar damage. Dizziness after spinning is the *semicircular canals*, which detect rotation in three planes, not static position or linear motion (otolith organs and their crystals), limb position (proprioceptors) or pitch (cochlea).

Senses in concert and senses deprived ○

Sensory interaction is one sense shaping another: in the McGurk effect (Harry McGurk and John MacDonald, 1976), "ba" heard with lips saying "ga" is perceived as "da". Combining senses aids learning, so teaching for learning difficulties uses multisensory input with regular repetition. In **synaesthesia** a minority automatically and consistently experience a second sensation, such as coloured letters. Remove input and the brain makes its own: in the McGill laboratory of **Donald Olding Hebb** (Bexton, Heron and Scott, 1954), students in goggles and gloves quit within days, thought poorly and hallucinated, the classic sensory deprivation study. Likewise, **Charles Bonnet syndrome** (**Charles Bonnet**, 1760) is complex visual hallucinations in people with sight loss who are cognitively intact and keep insight.

> **! EXAM TRAP**
>
> Charles Bonnet syndrome is not auditory hallucinations in dementia or deafness, tactile ones in alcohol withdrawal, multimodal ones in delirium or olfactory auras before seizures.

> **SCAN YOUR BRAIN**
>
> Fill in the blanks. The answers are below; cover them first.
>
> 1. The absolute threshold is detected on _____ percent of trials.
> 2. SDT indexes sensitivity by _____.
> 3. Scotopic vision depends on the _____.
> 4. Mach bands are explained by lateral _____.
> 5. High frequencies peak near the _____ of the basilar membrane.
> 6. Rotational acceleration of the head is detected by the _____ canals.
>
> **Answers:** (1) 50, (2) d′ (d prime), (3) rods, (4) inhibition, (5) base, (6) semicircular.

1.4 Perception: Organising and Interpreting the World LOWER YIELD · 38 QS

Section 1.3 followed light as far as the retina. This section asks what the brain does with it next, and why seeing, as its illusions show, is far less direct than it feels.

Inference or direct pickup: the theories ●●

Perception is the selection, organisation and interpretation of sensory signals into meaningful objects and events: sensation gives you a pattern of light, perception your mother's face. **Bottom-up processing** (data-driven) builds the percept from features of the stimulus; **top-down processing** (conceptually driven) lets knowledge, expectation and context shape how those features are read. The theories differ in the weight they give each.

For **Hermann von Helmholtz** (1821 to 1894), perception was **unconscious inference**: rapid, automatic conclusions from sensory evidence plus past experience. **Richard Langton Gregory** (1923 to 2010) built this into the **constructivist approach**, the indirect or top-down theory (*Eye and Brain*, 1966), in which a percept is a best-guess hypothesis; that is why a hollow mask looks like a normal, convex face. The **Bayesian approach** puts the idea in probabilities: the percept combines prior beliefs with sensory evidence, and the weaker the evidence, the more the priors decide.

The classic bottom-up theory is the **direct perception** (ecological) approach of **James Jerome Gibson** (1904 to 1979). The light reaching a moving observer, he argued, specifies the world without inference: optic flow streams outwards from the point you are heading for, and objects offer **affordances**, possibilities for action perceived directly (a chair affords sitting). **Ulric Neisser** (1928 to 2012) joined the two directions in his perceptual cycle (1976): a schema directs exploration, and the information sampled modifies the schema.

> **! EXAM TRAP**
>
> Unconscious inference and probabilistic reasoning define the **constructivist** approach. The computational approach (Marr, below) asks what each stage of vision computes, and the information-processing approach describes a chain of stages; neither centres on inference. Bayesian perception rests on **prior beliefs and sensory evidence**, not genes, culture or emotion. The bottom-up theorist, and the coiner of affordances, is **James J. Gibson**: Helmholtz and Gregory are top-down, Wertheimer is a Gestalt psychologist, and Gibson's wife Eleanor built the visual cliff. The apperceptive mass (Herbart's term, often credited to Wundt; Section 1.1) is not unconscious inference.

Gestalt organisation: figure, ground and grouping ●●

No school has contributed more to perception than Gestalt psychology (Section 1.2), launched in 1912 by **Max Wertheimer** (1880 to 1943) with **Kurt Koffka** (1886 to 1941) and **Wolfgang Köhler** (1887 to 1967). Its claim is that the whole is different from the sum of its parts.

First the field splits into a **figure**, which stands out, and a ground behind it. As **Edgar Rubin** (1886 to 1951) described in 1915, the figure is thing-like (high in object character), clearly outlined, usually smaller and in front; the ground is shapeless, diffuse and seems farther away. Rubin's vase is a reversible figure: it flips into two faces although the stimulus never changes, so the perceiver does the organising.

Elements are then grouped (Figure 1.4). By the **law of proximity**, elements close together in space or time belong together. By **similarity**, alike elements group; by the **law of closure**, gaps are filled, so a broken outline is seen as a circle; by **good continuation** (continuity), elements on a smooth path group; by **common fate**, elements moving together group, like a

flock of birds. Symmetry, common region (a shared enclosure) and connectedness (joining lines) group elements too. The master law is **Prägnanz** (good figure, simplicity): every pattern is seen in the simplest, most complete organisation the conditions allow, so two overlapping rings look like two circles.

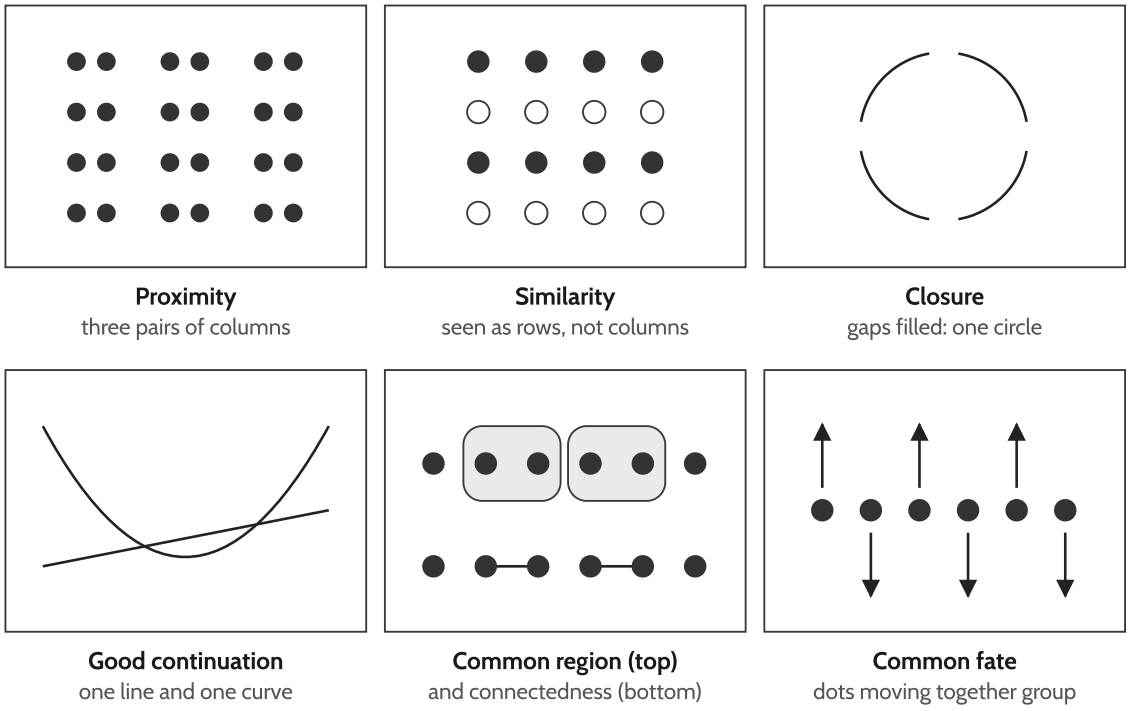


Figure 1.4 Six grouping laws you can watch in your own perception. In each panel the elements are physically identical or evenly spaced in some respect, yet you see definite groups; the grouping is supplied by your visual system.

> **! EXAM TRAP**
>
> Close in **space and time** is **proximity**. The tendency to **complete** a figure is **closure**; continuation and continuity both name the smooth-path law. Organised "in the best possible whole form" is **Prägnanz**. The school that contributed most to perception is **Gestalt**, not structuralism, behaviourism or psychoanalysis. Interposition is a depth cue, interpolation fills in between known points, extrapolation extends beyond them; one paper keyed **structuring** for working out a rule from incomplete evidence and picking the items that fit it.

Depth cues and the visual cliff ●●

How does a flat retinal image give you three dimensions? The brain combines depth cues, and only two are **binocular cues**, needing both eyes. In **retinal disparity**, the eyes, about 6 centimetres apart, receive slightly different views, and the nearer the object, the greater the difference; fusing the two gives the solid depth of stereopsis. In **convergence**, the eyes turn inwards to fixate a near object, and the brain registers how far they have turned.

All other cues are **monocular cues**, available to one eye (Table 1.5). Most are static pictorial cues, the ones a painter uses to put depth on a flat canvas. In the **texture gradient**, pebbles look smaller and more densely packed with distance; engineers exaggerate it by painting stripes ever closer together before a toll plaza or on an exit ramp, so that drivers feel their

speed and slow down. **Motion parallax** needs movement: from a train, nearby poles flash past while distant hills crawl.

Table 1.5 Depth cues classified.

Cue	Type	Depth signal
Retinal disparity	Binocular	Images differ more when near
Convergence	Binocular, oculomotor	Eyes turn in more when near
Accommodation	Monocular, oculomotor	Lens thickens for near objects
Motion parallax	Monocular, movement	Near objects sweep past faster
Interposition (overlap)	Monocular, pictorial	A partly hidden object is farther
Relative and familiar size	Monocular, pictorial	Smaller image, farther object
Linear perspective	Monocular, pictorial	Parallel lines converge
Texture gradient	Monocular, pictorial	Texture finer and denser
Aerial perspective	Monocular, pictorial	Distant objects hazy, bluish
Height in the field	Monocular, pictorial	Nearer the horizon, farther
Shading	Monocular, pictorial	Shadows reveal shape

KEY STUDY: THE VISUAL CLIFF (GIBSON AND WALK, 1960)

Question. Can infants perceive depth before learning it from falls? **Method. Eleanor Jack Gibson** (1910 to 2002) and **Richard Walk** (1920 to 1999) built a **visual cliff**: glass over a checkerboard lying just beneath one half (the shallow side) and far below the other (the deep side). Mothers called 36 infants aged 6 to 14 months across it. **Results.** Of the 27 who left the centre board, only three crept onto the deep side; newborn animals avoided it too. **Conclusion.** Depth perception is present by crawling age. **Criticism.** Crawlers have already had months of visual experience, so innateness is not proved.

! EXAM TRAP

The binocular cues are only **retinal disparity** and **convergence**. Interposition, accommodation, motion parallax and aerial perspective are all monocular. Assimilation (Piaget's term) is no depth cue. The cliff tested **depth perception**, not object permanence, colour or imprinting, and showed it present **by crawling age**, not fully formed at birth or only after age two. One paper keyed **180 degrees** for each eye's visual field; the better figure is about 160 degrees, with about 200 for both eyes together.

Perceptual constancy ○

Perceptual constancy is the perception of objects as stable despite changing sensory input. Hold one hand at arm's length and the other halfway to your face: the far hand's retinal image is half the size, yet both look alike. This **size constancy** follows **size-distance invariance**: perceived size is retinal size scaled by perceived distance, a rule that **Emmert's law** (1881) shows with afterimages, which look larger on a farther wall. Shape constancy keeps an opening door rectangular; brightness constancy keeps coal black in sunlight, because lightness depends on the ratio of an object's light to its surroundings'; colour constancy keeps a

banana yellow under any lamp; location constancy keeps the world still as the eyes jump in saccades. Tiny saccades also refresh the retinal image, which would otherwise fade, and so ensure constant perception.

> **! EXAM TRAP**
>
> The outstretched hand is **size constancy**, not shape constancy, motion parallax or depth perception. Lateral inhibition (Section 1.3) **enhances** edges; it does not dull them. A fixation is the pause between saccades, and sensory adaptation the fading they prevent.

Illusions and what they reveal ●●

An **illusion** is a misinterpretation of a stimulus that really exists. The error lies in perception, not sensation: the retinal image is accurate. Indian textbooks contrast **personal illusions**, which vary with observer and occasion (a rope seen as a snake at dusk), with permanent or universal illusions, which everyone sees every time (railway tracks converging).

In the **Müller-Lyer illusion**, published in 1889 by **Franz Carl Müller-Lyer** (1857 to 1916), a shaft ending in outward fins looks longer than an equal shaft ending in arrowheads (Figure 1.5). Gregory (1963) blamed **inappropriate constancy scaling**: arrowheads suggest the near outside corner of a building, fins the far inside corner of a room, and size constancy enlarges the "farther" shaft on a flat page. **Marshall Segall**, **Donald Thomas Campbell** (1916 to 1996) and **Melville Jean Herskovits** (1895 to 1963) showed that experience matters (1963, 1966): by their **carpentered-world hypothesis**, people raised among rectangular buildings are more susceptible than people raised among round huts.

The Ponzo illusion, the Ames room (a distorted room that looks rectangular through a peephole, shrinking a person in its far corner) and the moon illusion (terrain makes the horizon moon seem farther) are misapplied size scaling too: whatever is registered as farther is seen as bigger. The Ebbinghaus illusion (Titchener circles) is size contrast, and the Zöllner, Poggendorff and Hering figures distort lines. The Necker cube is an ambiguous figure that flips in depth; the Penrose triangle is an impossible figure.

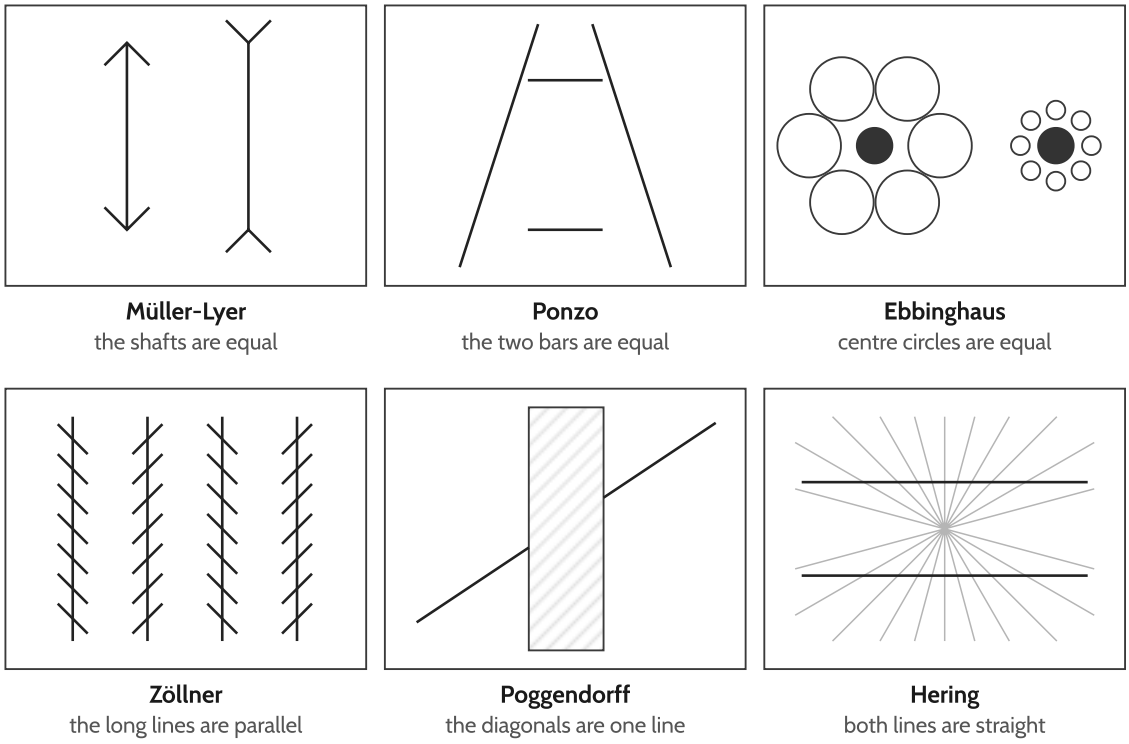


Figure 1.5 Six classic geometric illusions. In every panel the caption underneath is physically true, yet you see something else: the finned shaft looks longer, the upper bar looks longer, the circle among small circles looks larger, the long lines look tilted, the diagonal looks broken and the straight lines look bowed. Check them with a ruler.

> **! EXAM TRAP**
>
> Rails converging for everyone is a **permanent** illusion (visual too, but the item tests the classification), not a personal one or the Müller-Lyer. Illusions distort **perception**, not sensation, adaptation or "translation" (transduction). The Müller-Lyer is strongest with **buildings with corners** and best explained by **misapplied size constancy**, not disparity (it works with one eye), lateral inhibition or figure-ground reversal.

Illusion, hallucination and delusion ○

An illusion misinterprets a real external stimulus. A **hallucination** is a perception without an adequate external stimulus, such as a voice in an empty room. A delusion is not a perception at all but a false belief held with unshakeable conviction. Psychopathology adds the sensory distortion, a real object perceived but altered in size, intensity or quality (micropsia), and imagery, a picture in the mind's eye under voluntary control. Déjà vu is the illusion of having seen something before; déjà entendu, of having heard it before.

! EXAM TRAP

Illusions are a **misinterpretation of stimuli arising from an external object**, or **distortions of existing stimuli**. "Without an adequate external stimulus" and "not based on external reality" are **hallucinations**; "a real object perceived in a distorted way" is a sensory distortion; "vivid mental images" is imagery; "innate mechanisms" fails because learning shapes illusions. Clear memory of the past is hypermnesia; involuntary repeated speech is palilalia (Psychiatric Disorders chapter).

Motion perception

Real motion is signalled by direction-selective neurons, concentrated in area MT (V5). Apparent motion is seen where nothing moves: in Wertheimer's phi phenomenon (1912) two lines flashed in quick succession look like one line moving (strictly, smooth object motion is beta movement), and films give stroboscopic motion. In induced motion a still object seems to move because its frame moves, like the moon behind racing clouds. In the **autokinetic effect** a still light in darkness seems to wander for lack of a frame of reference; **Muzafer Sherif** (1906 to 1988) used it in 1935 to study how group norms form. The motion aftereffect (waterfall illusion) is a still scene drifting opposite to a motion just watched.

Pattern and object recognition ○

Recognising an A in any handwriting defeats **template matching**, comparison with exact stored copies, so feature analysis uses parts instead. In the **Pandemonium** model of **Oliver Selfridge** (1959), image demons record the pattern, feature demons shout when they find their line or curve, cognitive demons (one per letter) shout in proportion to their matching features, and a decision demon picks the loudest. **David Hubel** and **Torsten Wiesel** found such feature detectors, cells tuned to lines of one orientation, in cat visual cortex from 1959. A **prototype**, in the work of **Eleanor Rosch** (born 1938), is an idealised average member of a category. The **recognition-by-components** theory of **Irving Biederman** (1987) builds objects from about 36 three-dimensional geons, and **David Marr** (1945 to 1980) traced vision from a primal sketch through a viewer-centred 2.5-D sketch to an object-centred 3-D model. Prosopagnosia (face blindness) is a selective loss of face recognition. In the word superiority effect (**Gerald Reicher**, 1969), a letter is identified better in a word than alone: knowledge helps from the top.

Set, culture, subliminal perception and ESP ○

A **perceptual set** is a readiness to perceive in a particular way. **Jerome Seymour Bruner** (1915 to 2016) and **Leigh Minturn** (1955) found one shape read as B among letters and as 13 among numbers, a context effect. Bruner's New Look stressed needs: with **Cecile Goodman** in 1947 he found that poor ten-year-olds overestimated coin sizes more than rich ones, and **Elliott McGinnies** (1949) found that taboo words needed longer exposures (perceptual defence). Experience reshapes perception. Perceptual learning sharpens discrimination; **George Malcolm Stratton** (1865 to 1957) showed perceptual adaptation by adjusting to lenses that turned his world upside down (1896, 1897); and S.B., given sight at 52 and studied by Gregory and **Jean Wallace** (1963), could not at first judge distance. Culture matters too, as the carpentered world showed. **Herman Witkin** (1916 to 1979) found a perceptual style, **field de-**

pendence-independence (rod-and-frame and embedded figures tests): field-dependent people are swayed by the surrounding field, field-independent people separate items from it.

Subliminal perception is the processing of stimuli too faint or brief to be noticed. **James Vicary**'s 1957 "Eat Popcorn" cinema study was a hoax, admitted in 1962, but in masked semantic priming (**Anthony Marcel**, 1983) an unseen word speeds responses to a related one, and in blindsight (**Lawrence Weiskrantz**, 1974) patients with damaged primary visual cortex deny seeing yet locate targets above chance. **Extrasensory perception** claims perception with no sensory input: telepathy (mind to mind), clairvoyance (hidden objects) and precognition (the future), studied with psychokinesis (mind over matter) by parapsychology. **Joseph Banks Rhine** (1895 to 1980) used Zener cards at Duke (chance is 5 hits in 25), and later researchers the Ganzfeld procedure, guessing a target picture in a uniform sensory field. The results do not replicate, so most scientists remain sceptical.

> **! EXAM TRAP**
>
> Semantic priming studies **subliminal perception**, not field dependence-independence (Witkin), locus of control (a personality variable) or signal detection (Section 1.3). Failing to notice obvious changes is change blindness, an attention topic (Section 1.5), not a failure of constancy. Deprivation and coin size is **Bruner and Goodman**, not Segall, Herskovits, or **K. D. Broota** and **H. C. Ganguli** (Indian work on cultural differences in perceptual selectivity, 1970s).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Processing driven by knowledge, expectation and context is called _____ processing.
2. Gibson's term for the possibilities for action an object offers is _____.
3. The Gestalt tendency to fill the gaps in a figure is the law of _____.
4. The two binocular depth cues are retinal disparity and _____.
5. Gregory explained the Müller-Lyer illusion as inappropriate constancy _____.
6. A perception without an adequate external stimulus is a _____.
7. Biederman's recognition-by-components theory builds objects from about 36 _____.
8. Bruner and Goodman found that poor children overestimated the size of _____ more than rich children did.

Answers: (1) top-down, (2) affordances, (3) closure, (4) convergence, (5) scaling, (6) hallucination, (7) geons, (8) coins.

1.5 Attention and Consciousness

MEDIUM YIELD · 61 QS

Attention selects the small part of the world that reaches awareness, and consciousness, the awareness itself, changes state many times a day and night: in sleep and dreams, under hypnosis, in meditation and on drugs.

What attention is: span and determinants ○

William James (1842 to 1910) defined attention in 1890 as the mind "taking possession … in clear and vivid form, of one out of what seem several simultaneously possible objects or trains of thought": selection plus concentration, needed because the brain's capacity is limited. Classical texts give it five characteristics: *selectivity*, *shifting* to a new object, a limited *span*, *fluctuation* (attention to an unchanging object waxes and wanes, as a faint ticking watch seems to fade and return) and *division* between tasks.

The **span of attention** is the number of objects apprehended in one brief glance. **William Stanley Jevons** (1871), glancing at handfuls of beans, was never wrong with three or four; **tachistoscope** studies, using displays flashed for a fraction of a second, put the span at about six to eight unrelated items. **External determinants** of attention are properties of the stimulus (intensity, size, contrast, change, movement, novelty, repetition); **internal determinants** are properties of the person (motives, interests, attitudes and **mental set**, a temporary readiness to perceive something).

> **! EXAM TRAP**
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> Number plates carry about four digits plus a few letters because of the **span of attention**, not selective, divided or sustained attention. Miller's memory span of seven (Section 1.7) is a different process.

Selective attention: filters, attenuators and spotlights ●●

The British engineer **Colin Cherry** (1953) named the cocktail party problem: how do we follow one conversation among many? Hearing your own name from a conversation you were ignoring is the **cocktail party phenomenon**.

KEY STUDY: CHERRY'S SHADOWING EXPERIMENTS (1953)

In **dichotic listening** a different message goes to each ear, and the listener repeats one aloud as it arrives (**shadowing**). People recalled almost nothing of the unattended message's meaning, missing even a switch from English to German, yet noticed physical changes such as a male voice turning female. **Neville Moray** (1959) then found that a word list repeated 35 times in that ear left no memory, yet a third of listeners noticed their own name there.

Donald Broadbent's **filter model** (1958), the first major information-processing theory, passes one message on a physical feature (ear, pitch) and blocks the rest before meaning is analysed: **early selection**, with the filter as a **bottleneck**. **Anne Treisman**'s **attenuation model** (1964) turns unattended messages down, not off. Each word in a mental dictionary has a threshold, and important words such as your name have permanently low thresholds, so they fire even from the weakened channel. **J. Anthony Deutsch** and **Diana Deutsch** (1963) instead proposed **late selection**, at the response, after every message is analysed for meaning (Table 1.6).

Table 1.6 Theories of selective auditory attention.

Theory	Theorist and year	Where selection happens
Filter (switch)	Broadbent, 1958	Early; the rest is blocked

Theory	Theorist and year	Where selection happens
Attenuation	Treisman, 1964	Early but leaky
Late selection	Deutsch and Deutsch, 1963	At the response, after meaning
Pertinence	**Donald Norman**, 1968	Late; signal strength plus relevance
Multimode	**William Johnston** and **Steven Heinz**, 1978	Flexible; later costs more effort
Perceptual load	**Nilli Lavie**, 1994 to 1995	Early if the task is demanding, late if easy

Vision has the same limits. **Michael Posner** likened covert attention, moved without moving the eyes, to a **spotlight** (others prefer a **zoom lens**, wider but less detailed). In his cueing task (1980) a central arrow directs voluntary **endogenous** attention, a peripheral flash captures reflexive **exogenous** attention, and after about 300 milliseconds the flashed spot is answered more slowly (**inhibition of return**). In Treisman and **Garry Gelade**'s **feature integration theory** (1980), features register in parallel at a preattentive stage, so a red letter pops out among green ones, but binding them into objects needs serial focused attention, so conjunction search slows with every added item; unattended features can combine wrongly (illusory conjunctions). Missing a person in a gorilla suit while counting basketball passes (**Daniel Simons** and **Christopher Chabris**, 1999) is **inattentional blindness**, missing an unexpected object in plain view; missing a large change made across a blink or flicker is **change blindness**; missing a second target 200 to 500 milliseconds after a first is the **attentional blink**.

> **! EXAM TRAP**
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> "Filter-attenuation theory" is **Treisman** (1964), not Broadbent, Johnston or Heinz, and her attenuator model, not Broadbent's switch model, best explains the cocktail party effect. A bottleneck is a narrow processing stage, not the cocktail party phenomenon. Failing to notice obvious changes is **change blindness**, not inattentional blindness or perceptual constancy (Section 1.4).

Divided attention, automaticity, Stroop and vigilance ●●

Divided attention asks how far we can do two things at once. **Daniel Kahneman**'s **capacity model** (*Attention and Effort*, 1973) treats attention as one limited pool of mental effort; **Christopher Wickens**'s **multiple resource theory** (1984) adds separate pools by modality, code and stage, so dissimilar tasks interfere less: driving combines better with the radio than with texting, though even a hands-free phone conversation impairs it.

Practice helps because some processes stop needing capacity. An **automatic process** runs without intention, awareness or interference with other activity (Posner and Charles Snyder, 1975). **Richard Shiffrin** and **Walter Schneider** (1977) contrasted fast, parallel, effortless and inflexible automatic processing, learned through consistent practice, with slow, serial, effortful and flexible **controlled processing**. In **Patricia Devine**'s model (1989), stereotypes are activated automatically in everyone, but low-prejudiced people use controlled processing to refute them (Chapter 3).

KEY STUDY: STROOP'S COLOUR-WORD EXPERIMENTS (1935)

John Ridley Stroop (1897 to 1973) asked people to read colour words printed in a conflicting ink (RED printed in green) or to *name the ink colour* of such words. Reading was barely slowed, but naming the inks took about 74 per cent longer than naming plain colour patches, and errors rose.

The **Stroop effect** is this slowing of colour naming when word and ink conflict. Reading is automatic, so the word cannot be shut out, and its name competes with the controlled naming of the ink at the response stage. The effect thus shows a limit of **selective attention**: interference in reaction time from conflicting information, one attribute of the stimulus (the word) dominating the other (the ink).

Watching radar for hours for a rare blip demands **sustained attention**, or **vigilance**. **Norman Mackworth** (1948) had observers watch a clock pointer for occasional double jumps; detection fell across the two-hour watch, the **vigilance decrement**. In an oddball task, where a rare target appears among frequent non-targets, the brain produces a positive event-related potential about 300 milliseconds later, the **P300**, which indexes the allocation of attention to the target and is reduced in schizophrenia and Alzheimer's disease. Attention fails in ADHD, in hemispatial neglect (ignoring the left side of space after right parietal damage) and in delirium (Chapter 8).

! EXAM TRAP

The Stroop effect is an **attention** phenomenon (selective attention; interference in reaction time from conflicting information), not long-term memory retrieval, working memory, procedural memory, perception or language development. RML 2024 asks what it is "an account of" and keys **stimulus dominance**: the printed word dominates its ink colour. "Semantic dominance" tempts because the word is meaningful; "instructional" and "conceptual" dominance are not terms used for it. Air traffic controllers and radar readers exemplify **sustained attention**, not selective, divided or "perceptual" attention.

Consciousness: James's stream and Freud's levels ●●●

Consciousness is our awareness of ourselves and our environment at a given moment. Its level, from coma to alert waking, depends on the brainstem reticular activating system and is scored on the **Glasgow Coma Scale** (eye, verbal and motor responses; 3 to 15). The structuralists analysed its contents into elements by introspection (Section 1.1); James objected that consciousness "does not appear to itself chopped up in bits … it flows": it is a **stream of consciousness** (1890), personal, ever changing, continuous and selective. Its most researched feature is **daydreaming**, which **Jerome L. Singer** showed to be normal and nearly universal.

Sigmund Freud (1856 to 1939) held that most mental life never enters the stream. His **topographic model** (1900) has three levels: the conscious, the small tip of present awareness; the **preconscious**, accessible memories recallable at will (your mobile number); and the much larger **unconscious**, repressed wishes known only indirectly, through dreams, symptoms and **Freudian slips**, errors of speech that betray unconscious impulses. By **psychic determinism**, no mental event is accidental. In his structural model (1923) the id works on the pleasure principle and the ego on the reality principle (Chapter 4). The **psychodynamic perspective** holds that behaviour is driven by unconscious inner forces largely beyond our con-

trol and shaped in early childhood. Research has not supported the Oedipus complex, but it does support the concept of an unconscious mind.

Modern psychology adds the **cognitive unconscious**, processing outside awareness that is automatic, not repressed, such as the unconscious stereotyping measured by the Implicit Association Test (Anthony Greenwald and colleagues, 1998). Processes never open to awareness, such as blood-pressure control, are **nonconscious**. **David Chalmers** (1995) called the question of why brain processing is accompanied by subjective experience at all the **hard problem of consciousness**. An **altered state of consciousness**, a term made standard by **Charles Tart** (1969), is a temporary, qualitative change in the overall pattern of mental functioning, arising naturally (sleep), psychologically (hypnosis, meditation) or chemically (drugs).

> **! EXAM TRAP**
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> "Stream of consciousness" is **William James** (1890), not Carl Jung, Wilhelm Wundt or B. F. Skinner; its most researched feature is **daydreaming**, not gender differences, brain-wave patterns or eye movements. Memories recallable at will are **preconscious**, not unconscious. The three levels form the **topographic** model (1900); id, ego and superego the **structural** model (1923).

> **INDIAN CONTEXT**
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> The *Mandukya Upanishad* names four states: jagrat (waking), svapna (dreaming), sushupti (dreamless sleep) and **turiya**, "the fourth", pure consciousness witnessing all three, identified with the **Atman**, the true self or soul, eternal consciousness (Section 1.2).

Biological rhythms and the suprachiasmatic nucleus ○

A **circadian rhythm** is a biological cycle of about 24 hours, such as sleep and waking or body temperature. Without time cues, in caves, human rhythms persist but drift slightly beyond 24 hours, so they come from an internal clock, reset daily by **zeitgebers** ("time givers"), above all light. The clock is the **suprachiasmatic nucleus** (SCN) of the hypothalamus, just above the optic chiasm: animals without it sleep the usual total amount but at random times. Light reaches it directly along the **retinohypothalamic tract**, and it drives the pineal gland to secrete **melatonin** in darkness. **Jet lag** and shift work put the clock out of step with the day. **Ultradian rhythms** are shorter than a day (the 90-minute sleep cycle), **infradian rhythms** longer (the menstrual cycle).

> **! EXAM TRAP**
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> The SCN is the **master circadian pacemaker**; its lesions disrupt circadian rhythms, not total sleep. It is not the hunger and satiety centre (lateral and ventromedial hypothalamus), the thermoregulation centre (preoptic area), the paraventricular stress nucleus, the controller of anterior pituitary hormones, or a seat of memory (hippocampus, mammillary bodies), language or pain perception.

The stages and cycles of sleep ●●

Sleep opened to study after **Hans Berger** published the first recordings of the human electroencephalogram (EEG) in 1929: neurons firing independently give small, fast, **desynchronised** waves, neurons firing together large, slow, **synchronised** ones. In 1953 **Eugene Aserinsky** and **Nathaniel Kleitman** found recurring bursts of rapid eye movement with a waking-like EEG, from which woken sleepers usually reported vivid dreams; **William Charles Dement** and Kleitman (1957) then described the cycle of stages. Sleep thus divides into **REM sleep** and NREM sleep, whose old stages 1 to 4 became N1, N2 and N3 in 2007 (Table 1.7).

Table 1.7 EEG waves and the stages of sleep.

State	EEG	What marks it
Alert waking	Beta, 13 to 30 Hz	Active thinking
Relaxed, eyes closed	Alpha, 8 to 12 Hz	Rest before sleep
N1	Theta, 4 to 7 Hz	Drowsy transition
N2 (about half the night)	Sleep spindles, K-complexes	Light but definite sleep
N3 (old stages 3 and 4)	Delta, 0.5 to 4 Hz, high amplitude	Deepest, **slow-wave sleep**; growth hormone peak
REM (20 to 25 per cent)	Fast, low voltage, desynchronised; sawtooth waves	Rapid eye movements, atonia, dreams

REM is also called **paradoxical sleep**, because the brain looks awake while the body is limp: the eyes dart, the core muscles are almost paralysed (**REM atonia**), breathing turns irregular and genital arousal accompanies nearly every REM period. One run of NREM stages followed by REM is a **sleep cycle** of about 90 minutes, so seven to eight hours hold **four to six cycles**; N3 dominates the early cycles, and REM periods lengthen toward morning (Figure 1.6). A newborn sleeps about 16 hours, half of it REM, and older adults about six, with little N3 (**Howard Roffwarg** and colleagues, 1966). Slow-wave sleep is switched on by GABA neurons of the **ventrolateral preoptic area** (VLPO), which silence the arousal systems; REM is generated in the pons.

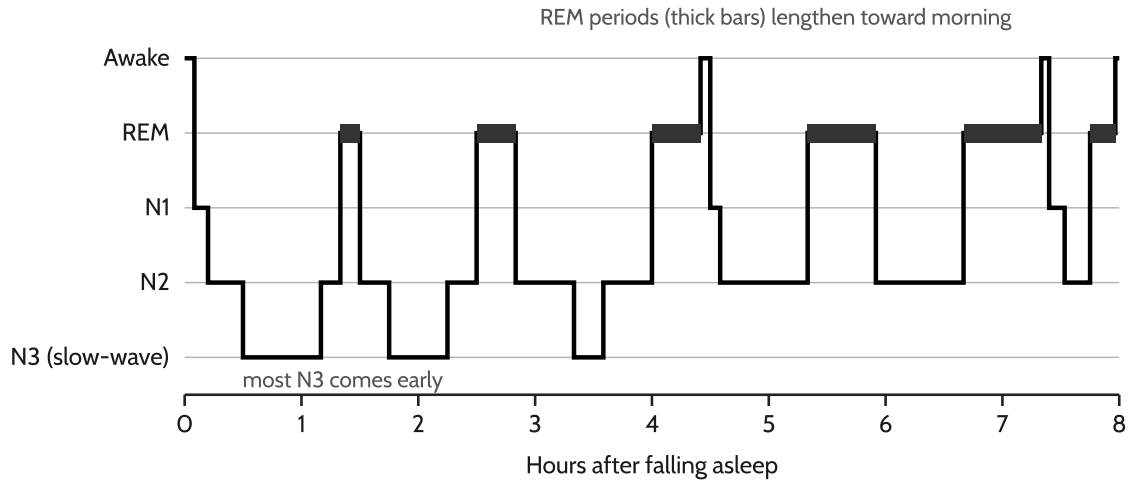


Figure 1.6 A typical night of young adult sleep (a hypnogram). Cycles of about 90 minutes repeat four to six times; N3 is concentrated early in the night, and REM periods become longer toward morning.

> **! EXAM TRAP**
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> REM sleep is **not** marked by EEG synchrony, delta waves or a growth hormone surge (all N3). Cycles per night: **four to six** (RML 2024); RML 2022, offering no 4 to 6, keys **4 to 7**, the option containing it. Slow-wave sleep begins when the **VLPO** becomes active; the SCN is the clock, and the sublaterodorsal nucleus (REM-on) and ventrolateral periaqueductal grey (REM-off) govern REM.

Why we sleep, and sleep disorders ○

Recuperation theories say sleep restores what waking runs down; **adaptation theories** say evolution timed sleep to keep animals safe and save energy; **memory consolidation** theories note that sleep after learning reduces forgetting (Section 1.7). Sleep loss brings sleepiness, low mood and lapses of sustained attention, including **microsleeps** of a few seconds. Volunteers woken at every REM onset tried to enter REM ever more often, then showed **REM rebound**, extra REM (Dement, 1960). In **cataplexy**, part of narcolepsy, strong emotion triggers a sudden loss of muscle tone in full wakefulness: REM atonia while awake. Sleep disorders fall into insomnia, **hypersomnia** (excessive sleepiness, as in narcolepsy), breathing-related disorders and the **parasomnias**, abnormal behaviours during sleep: NREM parasomnias (sleepwalking and sleep terrors from N3, and bedwetting) and REM parasomnias (nightmares, REM sleep behaviour disorder). Table 1.8 lists the main ones (Chapter 8).

Table 1.8 The main sleep disorders in brief.

Disorder	Key features	Stage or cause
Insomnia	Trouble falling or staying asleep	Stress, anxiety, poor habits
Narcolepsy	Daytime sleep attacks, cataplexy, sleep paralysis	REM intrudes into waking
Sleep apnoea	Breathing stops repeatedly; snoring	Airway collapse
Sleepwalking (somnambulism)	Sitting, walking or complex acts while asleep	N3, early in the night
Sleep terrors	Screaming fear, inconsolable, little recall	N3, early in the night

Disorder	Key features	Stage or cause
Nightmares	Frightening dream, vivid recall	REM, late in the night
REM sleep behaviour disorder	Acting out dreams	Loss of REM atonia
Restless legs syndrome	Leg discomfort, urge to move	Delays sleep onset

! EXAM TRAP

Somnambulism is sitting, walking or complex behaviour while asleep; screaming fear is sleep terrors, acting out nightmares is REM sleep behaviour disorder, and leg discomfort is restless legs syndrome. Sleepwalking arises from **N3**, not REM: a sleeper in REM is paralysed. Treatments for insomnia, such as stimulus control and Viktor Frankl's paradoxical intention (trying to stay awake), are in Chapter 7.

Dreams: Freud, Jung and the brain ●●

Freud's *The Interpretation of Dreams* (1900) called dreams "the royal road to a knowledge of the unconscious". Each dream is a **wish fulfilment**, the disguised fulfilment of a repressed wish. The **manifest content** is the dream as remembered, the **latent content** its hidden meaning, and the **dream work** disguises one as the other by condensation (several ideas in one image), displacement (feeling shifted to a trivial object), symbolisation (thoughts made into images) and secondary revision (tidying it into a story).

Carl Gustav Jung (1875 to 1961) saw dreams as undisguised symbolic messages that compensate for a one-sided conscious attitude. He held that the unconscious has two levels: a personal (individual) unconscious of forgotten and repressed experience and, beneath it, the **collective unconscious**, an inherited layer shared by all humanity, which he explained by evolution, through the similar brain structure of all human races. It holds **archetypes**, universal images recurring in myths: the persona (social mask), the shadow (disowned dark side), the **anima** (the feminine side of a man) and the **animus** (the masculine side of a woman). Jung also devised the word association test, in which hesitations reveal unconscious complexes (Chapter 4).

In the **activation-synthesis hypothesis** of **J. Allan Hobson** and **Robert McCarley** (1977), the pons fires randomly during REM and the cortex synthesises a story from the activation, which is why dreams are visual, bizarre and soon forgotten. Information-processing views link dreams to the overnight consolidation of recent experience, and **Antti Revonsuo**'s **threat-simulation theory** (2000) says dreams evolved to rehearse responses to danger. In a **lucid dream** the dreamer knows he is dreaming, confirmed by pre-arranged eye-movement signals from REM sleep (**Stephen LaBerge**). **Hypnagogic imagery** comes while falling asleep, **hypnopompic imagery** while waking.

! EXAM TRAP

The Interpretation of Dreams: **1900** (printed November 1899). The collective unconscious is from **Carl Jung**, not Sigmund Freud, Alfred Adler or Karen Horney; his two levels are **collective and individual**, not anima and animus or extraversion and introversion.

Hypnosis, meditation and psychoactive drugs ●●

Hypnosis is a social interaction in which one person's suggestions of changes in perception, memory or behaviour are experienced by the other as real and involuntary. Its crux is **suggestibility**, and it is not sleep. **Franz Anton Mesmer**'s **animal magnetism** (1770s) was put down to imagination by a royal commission in 1784. The Scottish surgeon **James Braid** coined "hypnotism" (*Neurypnology*, 1843). **Jean-Martin Charcot** took hypnotisability for a sign of hysteria; the **Nancy school** of **Ambroise-Auguste Liébeault** and **Hippolyte Bernheim** showed it was normal suggestion. Freud learned from both, published the mechanism of hysteria with **Josef Breuer** (1893; *Studies on Hysteria*, 1895), then replaced hypnosis with **free association**, the patient on the couch reporting whatever comes to mind.

Susceptibility is measured by the **Stanford Hypnotic Susceptibility Scales** (**André Weitzenhoffer** and **Ernest Ropiequet Hilgard**, 1959). Hypnosis can relieve pain and produce posthypnotic suggestions (acted on at a cue after hypnosis ends) and posthypnotic amnesia, but it cannot give superhuman strength or recover memories reliably, since it raises false recall and confidence together. Hilgard's **neodissociation theory** holds that hypnosis splits consciousness, a **hidden observer** registering pain that the hypnotised person denies; the **social-cognitive theory** of **Theodore Xenophon Barber** and **Nicholas Spanos** says hypnotised people enact an expected role.

Meditation trains attention. **Concentrative meditation** holds it on one object, such as the repeated mantra of **transcendental meditation**, introduced by **Maharishi Mahesh Yogi** in the 1950s; **mindfulness meditation** (opening-up) observes whatever arises without judging it, from Buddhist vipassana. **Herbert Benson** showed that meditation's slowed heart rate and breathing, the **relaxation response** (1975), follow from any similar technique. Surveys showing that regular meditators are less stressed are correlational, however: without random assignment, calmer people may simply meditate more, and only an experiment can show cause (Chapter 9). In McGill University's **sensory deprivation** studies (1954), students lying in a cubicle with goggles and gloves mostly quit within two or three days, and many hallucinated. A **psychoactive drug** alters perception, mood or consciousness by acting on the nervous system (Table 1.9). With repeated use, **tolerance** means a larger dose is needed for the same effect, **dependence** means the drug is needed to function, and **withdrawal** is the unpleasant reaction, usually opposite to the drug's effects, on stopping (Chapter 8).

Table 1.9 The main classes of psychoactive drugs.

Class	Examples	Main effects
Depressants	Alcohol, barbiturates, benzodiazepines	Enhance GABA; sedation, sleep
Stimulants	Caffeine, nicotine, amphetamines, cocaine	Arousal, euphoria, then a crash
Opiates	Morphine, heroin, codeine	Pain relief; mimic endorphins
Hallucinogens	LSD, psilocybin, mescaline	Distorted perception
Cannabis	Ganja, charas, bhang (THC)	Relaxation, slowed time sense

! EXAM TRAP

The crux of hypnosis is **suggestibility**, not the analytical mind or the logical mind. Freud and Breuer, not Freud and Charcot, published the mechanism of hysteria; Freud, not Bernheim, fathered psychoanalysis. Valium, Xanax, Rohypnol and Halcion are **benzodiazepines**: "sedatives" and "hypnotics" name uses, not the class, and they are not barbiturates.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The number of objects apprehended in one brief glance is the _____ of attention.
2. In the Stroop task, automatic _____ slows naming of the ink colour.
3. In Freud's model, memories recallable at will are _____.
4. The master circadian clock is the _____ nucleus.
5. Delta waves mark stage _____.
6. Jung's inherited layer of the psyche is the _____ unconscious.

Answers: (1) span, (2) reading, (3) preconscious, (4) suprachiasmatic, (5) N3, (6) collective.

1.6 Learning: Conditioning, Cognition and Observation HIGH YIELD · 143 QS

Almost everything that makes you a particular person was learned, from your language to the way your stomach tightens in an exam hall. This section runs from the simplest learning, through conditioning, to the maps, insights and imitation that conditioning alone could not explain.

What learning is ○

The standard definition descends from **Gregory Kimble** (1961): **learning** is a relatively permanent change in behaviour, or in the potential for behaviour, that results from experience. "Potential" matters because learning is inferred from *performance*, what an organism actually does, and the two can come apart. "Experience" rules out *maturation*, change on a biological timetable (a baby walks when her body is ready), as well as fatigue, drugs, motivation and instinct. **Nikolaas Tinbergen** (1907 to 1988) showed that a male stickleback attacks any crude model with a red belly: a *fixed action pattern*, an unlearned, stereotyped sequence triggered by a sign stimulus and run to completion.

The simplest learning involves one stimulus and is called non-associative. In **habituation** the response to a repeated, harmless stimulus decreases: trains outside a new hostel room wake you the first night but not by the end of the week. It is stimulus-specific and shows dishabituation (a novel strong stimulus restores the response), proof that it is learning, not fatigue. Its near opposite, **sensitisation**, is an increased response to many stimuli after a strong or noxious one. **Eric Kandel** (born 1929) traced both to synapses of the sea slug *Aplysia* (Physiological Psychology chapter). In infants, habituation reveals discrimination before speech (**Robert Fantz**, 1964): a baby bored with one picture who looks longer at a new one can tell them apart.

Between instinct and ordinary learning lies **imprinting**, the rapid, unrewarded and largely irreversible attachment to the first suitable moving object, possible only within a *critical period*: in 1935 **Konrad Lorenz** (1903 to 1989) reported incubator-hatched greylag goslings following him as their mother. Everything else here is associative learning: learning that two events go together.

> **! EXAM TRAP**
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> Fixed action patterns are **innate**, not products of classical or operant conditioning. The near opposite of habituation is **sensitisation**; "habitation" and "inhabitation" are sound-alikes.

Pavlov and classical conditioning ●●●

Ivan Petrovich Pavlov (1849 to 1936), a physiologist who won the 1904 Nobel Prize for work on digestion, collected his dogs' saliva through a tube in the cheek. In the late 1890s he noticed that they salivated before food arrived, at the sight of the dish or the attendant's footsteps. From about 1897 he studied these "psychic secretions": a metronome was sounded, meat powder followed, and after several pairings the metronome alone produced salivation. This is **classical conditioning** (Pavlovian or respondent conditioning): learning in which a neutral stimulus, repeatedly paired with a stimulus that already produces a response, comes to produce a similar response itself.

Five terms describe it. The **unconditioned stimulus** (UCS) produces a response without learning (food in the mouth), and that unlearned response is the unconditioned response (UCR: salivation to food). A neutral stimulus (NS), the metronome on day one, does not produce it. After pairing, the metronome is a **conditioned stimulus** (CS), and its learned response is the conditioned response (CR: salivation to the metronome). So the metronome is an NS before conditioning and a CS after it, and salivation is a UCR to food but a CR to the metronome. The fundamental element is association, the pairing of two stimuli close in time. Table 1.10 adds three cases; each response is involuntary.

Table 1.10 Three classical conditioning situations broken into their parts.

Situation	UCS	UCR	CS	CR
Little Albert	Loud clang	Startle and fear	White rat	Fear of the rat
Student who lost a match	Defeat and humiliation	Distress	The sports ground	Anxiety at the ground
Cancer patient	Chemotherapy drug	Nausea	Hospital smell	Nausea on arrival

Timing matters. In delay conditioning the CS begins first and is still on when the UCS arrives; this is the most effective arrangement, because the CS works as a warning. Trace conditioning, the other forward arrangement, leaves a gap between CS and UCS and is weaker; simultaneous conditioning gives no warning and is weak; backward conditioning (UCS first) is the weakest of all.

! EXAM TRAP

The bell is a **neutral stimulus before** pairing and a **conditioned stimulus after**; never a UCS, never an "operant stimulus". It becomes a CS by **repeated pairing with a UCS**, not by reward, punishment or random occurrence. "Delay" does not mean slow: delay conditioning is the most effective; trace has the gap.

The CR grows during acquisition, quickly at first and then more slowly. If the CS is then presented repeatedly without the UCS, the CR fades: **extinction**. (Pavlov called the UCS the "reinforcement", so withdrawing reinforcement means omitting the UCS.) Extinction suppresses rather than erases learning (Pavlov saw it as active inhibition): after a rest the CR returns with no new pairings (**spontaneous recovery**); a novel distracting stimulus restores it briefly (disinhibition); so does the UCS alone (reinstatement) or a return to the original context (renewal). Hence fears can relapse after therapy.

Responding to stimuli similar to the CS, the more so the more similar they are (a generalisation gradient), is **stimulus generalisation**; responding only to the CS, learned when a similar stimulus is never followed by the UCS, is **stimulus discrimination**. A dog whose circle-versus-ellipse discrimination **Shenger-Krestovnikova** (1921) made impossible became agitated and lost even easy discriminations: *experimental neurosis*, the first laboratory model of anxiety. In *higher-order conditioning* an established CS acts as the UCS for a new stimulus (a black square paired only with the metronome comes to produce salivation), which is how the word "exam" alone can make a heart race. In *sensory preconditioning* (**W. J. Brogden**, 1939) two neutral stimuli are paired first; when one is later conditioned, the other also produces the CR.

! EXAM TRAP

Omitting the UCS produces **extinction**. Spontaneous recovery needs only time, disinhibition a novel stimulus, reinstatement the UCS alone and reconditioning new pairings; sensitisation is no comeback. Avoiding a black cat but playing with a tabby is **discrimination**, not extinction, successive approximation (shaping) or negative transfer.

Conditioned fear and its therapies ●●●

John Broadus Watson (1878 to 1958), the founder of behaviourism, set out with **Rosalie Rayner** to show that emotions are conditioned. In 1920 they paired a white rat seven times with the clang of a steel bar behind the head of an eleven-month-old infant, "Albert B.". Afterwards the rat alone made him cry and crawl away, and the fear generalised to a rabbit, a dog and a fur coat but not to his wooden blocks: a conditioned emotional response. With one infant and no control, the study would be unethical today.

Can fear be unconditioned? In 1924 **Mary Cover Jones** (1897 to 1987) treated Peter, a boy who feared rabbits: while he ate a favourite food, a caged rabbit was brought a little closer each day until he let it nibble his fingers. This is **counterconditioning**: replacing a conditioned response with an incompatible one, so that the client learns the stimulus is no longer threatening. **Joseph Wolpe** (1915 to 1997) borrowed a physiologist's term for the principle, **reciprocal inhibition**: a response incompatible with anxiety, made to occur in the presence of the feared stimulus, weakens the bond between them. His **systematic desensitisation**

(*Psychotherapy by Reciprocal Inhibition*, 1958) has three steps: relaxation training, usually the progressive muscle relaxation of **Edmund Jacobson** (1888 to 1983); an anxiety hierarchy, its items rated on Wolpe's 0 to 100 Subjective Units of Disturbance Scale (SUDS); and desensitisation proper, in which the relaxed client imagines each item in turn, moving up only when it no longer arouses anxiety. A relaxed student who pictures himself on the stage, then introducing himself, then giving his feared talk, is being desensitised.

In **flooding**, another conditioning therapy, the client stays in the most feared situation at full intensity, with no steps or relaxation, until anxiety subsides: extinction with escape blocked. Implosive therapy (**Thomas Stampfl**) is flooding in imagination. Aversion therapy pairs a cue such as the taste of alcohol with a nausea-producing drug, and in covert sensitisation (**Joseph Cautela**, 1967) the behaviour and an unpleasant outcome are both imagined. Exposure and response prevention (**Victor Meyer**, 1966) exposes a person with obsessive-compulsive disorder (OCD) to the feared cue and blocks the compulsion, so that anxiety falls by habituation and extinction (Clinical Psychology chapter).

> **! EXAM TRAP**
>
> Systematic desensitisation is **Wolpe's** and rests on **reciprocal inhibition**, not operant extinction, shaping, habituation, the law of effect, catharsis or vicarious reinforcement. Relaxation is **Jacobson's**; hierarchy ratings use **SUDS**, not the Fear Survey Schedule, State-Trait Anxiety Inventory or Hamilton Anxiety Rating Scale. Graded steps with relaxation is desensitisation, full intensity is flooding, and imagined behaviour plus imagined unpleasantness is covert sensitisation. Exposure and response prevention rests on **habituation and extinction**; "graded exposure" is a procedure, not a principle. **Symptom substitution** is not a technique but the psychoanalytic prediction that a removed symptom will be replaced.

Prediction and preparedness ○

Conditioning was long thought to need only contiguity, the CS and UCS close together in time. In 1968 **Robert Rescorla** (1940 to 2020) showed that rats did not fear a tone if shocks were as likely without it as with it: conditioning depends on contingency, how well the CS predicts the UCS. In blocking (**Leon Kamin**, 1969), rats already afraid of a noise learned almost nothing about a redundant light added to it. In overshadowing the most salient part of a compound CS takes most of the learning; in latent inhibition (**Robert Lubow** and **A. U. Moore**, 1959) a stimulus often met alone beforehand conditions slowly.

The Rescorla-Wagner model (Rescorla and **Allan Wagner**, 1972), $\Delta V = \alpha\beta(\lambda - \Sigma V)$, says that an animal learns only when surprised. A CS gains associative strength (ΔV) in proportion to the prediction error, the gap between the most the UCS can support (λ) and what all stimuli present already predict (ΣV), scaled by salience and learning rate ($\alpha\beta$). The error shrinks as the UCS becomes expected, so each trial adds less; in blocking it is zero from the start. In the 1990s **Wolfram Schultz** found dopamine neurons of the ventral tegmental area signalling exactly this error. The CS is thus a signal that predicts the UCS, not, as Pavlov's stimulus substitution theory held, a substitute for it.

KEY STUDY: GARCIA AND KOELLING'S "BRIGHT, NOISY, TASTY WATER" (1966)

Question. Are all stimuli equally easy to associate with all consequences? **Method. John Garcia** (1917 to 2012) and **Robert Koelling** let rats drink saccharin-flavoured water whose every lick also produced a flash and a click; then some were made ill, others shocked. **Result.** Ill rats avoided the taste but not the light and click; shocked rats avoided the light and click but not the taste. **Conclusion.** Taste "belongs" with illness, and sights and sounds with pain; the finding so contradicted theory that Garcia struggled to publish it.

Such **conditioned taste aversion** forms in a single trial, even when illness follows the taste by hours, and refutes equipotentiality, the assumption that any stimulus can be associated with any consequence equally easily. **Martin Seligman** (born 1942) called the principle **preparedness** (1970): each species is biologically prepared to learn some associations easily, which is why phobias of snakes and heights are common and phobias of electric sockets rare. Chemotherapy patients are given a novel "scapegoat" food to protect their usual diet, and advertisers pair a brand with a beloved film star so that the liking transfers.

! EXAM TRAP

Taste aversion is **classical conditioning** (taste = CS, illness = UCS), not operant, second-order conditioning or a "spurious association" (a statistical term). It illustrates **biological preparedness**, defying equipotentiality and contiguity. Prediction error is signalled by **dopamine neurons in the ventral tegmental area**, not serotonin, GABA or acetylcholine. Latent *inhibition* is not Tolman's latent *learning*.

Thorndike, Skinner and consequences ●●●

Most behaviour, however, is emitted rather than elicited, and its consequences decide whether it recurs. **Edward Lee Thorndike** (1874 to 1949) shut hungry cats in puzzle boxes (*Animal Intelligence*, 1898). Each cat scrambled until it hit the release by accident, and over trials its escape time fell gradually, not suddenly: trial-and-error learning. His **law of effect** (1911) states that responses followed by satisfaction are strengthened and those followed by discomfort weakened. His connectionism, which treats learning as forming stimulus-response bonds, added the law of exercise (use strengthens a bond), later abandoned, and the law of readiness (acting when ready is satisfying).

The radical behaviourist **Burrhus Frederic Skinner** (1904 to 1990), in *The Behavior of Organisms* (1938), separated respondent behaviour, elicited by a stimulus, from operant behaviour, emitted by the organism and operating on the environment. In **operant conditioning** the probability of a response is changed by its consequences (Thorndike's trial-by-trial version is often called instrumental conditioning). In the Skinner box a rat pressed a lever freely while a cumulative recorder plotted its rate. A discriminative stimulus, such as a chai stall's "Open" board, signals that a response will be reinforced, bringing it under stimulus control. Applied behaviour analysis and functional analysis, which ask what a problem behaviour gets the person or lets him escape, rest on these principles; merely recording a behaviour often changes it (reactivity).

Two questions classify every consequence (Table 1.11): does the behaviour increase (reinforcement) or decrease (punishment), and is a stimulus added (positive) or removed (negative)? Positive and negative mean plus and minus, not good and bad. A friend's warm

agreement with an opinion you voice is positive reinforcement, so the attitude strengthens. Pressing a lever to switch off a shock already on is **negative reinforcement**. A fine for jumping a red light is negative punishment. A parent who buys a sweet to end a tantrum positively reinforces the tantrum and is negatively reinforced for giving in.

Table 1.11 The four kinds of operant consequence.

Effect on behaviour	Stimulus added (positive)	Stimulus removed (negative)
Increases (reinforcement)	Positive reinforcement: something wanted is given (praise)	Negative reinforcement: something unpleasant is taken away (a headache ends)
Decreases (punishment)	Positive punishment: something unpleasant is given (a scolding)	Negative punishment: something wanted is taken away (time-out, response cost)

Food is a primary reinforcer, effective without learning. Money and praise are **secondary reinforcers**, neutral stimuli that became reinforcing through association with primary ones; one linked with many primary reinforcers, as money is, is a generalised reinforcer. (Harry Harlow's infant monkeys preferred a cloth mother to a wire one that fed them: attachment rests on contact comfort, not secondary reinforcement.) In the 1960s **Teodoro Ayllon** and **Nathan Azrin** built the **token economy** at Anna State Hospital, Illinois: long-stay psychiatric patients earned tokens for self-care and ward work and exchanged them for privileges (*The Token Economy*, 1968).

Escape learning terminates an aversive stimulus already present; avoidance learning prevents its arrival. **Orval Hobart Mowrer** (1907 to 1982) explained avoidance with his **two-factor theory** (1947): a warning signal comes to elicit fear by classical conditioning, and avoiding it is negatively reinforced by fear reduction, an operant process that also keeps the fear from extinguishing. Compulsive washing in OCD is such an escape or safety behaviour. Time-out (brief removal from a setting where reinforcement is available) and **response cost** (loss of a reinforcer already held) are negative punishment. As **William Kaye Estes** (1919 to 2011) showed in 1944, punishment suppresses a response without erasing it; it also breeds fear of the punisher and models aggression.

New behaviour is built by **shaping**, reinforcing successive approximations to a target: facing the lever, approaching it, touching it, finally pressing it. Shaping works on a response's speed and force and needs no words. Chaining links responses into a sequence; a prompt adds a cue and fading withdraws it. Operant extinction, the fading of a response no longer reinforced, often begins with an extinction burst of harder, more varied responding (jabbing at a lift button), after which responding falls back to its operant level, the rate before reinforcement.

> **! EXAM TRAP**
>
> The law of effect is **Thorndike's** (not Skinner's, Pavlov's or Watson's), not modelling, insight or unconscious drive. Operant conditioning works through **reinforcement and punishment**, not modelling or cognitive dissonance. **Negative reinforcement is not punishment**, and it too increases behaviour, though one real item keyed only positive reinforcement. Removing a desirable stimulus to reduce behaviour is **response cost**, not negative reinforcement, positive punishment or extinction; time-out is **negative punishment**. OCD compulsions are **escape (avoidance) behaviours**, not conditioned stimuli, unconditioned responses or positively reinforced acts.

> **! EXAM TRAP**
>
> The token economy is **operant** (secondary reinforcement), the therapy **not** derived from classical conditioning, and **Ayllon and Azrin's** (Amity 2023 keyed Skinner, who supplied only the principles). Stickers that buy a homework exemption are **tokens**, not primary or generalised reinforcers or "prepotent" (dominant) responses. Ignoring tantrums is **extinction**, not "operant escape" (another wrong Amity 2023 key). Shaping is not chaining, prompting, fading or the law of effect; RML 2024 keyed its principles as speed, power and non-verbal.

Schedules and other operant effects ●●●

A **schedule of reinforcement** is the rule deciding which responses are reinforced. Continuous reinforcement, after every correct response, teaches fastest but extinguishes fastest; partial reinforcement makes behaviour persist far longer once reward stops (the partial reinforcement extinction effect). Reward may depend on the number of responses (ratio) or on time elapsed (interval), fixed or variable, giving the four schedules that **Charles Ferster** and Skinner mapped in 1957 (Figure 1.7).

- Fixed ratio (FR), as in piece-rate pay: high, steady responding with a brief post-reinforcement pause.
- Variable ratio (VR), as in slot machines: the highest, steadiest rate and the greatest resistance to extinction.
- Fixed interval (FI), as in checking a dish due out of the oven: a pause, then faster and faster responding, the scallop.
- Variable interval (VI), as in checking for a reply to a message: moderate, very steady responding.

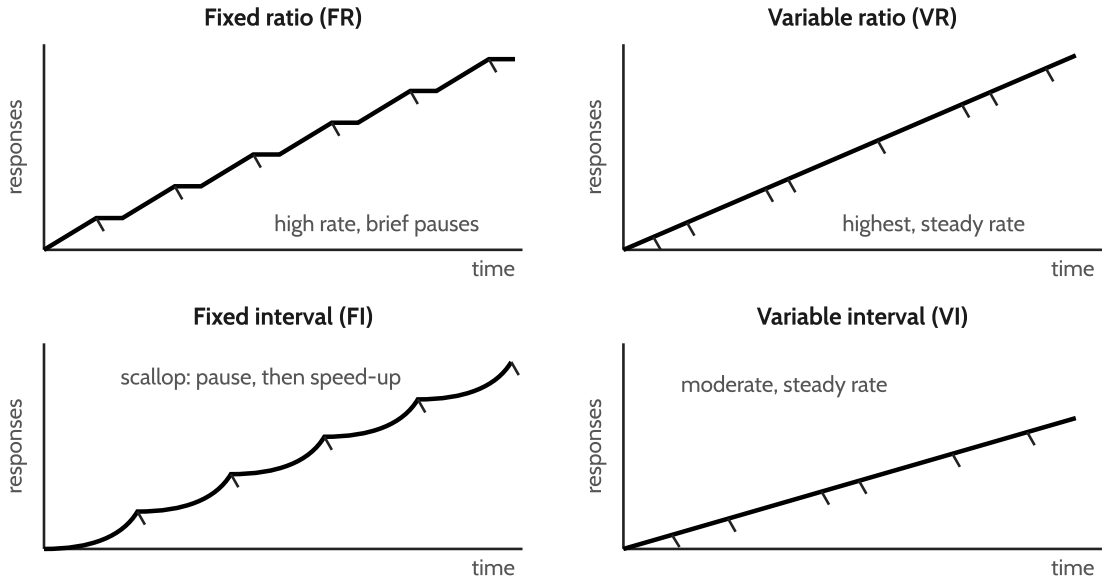


Figure 1.7 Stylised cumulative records for the four basic schedules. The steeper the line, the faster the responding; short diagonal ticks mark reinforcers. Notice the flat steps after each reinforcer on FR, the unbroken steep line on VR, the scallops on FI and the steady, shallower line on VI.

> **! EXAM TRAP**
>
> "Number" means ratio, "time" interval, "unpredictable" variable. Reinforcement after every response is **continuous**, the least resistant to extinction. Sheena's chocolate "for every 45 minutes" of study is keyed **fixed interval**.

Differential reinforcement extinguishes a problem behaviour while reinforcing other behaviour, that is, its absence (DRO), an incompatible (DRI) or alternative (DRA) behaviour, or a low rate of it (DRL). The **Premack principle** of **David Premack** (1959) holds that a more probable activity can reinforce a less probable one: homework first, then video games. Skinner's pigeons, fed every 15 seconds whatever they did (1948), developed odd rituals: superstitious behaviour, strengthened by accidental reinforcement.

KEY STUDY: LEARNED HELPLESSNESS (SELIGMAN AND MAIER, 1967)

Martin Seligman and **Steven Maier** gave dogs shocks they could switch off, identical shocks they could not control (the yoked group), or none. Later, in a shuttle box, most yoked dogs passively took shocks they could have escaped by jumping a barrier. Lack of control, not shock, produced **learned helplessness**: passive acceptance of aversive events, even avoidable ones, after repeated unavoidable ones. It became a model of depression, later recast in attributional terms (Psychiatric Disorders chapter).

Biology limits operant learning too: **Keller Breland** and **Marian Breland** (1961) found raccoons trained to bank coins rubbing them together as if washing food (instinctive drift). **Richard Herrnstein**'s matching law (1961) states that responses are divided among options in proportion to the reinforcement each yields. Rats that **Leo Crespi** (1942) shifted to a medium reward ran slower if shifted down, and faster if shifted up, than rats always given it: contrast effects, too abrupt to be learning, showing that reward size changes motivation. In **biofeedback** a bodily process is measured and displayed so that changes in the desired direction are reinforced, following **Neal Miller**'s (1909 to 2002) claim that autonomic responses can be operantly conditioned. Electromyographic (EMG) biofeedback suits tension headache, and **Arnold Kegel**'s perineometer (1948) was an early form.

> **! EXAM TRAP**
>
> Premack: **high-probability behaviour reinforces low-probability behaviour**, never the reverse. Learned helplessness is **Seligman and Maier's**, not Beck's, Sullivan's or Kahneman's. The Crespi effect is a **change in motivation when reward size changes**, not retroactive inhibition, extinction or insight. Autonomic control by operant conditioning is **biofeedback**, not hypnosis, sublimation or desensitisation.

Table 1.12 compares the two conditionings. They are not "entirely different" but interact constantly: Albert's fear was classical, but had he crawled away and felt relief, his avoidance would have been operant. The hallmark of behaviour therapy is the use of both to eliminate unwanted behaviour, not a search for underlying causes, insight, cognitive restructuring, dream analysis or free association.

Table 1.12 Classical versus operant conditioning.

Feature	Classical conditioning	Operant conditioning
Key figures	Pavlov, Watson	Thorndike, Skinner
What is associated	Two stimuli (CS and UCS)	A response and its consequence
Behaviour	Elicited, involuntary, emotional	Emitted, voluntary, goal-directed
Extinction	CS presented without the UCS	Response no longer reinforced

> **! EXAM TRAP**
>
> The main difference: **classical pairs two stimuli; operant relies on reinforcement and punishment**, not "classical relies on reinforcement" (backwards) or "only operant involves cognition".

Latent learning, insight and learning sets ●●

Against the view that learning is only stimulus-response connections stamped in by reward, **Edward Chace Tolman** (1886 to 1959), a purposive behaviourist, argued that animals learn knowledge.

> **KEY STUDY: TOLMAN AND HONZIK'S LATENT LEARNING EXPERIMENT (1930)**
>
> Tolman and **Charles Honzik** ran three groups of hungry rats through a maze daily for seventeen days: one always found food in the goal box, one never did, and one found food only from day 11. The third group made as many errors as the unrewarded rats for ten days, then at once matched the always-rewarded rats. It had learned the maze all along; reward merely gave it a reason to show it, so learning and performance differ.

This is **latent learning**: learning without obvious reinforcement that stays hidden until there is an incentive to show it. Tolman (1948) said the rats had formed a **cognitive map**, a mental representation of the layout; rats learn places and expectancies of what leads to what, not only turns. **Wolfgang Köhler** (1887 to 1967), a founder of Gestalt psychology (Section 1.2), studied chimpanzees on Tenerife (1913 to 1920). Sultan, unable to reach a banana with either of two short sticks, later fitted one into the other and at once raked it in; others stacked boxes. Such **insight learning**, the sudden grasp of a solution by perceiving the relations among a problem's elements, is abrupt, smooth once found, well retained and readily transferred; its flash of understanding is the aha experience. **Harry Harlow** (1905 to 1981) found that monkeys given hundreds of discrimination problems came to solve each new one in a single trial: a learning set, or learning to learn (1949), suggesting that apparent insight rests on earlier, gradual learning. Concept formation (Section 1.8) runs from observing instances, through discriminating them from non-instances and generalising to all that share the essential features, to abstracting the defining properties.

> **! EXAM TRAP**
>
> Latent learning and the cognitive map are **Tolman's**, not Bandura's, Skinner's, Köhler's or Harlow's (Gould studied honeybee navigation). Sultan means **insight**, and the realisation is the **"aha" experience**, not the "light bulb"; insight graded in a mental status examination is psychiatric. Concept formation runs **observation, discrimination, generalisation, abstraction**; assimilation is Piaget's term.

Observational learning ●●●

Most complex skills, from rolling a chapati to a cover drive, are learned largely by watching. This is **observational learning** (social learning, modelling): learning by watching others, called models, and the consequences of their behaviour. **John Dollard** (1900 to 1980) and Neal Miller (1941) held that imitation must itself be reinforced, the early social learning theory; **Albert Bandura** (1925 to 2021) showed that it need not be.

KEY STUDY: BANDURA, ROSS AND ROSS'S BOBO DOLL EXPERIMENT (1961)

Bandura, **Dorothea Ross** and **Sheila Ross** let 72 nursery-school children watch an adult punch, kick and hammer an inflatable Bobo doll. Left later with the doll, these children copied the model's specific acts and words far more than children who saw a quiet model or none, with no reinforcement at all. In 1965 children who saw the model punished imitated less, until offered a reward for showing what the model had done, when all groups did equally well. **Criticism.** Bobo is made to be hit, and only short-term effects were measured.

The model's consequences, then, affect performance, not learning: seeing a model rewarded increases imitation (vicarious reinforcement), and seeing one punished reduces it. Bandura's four processes are attention (noticing the model), retention (storing the act in images and words), motor reproduction (turning the memory into action) and motivation (a reason to perform). The first three govern learning and the fourth performance, so reinforcement is not necessary, and the model must actually demonstrate the target behaviour. Modelling is also a therapy (Clinical Psychology chapter).

With **Richard Walters** (1963) Bandura developed later social learning theory, renamed social cognitive theory in 1986. Its **reciprocal determinism** (1978) holds that the person (beliefs, expectations), behaviour and environment influence one another in both directions, against Skinner's view that environment alone shapes behaviour. Self-regulation means that most behaviour is guided by self-set standards and self-reward rather than external rewards. The theory's central motivational construct is **self-efficacy** (1977), a person's belief in his capability to carry out the actions a specific task requires. (**Julian Rotter**'s separate social learning theory of 1954 belongs to the Personality chapter: behaviour potential, the likelihood of a behaviour in a given situation in relation to a reinforcement, depends on expectancy and reinforcement value. Social learning frameworks in general explain behaviour by expectancies, situational demands and incentives rather than by reinforcement history alone.)

! EXAM TRAP

The four processes are **attention, retention, motor reproduction and motivation**, not sensory-perceptual or memory stages; the one NOT necessary is **reinforcement**, a view that is **Bandura's**, not Skinner's, Tolman's or Miller and Dollard's. Reciprocal causation belongs to **social cognitive theory**, not two-factor, drive-reduction or constructivist theory, and is neither reciprocal inhibition nor reciprocal altruism. Self-efficacy (Bandura, not Beck, Ellis or Skinner) is not self-esteem, external locus of control or Adlerian striving; it, not goal difficulty, rewards or drives, is the motivational core of social cognitive theory. Most behaviour stems from **self-regulation**, not "reciprocal dualism" or "behavioural determinism". Match: Dollard and Miller, early social learning; Bandura and Walters, later; Skinner, radical behaviourism; Maslow, self-actualisation.

Other theorists and learning in practice ○

The grand theories divided on two questions: is what is learned a stimulus-response connection, as the stimulus-response (S-R) behaviouristic approach holds, or knowledge (Tolman, Köhler); and does learning need only contiguity, or reinforcement too? **Edwin Ray Guthrie** (1886 to 1959) chose contiguity: a stimulus combination that accompanied a movement will tend to be followed by it again, at full strength after one trial. A habit is broken by attaching a new response to its cue: present the cue too weakly to trigger the habit and strengthen it gradually, as with a light blanket before a saddle (threshold method); repeat the response to exhaustion (fatigue method); or present the cue while a competing response occurs (incompatible response method). **Clark Leonard Hull** (1884 to 1952) chose reinforcement, defined as drive reduction: reaction potential equals habit strength multiplied by drive (${}_{S}E_{R} = {}_{S}H_{R} \times D$). Table 1.13 adds Estes and **Robert Mills Gagné** (1916 to 2002). Mowrer and his wife devised the bell-and-pad for bed-wetting (1938), and constructivists (Piaget, Vygotsky, Bruner) hold that learners actively build knowledge.

Table 1.13 Other learning theorists at a glance.

Theorist	Family	Central idea	Key work
Guthrie	S-R contiguity	One-trial learning; three ways to break habits	*The Psychology of Learning* (1935)
Hull	S-R reinforcement	Drive reduction; habit strength × drive	*Principles of Behavior* (1943)
Estes	Mathematical	Stimulus sampling: each trial conditions a fraction (θ) of stimulus elements	Stimulus sampling theory (1950)
Gagné	Instructional	Eight types in a hierarchy: signal, S-R, chaining, verbal association, multiple discrimination, concept, principle, problem solving	*The Conditions of Learning* (1965)

A learning curve plots performance against practice: usually negatively accelerated (fast early gains, then diminishing returns), sometimes positively accelerated (a slow start on a hard task) or S-shaped. **William Lowe Bryan** and **Noble Harter** (1897, 1899) found telegraphers stalling on a plateau, a temporary flat stretch while letter habits became automatic, before word habits lifted them again; it is not the final physiological limit. Overlearning, practice beyond the first errorless recall, aids retention (**William C. F. Krueger**, 1929). Distributed (spaced) practice usually beats massed practice; whole learning suits short, meaningful material and part learning long, difficult material. Knowledge of results (feedback) is essential,

and a specific goal makes practice purposeful (**Edwin Locke** and **Gary Latham**). Learners with learning difficulties need multisensory input and regular repetition.

Transfer of training, the effect of one task on learning another, is positive (cycling helps with a motorcycle), negative (an old habit interferes) or zero. **William James** (1890) and Thorndike refuted formal discipline, the belief that Latin trains the mind; Thorndike's identical elements theory ties transfer to elements the tasks share. **Sidney Leavitt Pressey** built the first teaching machine (1920s); Skinner's programmed instruction (1954) used small steps, active responses, immediate feedback and self-pacing.

> **! EXAM TRAP**
>
> Learning through S-R associations is the **stimulus-response behaviouristic approach**, not the cognitive, psychoanalytic or existential approach. "Learning always leads to efficacy" given **a specific goal**, not ability, aptitude or knowledge. A plateau is not the physiological limit or extinction.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The near opposite of habituation is _____.
2. Pavlov's bell was a neutral stimulus before pairing and a _____ stimulus after it.
3. Wolpe's systematic desensitisation rests on _____.
4. Removing an unpleasant stimulus to increase a behaviour is _____ reinforcement.
5. The schedule with the highest rate and greatest resistance to extinction is _____.
6. Bandura's four processes are attention, retention, _____ and motivation.

Answers: (1) sensitisation, (2) conditioned, (3) reciprocal inhibition, (4) negative, (5) variable ratio and (6) motor reproduction.

1.7 Memory: Encoding, Storage, Retrieval and Forgetting HIGH YIELD · 111 QS

Memory, the capacity to retain information and use it later, is not one thing. After encephalitis destroyed both his hippocampi in 1985, the British musician **Clive Wearing** could follow a conversation only while it lasted and had lost almost every episode of his past, yet he could still sight-read and conduct. To see how, begin with how memory was first measured.

Ebbinghaus and the measurement of memory ●●

Cognitive psychology treats the mind as an information processor (Section 1.2) and divides remembering into *encoding* (turning information into a code the system can hold), *storage* and *retrieval*. Encoding needs attention: fewer than half of the adults tested by **Raymond Nickerson** and **Marilyn Adams** (1979) could pick out the correct drawing of a coin they had handled thousands of times. Details never attended to are never encoded (*encoding failure*) and so cannot be retrieved.

The first memory experiments were made by **Hermann Ebbinghaus** (1850 to 1909), his own sole participant. To escape the associations of real words he invented the *nonsense syllable* (DAX, BUP), learned lists to one perfect recitation and relearned them later. Relearning was faster even when nothing could be recalled. The **savings method** expresses this as original learning time minus relearning time, divided by original learning time, multiplied by 100: a list learned in 20 readings and relearned in 12 shows (20 minus 12) divided by 20, or 40 per cent, savings. His *forgetting curve* (*Über das Gedächtnis*, 1885) falls steeply, losing more than half within an hour and about two thirds within a day, then levels off without reaching zero. He also first noted the spacing effect and the advantage of a list's beginning and end.

Retention is measured by *recall*, which is free (any order), serial (original order) or cued (from a hint); *paired-associate learning*, introduced by **Mary Whiton Calkins** in 1894, is cued recall of pairs, such as a Kannada word and its English meaning. *Recognition*, picking old items from new ones as in a multiple-choice question, is easier than recall. Relearning is the most sensitive measure. The *reconstruction method* hands back the presented items shuffled for the person to put in order.

> **! EXAM TRAP**
>
> Items presented, scrambled and put back in order (RML 2024) is the reconstruction method, a measuring technique, not Bartlett's reconstructive memory. Ebbinghaus, **Francis Galton** and Sigmund Freud were independent pioneers, not laboratory founders, and each devised a new method (nonsense syllables; questionnaires and word association; free association). RML 2020 keys "independent pioneers" alone, as Ebbinghaus's method was experimental, not observational; a bank version keys "(a) and (c)", adding "novel observational methods".

The multi-store model ●●●

In 1968 **Richard C. Atkinson** and **Richard M. Shiffrin** proposed the **multi-store model**, also called the modal or stage model: attended information passes from sensory memory to short-term memory, and rehearsed information passes on to long-term memory, from which it is retrieved when needed. Attention and rehearsal are *control processes* that the person chooses.

The first store, **sensory memory** (the sensory register), briefly holds a large amount of raw, pre-categorical information, one store per sense. **Ulric Neisser** named the visual store *iconic memory* and the auditory store *echoic memory* in 1967; a touch store, haptic memory, is little studied.

> **KEY STUDY: SPERLING'S PARTIAL REPORT (1960)**
>
> Shown twelve letters for 50 milliseconds, people report only four or five but insist they saw more. **George Sperling** sounded a tone just after the flash to signal which of the three rows to report. People gave about three of the four cued letters, so about nine of the twelve had been briefly available. Delaying the tone by about one second removed the advantage: the icon fades within roughly 250 to 500 milliseconds.

Echoic memory lasts two to four seconds, as the "three-eared man" study of **Christopher Darwin**, **Michael Turvey** and **Robert Crowder** (1972) showed.

Short-term memory (STM) holds a little information for seconds unless it is rehearsed. **George Armitage Miller** put its span at "the magical number seven, plus or minus two" (1956) meaningful units, or chunks. Grouping items into larger meaningful units is **chunking**: I N D I A I I T I S R O is twelve letters but three chunks (INDIA, IIT, ISRO). Without chunking and rehearsal, capacity is nearer four chunks (**Nelson Cowan**, 2001). In the Brown-Peterson technique (**John Brown**, 1958; **Lloyd Peterson** and **Margaret Peterson**, 1959), counting backwards blocked rehearsal, and recall of a consonant trigram fell from about 80 per cent at 3 seconds to about 10 per cent at 18, hence a duration of 18 to 30 seconds; **Geoffrey Keppel** and **Benton Underwood** (1962) traced the loss to interference, not decay. STM codes mainly by sound: **Reuben Conrad** (1964) found people writing V for B, and **Alan Baddeley** showed in 1966 that sound-alike words confuse immediate recall and meaning-alike words delayed recall. *Long-term memory* (LTM) codes mainly by meaning (Table 1.14).

Table 1.14 The three stores of the multi-store model compared.

Feature	Sensory memory	Short-term memory	Long-term memory
Capacity	Large	7 ± 2 items (Miller); about 4 chunks (Cowan)	No known limit
Duration	Icon 0.25 to 0.5 s; echo 2 to 4 s	About 18 to 30 s unrehearsed	Minutes to a lifetime
Main code	Raw sensory	Acoustic	Semantic
Main cause of loss	Decay and masking	Displacement and interference	Interference and retrieval failure

> **! EXAM TRAP**
> "Largest capacity but shortest duration" (JEMAS 2025) is sensory memory; "largest capacity" alone in the Atkinson and Shiffrin model is long-term memory. "Icon and echo" are always sensory memory, and the store that "lasts only a few seconds" is sensory memory, not eidetic imagery.

Free recall gives the U-shaped **serial position effect**, or primacy-recency effect (**Bennet Murdock**, 1962; Figure 1.8). The *primacy effect* reflects extra rehearsal of early items into LTM; the *recency effect* reflects the last items still sitting in STM. **Murray Glanzer** and **Anita Cunitz** (1966) found that 30 seconds of counting before recall abolished recency but spared primacy.

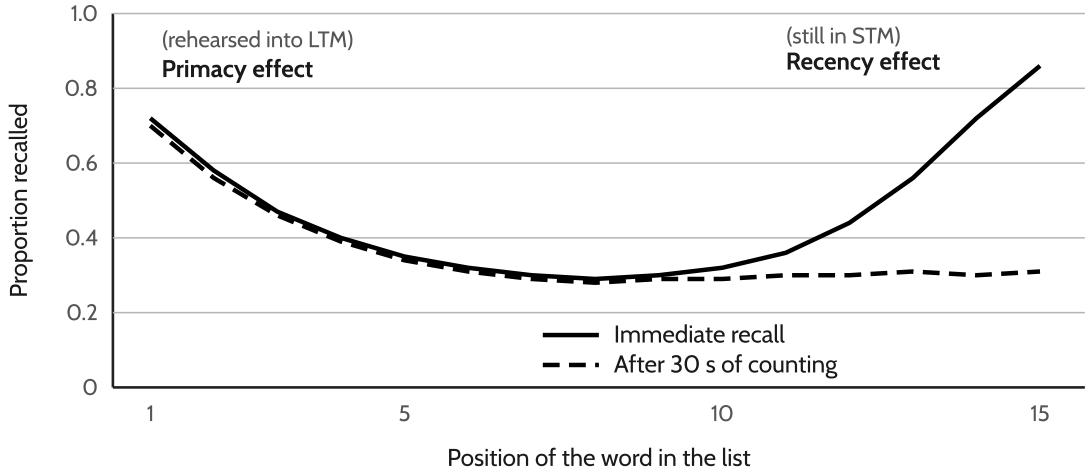


Figure 1.8 The serial position curve in free recall (schematic, with the pattern reported by Glanzer and Cunitz, 1966). Immediate recall (solid line) shows both primacy and recency. A 30-second filled delay (dashed line) removes the recency effect but leaves primacy intact.

Amnesic patients such as H.M., met below, have a normal span and recency effect but no new long-term learning, while patient K.F. (**Tim Shallice** and **Elizabeth Warrington**, 1970) had a two-item span and normal long-term learning: a double dissociation that supports separate stores. Yet neither store proved unitary, and mere repetition does little for LTM.

> **! EXAM TRAP**
>
> The stage model has three stores; working memory is NOT one of them (RML 2025), and information must pass through short-term (sometimes "short-term (working)") memory to reach LTM. RML 2024 keys "McCray and Hunter" (**J. W. McCrary** and **W. S. Hunter**, 1953) for first observing the primacy-recency effect, although Ebbinghaus noted it in 1885. Better recall of the last items is recency, not the Flynn effect (rising IQ scores) or halo effect (a rating bias). As order effects in a psychophysics experiment, primacy and recency are controlled by counterbalancing (Section 1.3).

Working memory ●●●

Multiply 24 by 7 in your head and you must hold numbers, calculate and store a partial result at once. Alan Baddeley and **Graham Hitch** (1974) therefore recast STM as **working memory**, a limited-capacity system for holding and manipulating information during reasoning and comprehension. Using the dual-task method, they found that people holding six digits still reasoned accurately, so STM must have several parts (Figure 1.9).

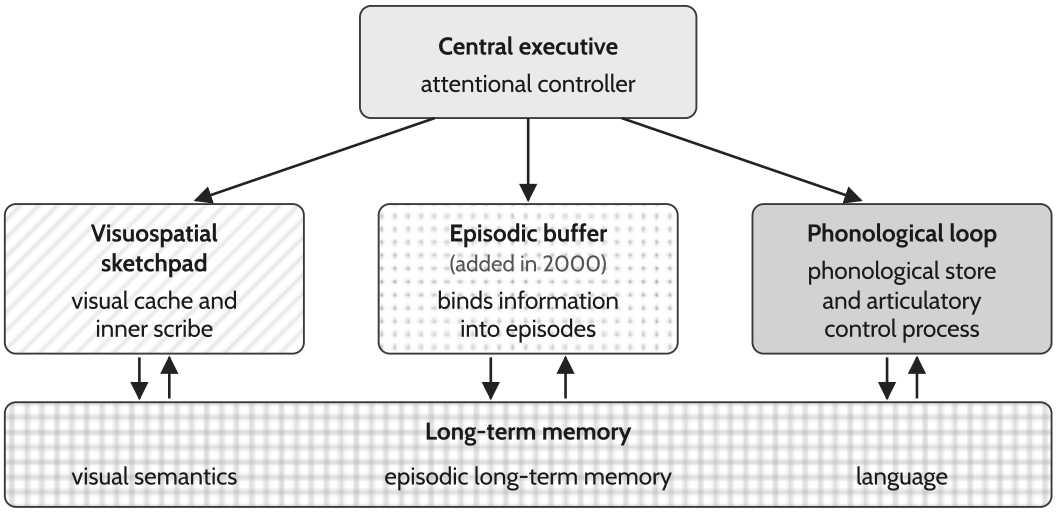


Figure 1.9 Baddeley's working memory model (Baddeley and Hitch, 1974; episodic buffer added by Baddeley, 2000). The central executive directs attention but stores nothing; the three subsidiary systems hold information and exchange it with long-term memory.

The **phonological loop** stores and rehearses auditory-verbal information: a phonological store (inner ear) holds speech sounds for about two seconds, and an articulatory control process (inner voice) refreshes them. Similar-sounding items are confused, spans are larger for short words (the word-length effect), and *articulatory suppression* (saying "the, the, the") abolishes the word-length effect. The **visuospatial sketchpad** holds visual and spatial information, such as a mentally rotated object; **Robert Logie** (1995) split it into a visual cache and an inner scribe.

The **central executive**, an attentional controller that stores nothing, focuses, divides and switches attention and inhibits unwanted responses. Random number generation loads it, and dorsolateral prefrontal damage causes a dysexecutive syndrome. In the Stroop task (**John Ridley Stroop**, 1935), naming the ink colour of RED printed in blue is slow because automatic reading competes with controlled colour naming, an interference in selective attention (Section 1.5). The **episodic buffer**, added by Baddeley in 2000, holds about four chunks and binds information from the loop, the sketchpad and LTM into single, coherent episodes. STM is storage alone (forward span); working memory is storage plus processing, tested by complex span and by backward digit span.

> **! EXAM TRAP**
> Random number generation loads the central executive (RML 2022); repeating a word loads the loop, a spatial configuration the sketchpad, irrelevant speech the phonological store. A "visual executive loop" (Amity 2023) does not exist, and the Stroop effect is selective attention, not memory.

Levels of processing ○

Fergus Craik and **Robert Lockhart** (1972) argued that memory is a by-product of how deeply information is processed. Their **levels of processing** run from shallow structural processing (is the word in capitals?) through phonemic processing (does it rhyme with WEIGHT?) to

deep semantic processing (does it fit the sentence?), and deeper processing leaves more durable traces, as Craik and Endel Tulving (1975) confirmed. Maintenance rehearsal, mere repetition, adds little to LTM; elaborative rehearsal, linking an item to meaning, builds it.

Words judged for "Does this describe you?" are recalled about twice as well as words judged for meaning: the **self-reference effect** (**T. B. Rogers**, **Nicholas Kuiper** and **W. S. Kirker**, 1977). Words you generate also beat words you merely read (the generation effect). Critics called depth circular, since it is defined by the memory it explains, and **Donald Morris**, **John Bransford** and **Jeffery Franks** (1977) found that rhyme processing wins on a rhyming test: memory depends on the match between study and test (transfer-appropriate processing).

> **! EXAM TRAP**
>
> The self-reference effect is Rogers, Kuiper and Kirker's (keyed "Rogers and Kuiper", RML 2024), not Craik and Lockhart's, **Allan Paivio**'s (dual coding theory, 1971: verbal plus imagery codes) or **Michael Posner**'s (attention). It means better recall of self-related information (IHBAS 2020), not self-concept, self-schema or self-awareness (Personality chapter).

The systems of long-term memory ●●●

Clive Wearing kept skills and facts but lost his past, so long-term memory is not one store. In 1972 **Endel Tulving** (1927 to 2023) separated **episodic memory**, personal events with their what, where and when (your first day of college), from **semantic memory**, the network of facts and concepts that make up our knowledge of the world, also called generic memory. Episodic recall carries autonoetic consciousness, a sense of travelling back in time; semantic memory carries only noetic knowing. Knowing how (riding a bicycle) is **procedural memory**. Tulving's patient K.C. kept his general knowledge but lost every episode of his life.

Explicit memory involves conscious recollection, tested by recall and recognition; **implicit memory** is an influence of past experience on behaviour, feelings or thought without conscious recollection (**Peter Graf** and **Daniel Schacter**, 1985), roughly **Larry Squire**'s declarative and nondeclarative memory. Declarative memory needs the hippocampus; nondeclarative memory covers procedural skills (striatum), priming, classical conditioning (amygdala, cerebellum) and habituation. In *priming*, earlier exposure eases processing of the same item (repetition priming) or a related one: BUTTER is recognised faster after BREAD (semantic priming; **David Meyer** and **Roger Schvaneveldt**, 1971), even with a masked, unseen prime, so semantic priming is a tool for studying subliminal perception.

> **! EXAM TRAP**
>
> Implicit memory does NOT involve conscious recollection (RML 2020); it "affects performance without conscious awareness" (RML 2025), and limited capacity describes STM. Procedural memory is one kind of implicit memory.

Autobiographical memory holds the specific episodes of one's own life; older adults recall a surplus from ages 10 to 30, the *reminiscence bump* (**David Rubin** and colleagues, 1986). *Prospective memory*, remembering to act in future (a tablet at 8 p.m.), contrasts with retrospective memory. **Roger Brown** and **James Kulik** (1977) named the **flashbulb memory**, a vivid memory of the circumstances in which one learned of a shocking event, and credited it to a "Now Print!" mechanism.

KEY STUDY: NEISSER AND HARSCH, THE CHALLENGER DISASTER (1992)

The morning after the Challenger shuttle exploded in January 1986, Ulric Neisser and **Nicole Harsch** had students write down how they heard the news. Two and a half years later, a quarter of 44 were wrong on every major detail, yet confident. After 11 September 2001, **Jennifer Talarico** and David Rubin (2003) found that flashbulb and everyday memories lost consistency equally, but flashbulb memories stayed more vivid and more confidently believed.

Flashbulb memories are thus special in vividness and confidence, not accuracy, and arise from emotional arousal at encoding plus rehearsal. *Eidetic imagery*, a visual image that lingers half a minute or more, occurs in a few children and rarely in adults. *Metamemory* (**John Flavell**, 1971), knowledge and control of one's own memory, is part of metacognition and includes the feeling of knowing, a sense that one could recognise an answer one cannot recall. In the **tip-of-the-tongue state**, which Roger Brown and **David McNeill** (1966) produced with definitions of rare words (sextant), people knew the first letter and syllable count: the word was stored but not retrievable.

! EXAM TRAP

A vivid recollection of an important event is a flashbulb memory (RML 2025), not an engram, hindsight bias (the "knew it all along" distortion of memory) or equipotentiality. Its confidence stays very high as accuracy falls (NIMHANS 2026 recall), and arousal plus rehearsal explains it, not Now Print!, lack of interference or eidetic imagery. Tip-of-the-tongue is a retrieval failure (RML 2024), not storage, repression or aphasia (Wernicke's or Broca's, where language itself is damaged). Planning, monitoring and controlling memory is metacognition (IHBAS 2020), not attention, inhibition or a knowledge base.

Networks, schemas and reconstruction ●●●

In the hierarchical network of **Allan Collins** and **Ross Quillian** (1969), concepts are nodes (ANIMAL, BIRD, CANARY) and, by cognitive economy, each property is stored once at the highest node it fits, so "A canary has skin" takes longer to verify than "A canary can sing". In the spreading activation model of Collins and **Elizabeth Loftus** in 1975, activation spreads to related concepts and primes them. In **connectionism**, or parallel distributed processing (**David Rumelhart**, **James McClelland** and colleagues, 1986), a concept is a pattern of simultaneous activation across many interconnected units, and knowledge lies in the connection weights. A proposition is the smallest unit of knowledge that can be true or false.

! EXAM TRAP

"A network of interconnected nodes" (IHBAS 2024) keys the connectionist model, although Collins and Quillian's network also has nodes; "associative" and "linkage" describe any network rather than name a model. Parallel distributed processing means associative networks and simultaneous activation (IHBAS 2025), not serial processing.

A **schema** is organised knowledge, built from experience, that guides perception and memory; the schema approach studies how such knowledge is used to understand behaviour. A script, a schema for a routine (the restaurant script of **Roger Schank** and **Robert Abelson**,

1977: enter, order, eat, pay, leave), belongs to generic memory. The schema idea came from **Frederic Charles Bartlett** (1886 to 1969). In *Remembering* (1932), English students retelling the Native American tale "The War of the Ghosts" shortened it and emphasised a few details (levelling and sharpening, in **Gordon Allport** and **Leo Postman**'s later terms), explained away puzzling events (rationalisation) and turned canoes into boats. Remembering is an "effort after meaning": *reconstructive memory* rebuilds the past from fragments and schemas.

Asked how fast filmed cars were going when they "smashed" rather than "hit" each other, students gave higher speeds and a week later more often recalled broken glass that was never shown (32 against 14 per cent; Elizabeth Loftus and **John Palmer**, 1974). This is the **misinformation effect**: memory distorted by leading questions or misleading later information. Whole false events can be implanted (being lost in a shopping mall as a child), and people who study bed, rest, tired and dream recall the absent "sleep" (the Deese-Roediger-McDermott paradigm). In **source monitoring** (**Marcia Johnson** and colleagues, 1993) we attribute a memory to its origin from its contextual details; reality monitoring, perceived versus imagined, is one kind. Source errors cause source amnesia (a fact remembered without its source), cryptomnesia (another's idea recalled as one's own) and false eyewitness identifications. *Déjà vu* is the illusion of recognition, jamais vu its reverse, and pseudologia fantastica pathological lying. In the recovered memory controversy Loftus warned that suggestive therapy can implant false memories of abuse, though real abuse can also be forgotten and recalled.

> **! EXAM TRAP**
>
> Failing to recall the relevant elements of an autobiographical experience is a source-monitoring deficit (RML 2022); "personal-" and "social-monitoring" are invented. Defence lawyers fear source errors (Amity 2023), not forgetting or poor eyesight; confidence does not mean accuracy. Schema theory is Bartlett's, not Wundt's, Titchener's or Watson's, and his rationalisation is not the defence mechanism. Scripts come from generic memory (RML 2025).

Retrieval and forgetting ●●●

Much forgetting is a failure to find what is still stored: given category names as cues, people recall far more words (Endel Tulving and **Zena Pearlstone**, 1966). By the **encoding specificity principle** (Tulving and **Donald Thomson**, 1973), a cue works to the extent that it was encoded with the memory; recognition beats recall because the test supplies the item. In **context-dependent memory** the physical setting must match: divers recalled words best where they had learned them, underwater or on the beach (**Duncan Godden** and Alan Baddeley, 1975). In **state-dependent memory** the internal state must match: material learned while drunk is recalled better when drunk again (**Donald Goodwin** and colleagues, 1969). **Gordon Bower** (1981) added mood-dependent memory and mood-congruent memory, the easier recall of material matching one's current mood.

> **! EXAM TRAP**
>
> Rahul, who expects to do better in his usual classroom (IHBAS 2024), shows encoding specificity, keyed because context-dependent memory is not offered; "the principle of congruity" is a distractor. Offered both (JEMAS 2025), choose context-dependent for surroundings and state-dependent for internal states. Alcohol-induced amnesia is state-dependent memory (RML 2022).

Forgetting is useful: it clears out unneeded detail and reduces interference, and the more is retained, the less is forgotten. First, **decay theory**, credited to **Edward Lee Thorndike**'s (1874 to 1949) law of disuse, says unused traces fade with time. But **John Jenkins** and **Karl Dallenbach** (1924) found that after eight hours of sleep students recalled over half of their nonsense syllables, and after eight waking hours about a tenth: events, not time, caused the loss. Table 1.15 lists other named effects on retention.

Interference theory blames other memories. In **proactive interference** earlier learning disrupts newer learning; in **retroactive interference** newer learning disrupts older memories (PRO: old disrupts new; RETRO: new disrupts old). Dialling your old phone number instead of the new one is proactive, as is earlier psychology intruding on a later sociology test; losing the old number to the new one is retroactive. After lists A-B and then A-C, recalling A-B measures retroactive and recalling A-C proactive interference. **Arthur Melton** and **J. M. Irwin** (1940) blamed response competition and unlearning.

> **! EXAM TRAP**
>
> RML 2024 keys an English speaker applying English grammar to Spanish as "retroactive". The key is wrong: old disrupts new, so it is proactive. "Interference from new information" (IHBAS 2024) is retroactive. Decay theory is keyed to Thorndike (RML 2024), not Ebbinghaus, Tulving or Melton and Irwin.

In retrieval failure, or cue-dependent forgetting (Tulving, 1974), the trace survives but the cues do not match. Retrieval can even cause forgetting: practising some items of a category (FRUIT-orange) makes the unpractised ones (FRUIT-banana) harder to recall than items of unpractised categories (**Michael Anderson**, **Robert Bjork** and **Elizabeth Bjork**, 1994), through inhibition of competitors. **Sigmund Freud** (1856 to 1939) proposed motivated forgetting through **repression**, the unconscious exclusion of threatening memories from awareness; suppression is conscious, and evidence for repression is weak. Last, consolidation (**Georg Elias Müller** and **Alfons Pilzecker**, 1900) can fail: a head injury or electroconvulsive therapy erases recent memories, and by Ribot's law (**Théodule Ribot**, 1881) the new perishes before the old.

> **! EXAM TRAP**
>
> Retrieval-induced forgetting (RML 2020) is keyed "practised information remembered better than unpractised", but its defining half is that related unpractised items suffer. Repression is unconscious, suppression conscious (RML 2025); motivated forgetting covers both, and inhibition weakens competitors. "Less retention leads to less forgetting" is the incorrect statement (RML 2025).

Table 1.15 Classic named effects on retention.

Effect	Who and when	What happens
Zeigarnik effect	**Bluma Zeigarnik**, 1927	Interrupted tasks are recalled better
Ovsiankina effect	**Maria Ovsiankina**, 1928	Interrupted tasks tend to be resumed
Reminiscence	**Philip Ballard**, 1913	Recall rises after a delay without study
Von Restorff effect	**Hedwig von Restorff**, 1933	A distinctive list item is remembered better
Spacing effect	Ebbinghaus, 1885	Distributed practice beats massed practice
Testing effect	**Arthur Gates**, 1917	Retrieval practice beats restudy

Amnesia and the biology of memory ●●●

Amnesia is a pathological loss of memory. In **anterograde amnesia** new memories cannot be formed after an injury; in **retrograde amnesia** memories from before it are lost, usually sparing the remote past. Of the other "a-" disorders, aphasia (with alexia and acalculia) affects language, apraxia skilled movement and agnosia recognition; dementia is a progressive decline in memory, judgement and abstraction.

In 1953 the neurosurgeon **William Beecher Scoville** removed both medial temporal lobes, with most of the hippocampus and amygdala, from **Henry Gustav Molaison** (1926 to 2008), known as H.M., to relieve his epilepsy. Scoville and **Brenda Milner** reported in 1957 a profound anterograde amnesia: he reread magazines with fresh interest and never learned his new address. Yet his intelligence, digit span and childhood memories were intact, and over three days he improved at tracing a star in a mirror while denying that he had ever done it.

> **! EXAM TRAP**
> H.M.'s main deficit was anterograde amnesia for declarative memory (Real PYQ 2024); procedural, short-term and old memories were spared. Both medial temporal lobes were removed, not frontal, parietal or occipital cortex. The hippocampus is the structure damaged in anterograde amnesia and critical for consolidation (JEMAS 2025, RML 2024, RML 2023), not the hypothalamus, pituitary, cortex, amygdala (emotional memory), prefrontal cortex (working memory) or cerebellum (motor learning).

Korsakoff's syndrome (Korsakoff's psychosis; **Sergei Korsakoff**, late 1880s) results mainly from thiamine (vitamin B1) deficiency in chronic alcoholism, often after Wernicke's encephalopathy (**Carl Wernicke**, 1881: confusion, ataxia, nystagmus); together they form the Wernicke-Korsakoff syndrome, which damages the mammillary bodies and mediodorsal (dorsomedial) thalamus. It brings dense anterograde amnesia, extensive retrograde amnesia and **confabulation**, sincere false stories that fill memory gaps, unlike malingering (deliberate faking), delirium (acute, fluctuating consciousness) or confusion. Alzheimer's disease, by contrast, first erodes recent episodic memory, and Huntington's and Parkinson's diseases impair procedural learning.

> **! EXAM TRAP**
>
> Memory deficits, confabulation and nystagmus in a chronic drinker (Real PYQ 2024) is Wernicke-Korsakoff syndrome, not Alzheimer's, vascular or frontotemporal dementia. The alcohol-related illness with profound retrograde amnesia (RML 2024) is Korsakoff's psychosis; the vitamin is B1, not B12, D or folate.

Infantile amnesia, the lack of memories from the first three or four years, reflects an immature hippocampus and the lack of language. Transient global amnesia is a sudden anterograde amnesia lasting hours. Dissociative amnesia, a loss of personal memories with no medical cause (generalised when it covers the whole life), belongs to the Psychiatric Disorders chapter.

Richard Semon (1904) called the physical memory trace the **engram**. **Karl Spencer Lashley** (1890 to 1958) cut away rats' cortex after maze learning and found that memory loss depended on how much was removed (mass action), not where, with areas substituting for one another (equipotentiality): memories are distributed across the cortex ("In search of the engram", 1950). **Donald Olding Hebb** (1904 to 1985) proposed in 1949 that neurons active together strengthen their links, and **Timothy Bliss** and **Terje Lømo** (1973) found such **long-term potentiation** (LTP), a lasting strengthening of hippocampal synapses after high-frequency stimulation and the leading cellular candidate for memory. The hippocampus consolidates new memories into the neocortex, partly during sleep, so remote memories survive its damage.

Improving memory ○

A mnemonic adds organisation, meaning or imagery. The method of loci, credited to **Simonides of Ceos**, places vivid images along a familiar route; the peg-word system links items to rhyming number pegs (one is a bun); the keyword method (Richard Atkinson, 1975) links a foreign word to a sound-alike image (Spanish "pato", duck: a duck in a pot). An acronym is a word of first letters (VIBGYOR); an acrostic is a sentence of first letters ("Every Good Boy Deserves Fruit"). SQ3R (**Francis P. Robinson**, 1946) is Survey, Question, Read, Recite, Review; PQRST is Preview, Question, Read, Self-recitation, Test. Testing and spacing beat rereading, overlearning gives diminishing returns, and sleep consolidates.

> **! EXAM TRAP**
>
> A first-letter sentence is an acrostic, not an acronym, peg-word or keyword. The "S" of PQRST is self-recitation (RML 2024), not storing, sequencing or semantics.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Learning in 30 trials and relearning in 18 gives savings of _____ per cent.
2. A 30-second filled delay before recall abolishes the _____ effect.
3. Baddeley added the _____ to working memory in 2000.
4. Flashbulb memories keep their vividness and _____ as accuracy falls.
5. Distortion of a memory by a leading question asked afterwards is the _____ effect.
6. Dialling your old phone number instead of the new one is _____ interference.
7. Korsakoff's syndrome follows deficiency of _____ (vitamin B1).

Answers: (1) 40, (2) recency, (3) episodic buffer, (4) confidence, (5) misinformation, (6) proactive, (7) thiamine.

1.8 Thinking, Problem Solving, Reasoning, Decision Making and Language

LOWER YIELD · 53 QS

Thinking is the mental manipulation of internal representations to form new ones, as in solving, reasoning, deciding or creating. Seeing the chai stall is not thinking; working out whether you can afford a second cup is. Banished by Watson, thought returned with the cognitive revolution (Section 1.2): cognitive psychology studies the structuralists' topics (consciousness, imagery, thought) by objective measures.

Elements, Imagery and Types of Thought ○

Thought uses five elements: *images*, *concepts*, *symbols*, *propositions* (the smallest units of knowledge that can be true or false) and *language*. **Roger Newland Shepard** and Jacqueline Metzler (1971) found that the time to judge two rotated block figures identical rose linearly with the angle between them, as if the image were turned at a steady speed: **mental rotation**. **Stephen Michael Kosslyn**'s image-scanning studies (1978) suggested that images depict space, against **Zenon Walter Pylyshyn**'s view (1973) that knowledge is stored as propositions. **Allan Paivio**'s **dual-coding theory** (1971) gives pictures and concrete words both a verbal and an imagery code, so they are remembered better than abstract words. Table 1.16 lists types of thinking.

Table 1.16 Types of thinking.

Type	Associated with	What it is
Directed versus undirected	Würzburg school; Freud	Goal-guided and reality-checked versus drifting by association (daydreaming)
Autistic (dereistic)	**Eugen Bleuler**, 1911	Wish-driven fantasy that ignores reality
Convergent versus divergent	J. P. Guilford	One correct answer versus many varied answers
Critical versus creative	Robert Ennis; Guilford	Evaluating claims versus generating novel, useful ideas
Reflective	**John Dewey**, 1910	Careful weighing of a belief and its grounds

Type	Associated with	What it is
Productive versus reproductive	**Max Wertheimer**, 1945	Restructuring a problem versus reapplying a method

> **! EXAM TRAP**
>
> Dual coding favours **visual** information, not verbal, auditory or semantic. Autistic thinking is fantasy, not autism. Loosening of associations, circumstantiality and word salad are disorders of the *form* of thought (Psychiatric Disorders chapter).

Concepts: Rules and Prototypes ○

A **concept** is a mental category of things sharing properties, formed through observation, discrimination, generalisation and abstraction. The classical view defines it by necessary and sufficient features, which suits *artificial concepts* set by a rule: *conjunctive* (all attributes: a voter is a citizen, adult *and* enrolled), *disjunctive* (any one: "out" in cricket) or *relational* ("taller than"). **Jerome Seymour Bruner**, Jacqueline Goodnow and George Austin (1956) found that people learn them by testing hypotheses, by *scanning* or *focusing*; *conservative focusing*, changing one attribute at a time, works best, and *focus gambling* changes several.

Natural concepts are fuzzier. **Ludwig Wittgenstein** (1953) saw that games share only a **family resemblance**, and **Eleanor Rosch** showed in the 1970s that typicality is graded: "a robin is a bird" is verified faster than "a penguin is a bird". We classify by similarity to a **prototype**, the abstract, averaged best example, and learn the *basic level* (chair, not furniture) first; *exemplar theory* stores many instances instead. How concepts are networked in memory is told in Section 1.7.

> **! EXAM TRAP**
>
> Artificial concepts by hypothesis testing = **Bruner**, not Rosch (natural concepts), **Clark Leonard Hull** (gradual associative abstraction, 1920) or Watson. A prototype is **abstract and averaged**, not a template (exact copy), a feature (a part) or a geon (a simple solid).

Problem Solving: Strategies and Obstacles ●●

A problem is a gap between an initial state and a goal state, well defined (a sudoku) or ill defined ("choose a career"). **Allen Newell** and **Herbert Alexander Simon** (*Human Problem Solving*, 1972; the General Problem Solver) treated solving as search through a **problem space** of states linked by operators (legal moves). An **algorithm** guarantees the solution but may be slow (trying every padlock code); a **heuristic** is a quick rule of thumb with no guarantee. **Means-end analysis** sets subgoals that reduce the biggest difference from the goal (to reach Bengaluru, first reach the station); *working backward* starts from the goal; **hill climbing** always takes the step that gets closest, so it stalls when a backward step is needed; *analogy* reuses an earlier solution, though people rarely notice one unprompted (**Mary Gick** and **Keith Holyoak**'s radiation problem, 1980); *trial and error* is the most primitive. Stage models range from Dewey (1910) to Sternberg's problem-solving cycle.

The Gestaltists stressed **insight**, a sudden restructuring of the problem: **Wolfgang Köhler**'s chimpanzee Sultan suddenly joined two sticks to reach a banana (1917), and **Norman Maier**'s nine-dot problem (1930) defeats people until they let the lines leave the square. Janet

Metcalfe and David Wiebe (1987) found "warmth" ratings flat until just before an insight solution, but rising steadily on algebra. *Incubation*, setting a problem aside, often helps. Experience can also block a solution.

KEY STUDY: DUNCKER'S CANDLE PROBLEM (1945)

Karl Duncker asked people to fix a candle to a wall with matches and a box of drawing pins. Few thought of pinning up the emptied box as a shelf, but many more did when the box came empty. Knowing an object's usual use (top-down processing) blinded them to its other uses: **functional fixedness**.

Functional fixedness has a strategic cousin, **mental set** (Einstellung): in **Abraham Luchins**'s water jars (1942), people drilled on B minus A minus 2C kept using it when A minus C was far simpler. Confirmation bias and irrelevant information also obstruct.

! EXAM TRAP

A heuristic is "a general rule that is usually right": efficient, **not guaranteed**, not more accurate than an algorithm. Functional fixedness concerns **objects** (a matchbox seen only as a container, not a platform) and reflects overactive top-down processing; mental set concerns **strategies**. As fixedness is a special case of set, one paper keyed **mental set** "the most basic and common obstacle"; *fixation* is the umbrella word, "mental barrier" no term, trial and error a strategy. In a mental status examination, insight is awareness of illness, graded 1 to 6; **intellectual insight** is grade 5.

Reasoning: Deduction and Induction ●●●

Deductive reasoning moves from general premises to a conclusion that must be true if they are ("All Mysuru autos must use the meter; this is one; so it must"). **Inductive reasoning** moves from specific observations to a probable generalisation that the next case may overturn ("every auto I took overcharged, so all do"). An argument is *valid* if its conclusion follows from its form, *sound* if its premises are also true. **Charles Sanders Peirce** (1839 to 1914) called inference to the best explanation, as in diagnosis, **abductive reasoning**. In research, deduction tests theory against data (quantitative work), induction builds theory from data (qualitative work), and the hypothetico-deductive method of **Karl Raimund Popper** (1902 to 1994) is the full cycle of hypothesis, deduced prediction and test.

! EXAM TRAP

"Starts with a theory and tests it with observations" = **deductive**. "Gathers data, looks for patterns, then generates a theory", "specific observations to broader generalisations", "theory development" and "qualitative research" = **inductive**. Keys prefer "deductive reasoning" to "hypothetico-deductive method" and rest deductive validity on **the soundness of the initial theory**. **Heuristics** was keyed as NOT reflecting induction; probabilistic conclusions, decision making and concept formation do.

In syllogisms (two premises, one conclusion), **belief bias** makes people accept invalid arguments with believable conclusions, and the *atmosphere effect* (Robert Woodworth and Saul Sells, 1935) lets a "some" or "no" premise suggest a matching conclusion. For "if p, then q",

modus ponens (p, so q) and *modus tollens* (not q, so not p) are valid; *affirming the consequent* (q, so p) and *denying the antecedent* (not p, so not q) are fallacies. In **Peter Cathcart Wason**'s selection task (1966), under one person in ten turns the right cards (Figure 1.10), yet a permission rule ("if drinking beer, must be over 19") makes the same logic easy.

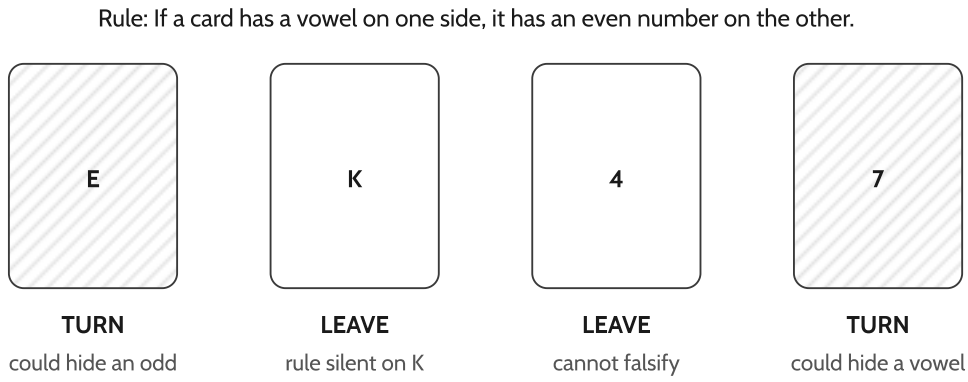


Figure 1.10 The abstract Wason selection task. The two cards with heavy outlines (E and 7) are the only ones that could reveal a counterexample. The popular choice of the 4 card tests for confirmation, which can never prove a general rule.

Philip Nicholas Johnson-Laird (1983) argued that we reason with *mental models* of possible situations, not formal rules. In Wason's 2-4-6 task (1960), people guessed "rising by two" and tested only fitting sequences (the rule was any increasing numbers): **confirmation bias**, seeking and favouring evidence for what one already believes.

> **! EXAM TRAP**
>
> Confirmation bias was keyed as "closely related to" **representativeness**, a debatable key (recognition is least related). Acting on an expectation until it comes true is a **self-fulfilling prophecy**, not confirmation bias, the base-rate fallacy or "motivational reaction" (no term). Piaget's formal operations bring hypothetico-deductive (a capacity, not a stage), counterfactual and metacognitive reasoning; **dialectical** reasoning, integrating contradictions, is postformal (Developmental Psychology chapter).

Judgement and Decision Making ○

Normative models say how a rational agent should decide, maximising expected utility; *descriptive* models say how people actually decide. Simon's **bounded rationality** holds that information, mental capacity and time limit choice, so people practise **satisficing**, taking the first good-enough option (Nobel Prize, 1978). **Gerd Gigerenzer** (born 1947) adds that fast and frugal heuristics, such as the *recognition heuristic* (choosing the option you recognise), are often ecologically rational.

Amos Tversky and **Daniel Kahneman** (Kahneman's Nobel Prize, 2002) catalogued the heuristics of judgement and their biases (Table 1.17). Representativeness invites *base-rate neglect*: a description that "sounds like" an engineer is judged an engineer whether engineers are 30 or 70 in 100. Most people judged Linda, a socially concerned philosophy graduate, likelier to be "a feminist bank teller" than "a bank teller", although a conjunction cannot be likelier than its parts: the **conjunction fallacy**.

Table 1.17 The classic heuristics and their typical biases.

Heuristic	Proposed	We judge by	Typical error
Availability	1973	Ease of recall	More words seem to begin with K than have K third
Representativeness	1972	Similarity to the typical case	Base-rate neglect; conjunction fallacy; gambler's fallacy (after many reds, black seems "due")
Anchoring and adjustment	1974	Adjusting from a starting value	Estimates stay near an arbitrary anchor

In the Asian disease problem (1981), most people preferred saving 200 of 600 lives for sure to a gamble, but chose the gamble when the sure option read "400 people will die": risk aversion for gains, risk seeking for losses. **Prospect theory** (1979) explains this *framing effect*, the same choice decided differently when worded as gains or as losses: outcomes are gains or losses from a reference point, and losses loom about twice as large as equal gains (**loss aversion**), which also drives the endowment effect (overvaluing what we own), status quo bias (preferring no change) and the sunk-cost fallacy (persisting because of past investment). **Hindsight bias**, the "I knew it all along" effect (**Baruch Fischhoff**, 1975), and overconfidence (confidence exceeding accuracy) distort self-knowledge. Dual-process theory contrasts fast, intuitive *System* 1, home of the heuristics, with slow, deliberate *System* 2.

> **! EXAM TRAP**
>
> Bounded rationality = **Herbert Simon**, not Kahneman, Tversky or Howard Gardner. Representativeness errs when people **ignore base rates and focus solely on similarity**; the other options describe availability, anchoring and confirmation bias. Overrating how well we would have predicted events = **hindsight**, not the just-world hypothesis or "decisiveness".

Creativity ○

Creativity is producing work that is both novel and appropriate. **Joy Paul Guilford**, whose 1950 American Psychological Association address launched its study, equated it with divergent production, scored by *fluency* (number of ideas), *flexibility* (number of categories), *originality* (statistical rarity) and *elaboration* (detail). **Graham Wallas** (*The Art of Thought*, 1926) described four stages: *preparation* (defining and working on the problem), *incubation* (setting it aside while work continues outside awareness), *illumination* (the sudden insight) and *verification* (testing and refining it).

> **! EXAM TRAP**
>
> Order: **preparation, incubation, illumination, verification** ("Please Incubate Ideas Very carefully"). Sensitivity, fluency, flexibility and originality are Guilford's abilities; analytical, creative and practical are Sternberg's intelligences; "problem-finding, brainstorming, selection, implementation" is not Wallas.

Creativity is measured by the *Torrance Tests of Creative Thinking* (**Ellis Paul Torrance**, 1966) and **Sarnoff Mednick**'s *Remote Associates Test* (1962): which word links cottage, Swiss and cake? (Cheese.) **Edward de Bono**'s *lateral thinking* (1967) challenges assumptions; **Alex Faickney Osborn**'s brainstorming (*Applied Imagination*, 1953) defers judgement and seeks

quantity. By the threshold hypothesis, creativity and IQ correlate only up to an IQ of about 120. In **Robert Jeffrey Sternberg** and Todd Lubart's investment theory (1991), creative people "buy low and sell high". The creative personality is open to experience, independent and tolerant of ambiguity.

The Structure of Language ○

Language combines arbitrary symbols by rules into unlimited messages. **Charles Francis Hockett** (1960) listed its design features, notably semanticity (meaning), arbitrariness (nothing in "dog" is doglike), displacement (talking about what is absent), generativity (endless new sentences) and duality of patterning (meaningless sounds build meaningful units). The **phoneme** is the smallest unit of sound that distinguishes words (/p/ and /b/ in pat and bat) and has no meaning; the **morpheme** is the smallest unit of meaning, free (cat) or bound (-s), so "cats" has two. Syntax governs how words combine, *semantics* is meaning, **pragmatics** is use in social context ("Can you pass the salt?" is a request) and **prosody** is rhythm and intonation. **Noam Chomsky**'s transformational grammar (*Syntactic Structures*, 1957; *Aspects*, 1965) separates *surface structure*, the words as spoken, from **deep structure**, who did what to whom: "Visiting relatives can be boring" has one surface and two deep structures. *Competence* is tacit knowledge of the rules, *performance* their imperfect use.

> **! EXAM TRAP**
>
> Smallest unit of meaning = **morpheme**; basic unit of sound = **phoneme**; a syllable is a unit of pronunciation, a consonant one kind of phoneme. Emotional prosody depends on **right-hemisphere regions homotopic to the language areas**, not Broca's area, motor cortex or occipital lobe. **Recursive thought** ("I think that you think...") is perspective taking, not reversibility (Piaget's mental undoing of an action), referential style (below) or reconstruction (rebuilding a memory).

Acquiring Language ●●

Burrhus Frederic Skinner's *Verbal Behavior* (1957) explained language by reinforcement of verbal operants such as the *mand* (a request) and the *tact* (naming). Chomsky's 1959 review replied that children produce sentences they never heard, make errors no adult models ("I goed") and are rarely corrected for grammar. He proposed an innate **language acquisition device** (LAD) holding a *universal grammar*, argued from the **poverty of the stimulus**: children's input is too sparse to explain the grammar they reach. Bruner's *language acquisition support system* (LASS, 1983) stresses adult scaffolding and child-directed speech (motherese): higher and wider pitch, slower rate, more pauses, shorter utterances. **Dan Isaac Slobin**'s *language-making capacity* (1985) is a set of cognitive and perceptual operating principles specialised for language. **Eric Heinz Lenneberg** (1967) proposed a **critical period** from about two to puberty, now usually called a sensitive period; Genie, isolated until 13, learned words but never grammar.

Infants coo at about 2 months and babble at about 6; at about 12 months a single word, a **holophrase**, does a sentence's work; at 18 to 24 months comes what **Roger Brown** (1925 to 1997) called **telegraphic speech**, two words without function words ("more milk"); and at 2 to 3 years **overregularisation** ("two foots"). Children overextend words ("dog" for every animal), underextend them ("dog" only for their own) and learn new words from one exposure

(fast mapping); **Katherine Nelson** (1973) found some children's first words mostly object names, a *referential* style. Language is most often compromised in autism spectrum disorder.

> ! EXAM TRAP
>
> Chomsky's innate capacity is the **LAD** (an inborn processor activated by verbal input, holding universal grammar, the rules common to all languages), not the phonological loop (working memory), the zone of proximal development (Vygotsky) or a schema; LASS = **Bruner**, LMC = **Slobin**. Irreversible deficits after early deprivation support **sensitive periods**, not semantics or syntax (levels of language) or nativism (the broader theory that language is innate).

Language and Thought ●●

Benjamin Lee Whorf (1897 to 1941), a student of **Edward Sapir** (1884 to 1939), argued that language shapes thought. The strong version, linguistic determinism, has failed: **Brent Berlin** and **Paul Kay** (1969) found colour terms entering languages in a near-universal order, and Rosch found that the Dani, with two colour terms, remember focal colours as English speakers do. The weak version, **linguistic relativity** (language influences thought), survives: Hindi kinship terms (*chacha*, *mama*, *phupha*) make speakers track which side an uncle is on. **Steven Pinker** (1994) holds that we think in "mentalese".

For **Jean Piaget** (1896 to 1980), thought comes first and language expresses it; preschoolers' egocentric speech simply fades. For **Lev Semyonovich Vygotsky** (*Thinking and Speech*, 1934), thought and speech have separate roots that merge at about two, after which thought becomes verbal; social speech becomes self-guiding private speech and then **inner speech**. He traced concepts from heaps through complexes to true concepts in adolescence, tied to society's tasks, social institutions and self-reflection.

> ! EXAM TRAP
>
> **Vygotsky: thought depends on language; Piaget: language depends on thought.** "One studied early, the other later language" is wrong: both studied the same years. Private speech **guides behaviour and is internalised**; it is not regression, imitation, disorder or fading egocentrism. Gender plays no part in Vygotsky's account. "Language and thought involve..." expects **simultaneous** processes; "independent" fits only their early roots. Linguistic relativity = **Whorf**, not Chomsky, Piaget or Pinker.

Bilingualism is the use of two languages, and a *polyglot* speaks many (a linguist studies language scientifically); bilingual children reach milestones on time. Apes such as Washoe, Nim Chimpsky and Kanzi learn meaningful symbols but little grammar. Speech perception is *categorical* (/ba/ switches abruptly to /pa/ as voice onset time rises) and audiovisual: in the McGurk effect, a heard "ba" dubbed onto lips saying "ga" is heard as "da". Reading builds most directly on spoken language, above all awareness of its sounds, not on theory of mind, imagination or mental representation. **Aphasia** is an acquired disorder of language from brain damage, not of intelligence, memory or abstract thinking (Broca's aphasia impairs production, Wernicke's comprehension: Physiological Psychology chapter).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Seeing a box only as a container shows _____.
2. Reasoning from specific observations to a probable generalisation is _____.
3. Wallas's second stage is _____.
4. The smallest unit of meaning is the _____.
5. "More milk" at 18 to 24 months is _____ speech.
6. For Vygotsky, private speech becomes _____ speech.

Answers: (1) functional fixedness, (2) induction, (3) incubation, (4) morpheme, (5) telegraphic, (6) inner.

1.9 Intelligence: Theories, Measurement, Heredity and Environment

MEDIUM YIELD · 63 QS

Everyone thinks they recognise intelligence, yet psychologists still argue over what it is, whether it is one thing or many, and how much of it a test can capture.

Defining Intelligence ○

Alfred Binet (1857 to 1911) put judgement at the centre: "to judge well, to comprehend well, to reason well." **Edwin Garrigues Boring** (1886 to 1968) retorted in 1923 that intelligence is "what the tests test", an **operational definition** (one framed by its measuring procedure). **David Wechsler** (1896 to 1981) gave the most quoted definition (1944): "the aggregate or global capacity of the individual to act purposefully, to think rationally and to deal effectively with his environment", aggregate because it comprises many abilities, global because they work as one. **Robert Jeffrey Sternberg** (born 1949), who re-polled the experts with Douglas Detterman in 1986, stresses adapting to, selecting and shaping real-world environments.

Achievement tests look back at what a specific course taught (a board examination); **aptitude** tests look ahead to the capacity to learn a specific skill with training (the Differential Aptitude Tests), so they anchor guidance; intelligence tests predict learning in general. Among lay (implicit) theories, **Carol Susan Dweck** (born 1946) contrasted an entity theory, or fixed mindset, with an incremental theory, or **growth mindset** (*Mindset*, 2006), the belief that effort develops intelligence. Fixed-mindset children avoid challenges; children praised for effort, not cleverness, persist after failure.

! EXAM TRAP

The WAIS is an *intelligence* test, not an aptitude, achievement or attitude (evaluative stance) test. RML 2022 keyed the aptitude test as "the heart of the guidance programme", over the case study (a life history), the interview and the cumulative record (a running school file); textbooks give the phrase to counselling, so choose it if offered.

From Galton to Binet: The First Tests and the IQ ●●

Francis Galton (1822 to 1911), author of *Hereditary Genius* (1869) and father of differential psychology, measured reaction time and sensory acuity in his London anthropometric laborato-

ry (1884). **James McKeen Cattell** (1860 to 1944) coined the term **mental test** (1890) for a similar battery, which failed to predict college grades (Clark Wissler, 1901). Asked by the French ministry to pick out children needing special schooling, Binet and the physician **Théodore Simon** (1873 to 1961) measured judgement and reasoning instead, keeping items that older children passed more often than younger ones. Their 1905 scale had 30 items of rising difficulty; the 1908 revision grouped them by age, a child's **mental age** being the age level of the items passed; the 1911 revision reached adults. (**Jean Piaget** standardised tests in Simon's laboratory around 1920 but did not build the first test.)

Henry Herbert Goddard (1866 to 1957) translated the scale into English in 1908. In 1912 **William Stern** (1871 to 1938) divided mental age by chronological age, coining the **intelligence quotient (IQ)**; **Lewis Madison Terman** (1877 to 1956) multiplied it by 100 in his 1916 **Stanford-Binet**. So the **ratio IQ** = (MA ÷ CA) × 100: a 10-year-old with a mental age of 8 scores 80. In 1917 **Robert Mearns Yerkes** (1876 to 1956) led the first group tests, for army recruits: the verbal Army Alpha and, for the illiterate, the pictorial Army Beta.

> **! EXAM TRAP**
>
> Stern coined IQ; Terman popularised it. Inverting the ratio (10 ÷ 8 × 100 = 125) is the usual slip: a mental age below chronological age always gives an IQ below 100. James McKeen Cattell coined "mental test"; Raymond Cattell, unrelated, gave us fluid and crystallised intelligence and the 16PF personality questionnaire.

Deviation IQ and What a Score Means ●●

The ratio IQ fails for adults: mental age stops growing in the late teens, so an ordinary 40-year-old with a mental age of 16 would score (16 ÷ 40) × 100 = 40. The Wechsler-Bellevue scale (1939) dropped mental age and compared each person with norms from a large representative sample of the same age. In this **deviation IQ** (the "standard IQ"; loosely, the actual score relative to the expected mean score for one's age) every age group has a mean of 100 and a standard deviation (SD), the typical distance of scores from the mean, of 15: IQ = 100 + 15z (z: distance from the mean in SDs).

Because IQs are roughly normally distributed, any IQ converts to a percentile, the percentage of people scoring below it (Figure 1.11): 100 is the 50th percentile and 85, one SD below, the 16th. Stanford-Binet deviation IQs used an SD of 16 from 1960 until the fifth edition (2003), so there an IQ of 136 gives z = 36 ÷ 16 = 2.25, about the 98.7th percentile. WAIS-IV bands: 130 or above very superior, 90 to 109 average, 80 to 89 low average, 70 to 79 borderline ("very low" in newer editions), 69 or below extremely low. DSM-IV-TR's borderline intellectual functioning (71 to 84) straddles two bands, so 70 to 84 is "low average or borderline, depending on the edition".

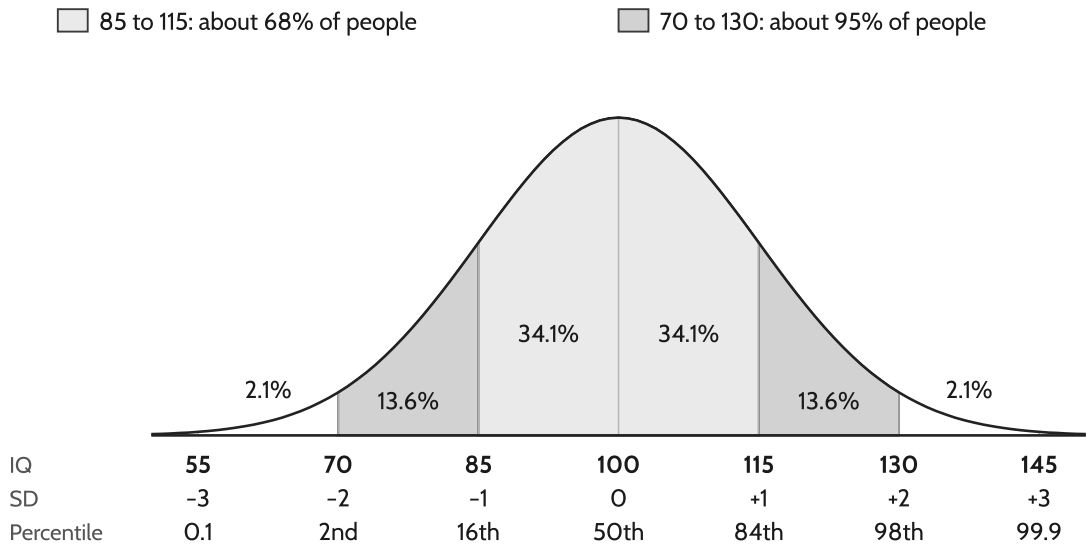


Figure 1.11 The distribution of deviation IQ (mean 100, SD 15). Read across the bottom rows to convert an IQ into standard deviations and percentiles: 85 is the 16th percentile, 115 the 84th, 130 about the 98th, and 70 about the 2nd.

> **! EXAM TRAP**
>
> The original Stanford-Binet used the ratio IQ. The deviation IQ is not "only for adults", not built from unstandardised raw scores, and no less reliable than the ratio IQ, whose meaning shifts with age.

The Wechsler Scales, the SB5 and Other Tests ●●●

Wechsler, at New York's Bellevue Hospital, set out to fix the Stanford-Binet's faults: little appeal for adults, too much word manipulation, too much weight on speed and reliance on mental age. His tests add non-verbal performance subtests and give a profile as well as a total. The family is the WAIS (1955; WAIS-IV 2008, ages 16 to 90), the WISC (1949; WISC-V 2014, 6 years to 16 years 11 months) and the WPPSI (from 2 years 6 months). In the WAIS-IV, four index scores (Table 1.18) replaced the Verbal and Performance IQs and combine into the Full Scale IQ (FSIQ), with an optional General Ability Index; the WISC-V and WAIS-5 (2024) have five.

Table 1.18 The WAIS-IV (2008): 10 core subtests in four indices.

Index	Core subtests	Supplemental
Verbal Comprehension (VCI)	Similarities, Vocabulary, Information (3)	Comprehension
Perceptual Reasoning (PRI)	Block Design, Matrix Reasoning, Visual Puzzles (3)	Picture Completion, Figure Weights
Working Memory (WMI)	Digit Span, Arithmetic	Letter-Number Sequencing
Processing Speed (PSI)	Symbol Search, Coding	Cancellation

> **! EXAM TRAP**
>
> The Working Memory Index reflects attention, concentration and mental control. "VIQ, PIQ, FSIQ and memory quotient" mixes the old WAIS with the old Wechsler Memory Scale (1945), a memory test whose single memory quotient became five WMS-IV (2009) indices: Auditory Memory, Visual Memory, Visual Working Memory, Immediate Memory and Delayed Memory. Letter-Number Sequencing, keyed as core in the WAIS-IV but not the WISC-V, is strictly supplemental in one and secondary in the other; it is still the best answer, as the other options are core in both.

The Stanford-Binet Fifth Edition (SB5; Gale Roid, 2003; ages 2 to 85 and over; mean 100, SD 15) measures five factors drawn from Carroll's model (below) and Alan Baddeley's working memory: Fluid Reasoning, Knowledge, Quantitative Reasoning, Visual-Spatial Processing and Working Memory, each verbal and non-verbal. The Kaufman Brief Intelligence Test (KBIT-2) has a Verbal (crystallised) and a Nonverbal (fluid) scale. For infants, the Bayley Scales (cognitive, language and motor development up to 42 months) yield a developmental quotient (DQ); the McCarthy Scales cover about 2½ to 8½ years.

The WAIS-IV FSIQ is highly reliable (about .98). Content validity asks whether items sample the domain, **concurrent validity** whether scores agree with a criterion measured at the same time (an established test given alongside), predictive validity whether they forecast a later one, and construct validity whether the test behaves as theory predicts. IQ correlates about .5 with school grades and job performance (Frank Schmidt and John Hunter, 1998). The WAIS documents intellectual disability or decline, as for a disability allowance; children's special educational needs call for the WISC.

Culture-Fair and Indian Tests ●●

Because vocabulary and information depend on schooling, psychologists sought a culture-free test; that proved impossible, so a **culture-fair test** instead minimises language, reading and specific knowledge. The classic is **Raven's Progressive Matrices** (Standard form 1938) by Spearman's student **John Carlyle Raven** (1902 to 1970): each of its 60 progressively harder items is a pattern with a piece missing, solved by inferring the rule. It is the classic test of fluid intelligence and the best single marker of g, widely used with Indian norms. Raymond Cattell's figure-only Culture Fair Intelligence Test works the same way.

In India, **V. V. Kamat**'s Binet-Kamat Test (1930s) adapted the Stanford-Binet and is heavily verbal; **Arthur J. Malin**'s Intelligence Scale for Indian Children (MISIC, 1969) adapts the WISC and is half verbal; **C. M. Bhatia**'s Battery of Performance Tests of Intelligence (1955) has five performance subtests (Kohs block design, Alexander's pass-along, pattern drawing, immediate memory, picture construction), needs no reading and has rural and urban norms; normed on 11- to 16-year-olds, it is also used with illiterate adults. So for an illiterate rural 10-year-old, where the WISC-IV and SB5 rest on schooling and foreign norms, Bhatia's Battery best minimises cultural and educational bias.

> **! EXAM TRAP**
>
> Raven's measures *fluid* intelligence: prefer it to crystallised, verbal, social or even "general intelligence". It is culture-fair, not culture-free. The Battery is C. M. Bhatia's, not that of Girishwar Misra, Durganand Sinha or J. B. P. Sinha (cultural, indigenous and organisational psychologists).

Factor Theories: From g to CHC ●●

Factor analysis asks how few underlying factors explain a table of correlations among tests. Its inventor, **Charles Edward Spearman** (1863 to 1945), found in 1904 that schoolchildren's marks and sensory scores all correlated positively, the **positive manifold**, and explained it by a **general factor (g)** common to every test plus a specific factor (s) peculiar to each: the **two-factor theory**. **Louis Leon Thurstone** (1887 to 1955) found seven **primary mental abilities** (1938): verbal comprehension, word fluency, number, space, associative memory, perceptual speed and reasoning. Because these correlate, g returned at the top of hierarchies such as **Philip Ewart Vernon**'s (1950) and **John Bissell Carroll**'s three strata (1993); Kevin McGrew merged Carroll's model with Cattell and Horn's into the CHC theory behind most modern tests (Table 1.19).

Joy Paul Guilford (1897 to 1987) went the other way. His **structure of intellect** (1967) crosses five operations (cognition, memory, divergent production, convergent production, evaluation), four contents (figural, symbolic, semantic, behavioural) and six products, giving 5 × 4 × 6 = 120 abilities, later 150 and then 180.

Raymond Bernard Cattell (1905 to 1998), later with **John Leonard Horn** (1928 to 2006), split g in two (1963). **Fluid intelligence (Gf)** is the ability to reason and solve novel problems independently of acquired knowledge; measured by matrices, it peaks in the early twenties and then declines. **Crystallised intelligence (Gc)** is the repertoire of information, cognitive skills and strategies acquired through education and experience; measured by vocabulary and information, it holds or rises into the sixties. In Cattell's investment theory, Gf invested in learning crystallises as Gc; Horn later extended the list to about nine broad abilities.

Table 1.19 Other factor theories and the hierarchies.

Theorist	Year	Theory	Key structure
Edward Thorndike	1920s	Multifactor ("anarchic")	Abstract, mechanical, social; CAVD test; no g
Godfrey Thomson	1916	Sampling theory	Tests sample overlapping sets of many small bonds
P. E. Vernon	1950	Hierarchical	g; v:ed and k:m; minor; specific
Arthur Jensen	1969	Level I and II	Associative (rote) versus conceptual
J. B. Carroll	1993	Three strata	g; 8 broad; about 70 narrow

> **! EXAM TRAP**
>
> g is general and s specific; neither means group factors (Thurstone, Vernon) or a genetic predisposition. Guilford's cube has three dimensions but 120 abilities; RML 2021 keyed "120 dimensions", so give 120 there. "Ability to deal with new problems" is Gf; block design, matrix reasoning and coding do not measure Gc, and processing speed and working memory are correlates, not definitions, of it.

Gardner, Sternberg, PASS and Emotional Intelligence ○

Howard Gardner (born 1943) built his theory from brain damage and prodigies, not factor analysis. *Frames of Mind* (1983) proposed seven relatively independent **multiple intelligences**: linguistic, logical-mathematical, spatial, musical, bodily-kinaesthetic, interpersonal and intrapersonal; naturalist was added in 1999 and existential only considered. Critics note that most correlate with one another and with g.

Sternberg's **triarchic theory** (*Beyond IQ*, 1985) has three parts. The componential or analytical part is the mental machinery: metacomponents that plan and monitor, performance components that execute, and knowledge-acquisition components that learn. The experiential or creative part copes with novelty and automatises the familiar. The contextual or practical part adapts to the environment, shapes it or selects a better one: a clerk who learns her office's unwritten rules adapts, one who changes a pointless procedure shapes, one who leaves for a better job selects. In the late 1990s this grew into **successful intelligence**, the ability to achieve success in life by one's own standards and within one's culture through a balance of analytical, creative and practical abilities; it rests partly on tacit knowledge, the unspoken know-how a setting demands.

> **! EXAM TRAP**
>
> Componential = analytical, experiential (not "experimental", and not Dweck's "incremental") = creative, contextual = practical. "Verbal, performance, full scale" is Wechsler; "fluid, crystallised" is Cattell; "linguistic, logical-mathematical, spatial" is Gardner.

The **PASS model** of the Odisha-born **Jagannath Prasad Das** (born 1931), **Jack A. Naglieri** and **John R. Kirby** (1994), built on **Alexander Romanovich Luria**'s (1902 to 1977) functional units of the brain, has four processes: Planning, Attention, Simultaneous (parts into a whole) and Successive (serial order), measured by the Cognitive Assessment System. **Stephen J. Ceci**'s bioecological view (1990) ties each cognitive potential to its context.

Edward Lee Thorndike's (1874 to 1949) social intelligence (1920), the ability to understand and manage people, foreshadowed **emotional intelligence**, defined by **Peter Salovey** and **John D. Mayer** (1990) as the ability to monitor and use one's own and others' emotions; their four-branch ability model (perceiving, using, understanding and managing emotions) is measured by the MSCEIT. **Daniel Goleman** (born 1946) popularised a mixed model (1995): knowing one's emotions, managing them, motivating oneself, recognising others' emotions and handling relationships (relabelled in 1998 as self-awareness, self-regulation, motivation, empathy and social skills). **Reuven Bar-On**'s EQ-i (emotional quotient, 1997) and trait EI are self-report versions; **Danah Zohar** proposed spiritual intelligence (2000).

> **! EXAM TRAP**
>
> Salovey and Mayer coined emotional intelligence; Goleman popularised it. His 1998 labels rename the same five parts, so match the question's wording, and if both lists appear choose the original 1995 one ("knowing one's emotions" to "handling relationships"). "IQ, EQ, SQ, CQ" is no one's model of EI.

Heredity, Environment and the Brain ○

Higher IQ goes with faster reaction and inspection times, larger working memory, slightly larger brains and, in Richard Haier's 1988 study, lower brain glucose use (neural efficiency); intelligence is polygenic. Family studies compare relatives of differing closeness, adoption studies compare children with their biological and adoptive parents, and **twin studies** compare monozygotic (MZ) twins, who share virtually all genes, with dizygotic (DZ) twins, who share half, most powerfully when reared apart. In **Thomas J. Bouchard Jr.** and Matt McGue's 1981 review, MZ twins correlated .86 on IQ when reared together and .72 when reared apart (about .70 in Bouchard's Minnesota Study of Twins Reared Apart, from 1979), DZ twins .60.

Heritability is the proportion of *variation* in a trait, within one population at one time, associated with genetic variation; it says nothing about one individual. It rises with age, and **Eric Turkheimer** (2003) found it near .10 in the poorest families, a gene-environment interaction. **Richard Lewontin**'s seeds (1970) show why it cannot explain group gaps: varied seed differs genetically within each tray, yet the gap between a rich-soil and a poor-soil tray is wholly environmental. In **Irving Gottesman**'s **reaction range** (1963), genes set limits and environment decides where within them a person lands.

Iodine, nutrition and schooling raise IQ; early intervention (Head Start, 1965; the Milwaukee Project; the Abecedarian Project, 1972, from infancy) brings IQ gains that often fade but lasting school and life benefits. **James Robert Flynn** (1934 to 2020) showed in 1984 that average scores rose about 3 points a decade, the **Flynn effect**, too fast for genes. Neither **Arthur Robert Jensen** (1923 to 2012), who argued in 1969 that group IQ gaps are partly genetic, nor *The Bell Curve* (**Richard J. Herrnstein** and **Charles Murray**, 1994) has direct support, and **Claude Steele** and **Joshua Aronson** (1995) showed that **stereotype threat**, fear of confirming a group stereotype, depresses scores.

> **! EXAM TRAP**
>
> Individual differences come from many genetic and environmental factors together, not from genes, environment or income alone. Linkage studies, which track DNA markers through families, are neither twin nor adoption designs.

Intellectual Disability and Giftedness ●●

Intellectual disability (DSM-IV-TR's mental retardation) requires deficits in intellectual functions, an IQ about two SDs below the mean (about 70; 85 is merely low average); deficits in **adaptive functioning**, the conceptual, social and practical skills of independent living; and onset in the developmental period. DSM-IV-TR graded severity by IQ (Table 1.20); DSM-5 grades it by adaptive functioning. **Global developmental delay** is a provisional DSM-5 label for children under 5 who cannot yet be fully assessed, so an assessed 7-year-old with an IQ of 65 and severe adaptive deficits has intellectual disability itself. Causes include Down syndrome, fragile X and phenylketonuria.

Table 1.20 Levels of intellectual disability (DSM-IV-TR).

Level	IQ range	Historical label
Mild (about 85% of cases)	50 to 55 to about 70	Moron (Goddard, 1910)

Level	IQ range	Historical label
Moderate	35 to 40 to 50 to 55	Imbecile
Severe	20 to 25 to 35 to 40	Imbecile or idiot
Profound	Below 20 or 25	Idiot

Adaptive behaviour is measured by **Edgar Arnold Doll**'s (1889 to 1968) **Vineland Social Maturity Scale** (1935), which yields a social age and a social quotient = (SA ÷ CA) × 100, and by the Vineland Adaptive Behavior Scales (communication, daily living skills, socialisation, motor). India's National Trust Act (1999) covers autism, cerebral palsy, mental retardation and multiple disabilities, but not specific learning disability.

At the other tail, an IQ of 130 or above is the usual cut-off for giftedness, though **Joseph Renzulli** (born 1936) required three rings (1978): above-average ability, task commitment and creativity.

KEY STUDY: TERMAN'S GIFTED CHILDREN (FROM 1921)

Question. Are bright children frail and odd? **Method.** Terman followed 1,528 California children with IQs of about 135 or more (mean about 151), the "Termites", for life. **Result.** They were healthier, better adjusted and more successful than average; social and behavioural problems clustered among those with childhood IQs of about 180 or more. **Criticism.** The mostly white, middle-class sample missed two future Nobel physicists.

Savant syndrome pairs a developmental disability, often autism, with an island of exceptional skill such as calendar calculation; a prodigy excels without disability. Creativity is treated with thinking (Section 1.8).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The term intelligence quotient was coined in 1912 by _____.
2. On a Wechsler scale, an IQ of 85 lies at the _____ percentile.
3. For an illiterate rural Indian child, the best choice is _____ Battery of Performance Tests.
4. Cattell called crystallised intelligence a repertoire of information, cognitive skills and _____.
5. Adapting to, shaping and selecting environments make up Sternberg's _____ subtheory.
6. Heritability is the proportion of _____ in a trait, within a population, associated with genetic variation.

Answers: (1) William Stern, (2) 16th, (3) Bhatia's, (4) strategies, (5) contextual, (6) variation.

1.10 Motivation MEDIUM YIELD · 74 QS

Why does a student study until 2 a.m., or a monkey dismantle a puzzle that hides no food? Each of psychology's answers below grew from the failures of the last.

What motivation is ●●

In psychology, **motivation** (Latin *movere*, to move) is the set of processes that start, direct, energise and sustain behaviour.

Take thirst. A **need** is a state of deficiency (the body lacks water, felt or not). A **drive** is the aroused, tense state the need produces, which energises behaviour. A **motive** is any internal state directing behaviour towards a goal, usually a learned one such as achievement. An **incentive** is an external object that attracts or repels; the **goal** is the end state pursued. The **motivational cycle** runs need, drive, instrumental behaviour, goal, relief, until the need returns.

The **primary motives** (biological, unlearned, mostly homeostatic) include hunger, thirst, sleep, pain avoidance, sex and the maternal drive; the **secondary motives** (psychosocial, learned), such as achievement, affiliation, power and gregariousness, grow from experience. Henry Murray's **viscerogenic** (bodily) and **psychogenic** (mental) needs draw the same line.

Doing something for its own sake is **intrinsic motivation**; doing it for a separable outcome such as marks or praise is **extrinsic motivation**. A toddler who struggles to open a box she knows is empty is intrinsically motivated: no bodily imbalance (homeostasis), payoff (extrinsic) or learned social motive (secondary) is involved. Malingering (feigning for an external reward) and factitious disorder (feigning without one) differ on this line.

> **! EXAM TRAP**
>
> **Gregariousness** is a **social motive**, though McDougall called it an instinct ("psychological" and "personal" motives are not standard classes). Among the perspectives (Section 1.2), inner forces beyond individual control are **psychodynamic**; free will and an innate tendency to grow to one's maximum potential are **humanistic** (Maslow, Rogers).

Instinct theories ○

An **instinct** is a complex, unlearned, stereotyped pattern of behaviour shared by a whole species. **William James** (1842 to 1910) held in 1890 that humans have more instincts than animals, not fewer. **William McDougall** (1871 to 1938), in *An Introduction to Social Psychology* (1908), made instincts, each paired with an emotion (flight with fear, pugnacity with anger), the prime movers of all behaviour; he later listed eighteen. His **hormic psychology** (Greek *horme*, urge) holds that behaviour is purposive.

Sigmund Freud (1856 to 1939) divided the instincts (*Trieb*) in 1920 into the **life instincts** (Eros), whose energy is the **libido**, and the **death instinct** (Thanatos), which, turned outwards, appears as aggression and war. Investing psychic energy in an object is **cathexis**; the **id** is the reservoir of the instincts.

The ethologists **Konrad Lorenz** and **Nikolaas Tinbergen** (Nobel Prize 1973) described **fixed action patterns** released by a **sign stimulus**: a male stickleback attacks anything with a red belly. Yet as an account of human motives, instinct theory had already collapsed by the late

1920s: naming is not explaining (a "gregarious instinct" is circular), the lists ran into thousands, and **Zing-Yang Kuo** (1930) found that kittens reared with rats seldom killed them.

> **! EXAM TRAP**
>
> The instinct theory of motivation is **McDougall's**, not James's, Freud's or Maslow's. Freud's "innate forces" are **instincts**, not needs (Maslow, Murray), drives (Hull) or the id (their reservoir). War and violence express the **death instinct**.

Drives, arousal and incentives ●●

Drive theory grew from physiology. **Walter Bradford Cannon** (1871 to 1945) named **homeostasis** (1926; *The Wisdom of the Body*, 1932), the body's thermostat-like correction around a set point, building on the *milieu intérieur* of **Claude Bernard** (1813 to 1878). On it **Clark Leonard Hull** (1884 to 1952) built **drive-reduction theory** (1943): a need creates a drive, a state of tension, and any response that reduces the drive is reinforced. In Hull's equation (${}_{S}E_{R} = {}_{S}H_{R} \times D \times K \times V$), K is **incentive motivation** (reward size) and V **stimulus-intensity dynamism** (stronger stimuli evoke the learned response more readily). **Primary drives** arise from tissue needs; **secondary drives**, such as learned fear, are acquired.

> **! EXAM TRAP**
>
> "Greater stimulus intensity, greater probability of the learned response" is **stimulus-intensity dynamism**, not incentive motivation, response amplitude (vigour) or reaction threshold (the minimum for any response). A deviation from homeostasis creates a **drive**, not equilibrium (the restored state), regulation (the correction) or dissonance (Festinger). Drive theory is about **reducing physiological needs**.

Yet **Harry Frederick Harlow**'s (1905 to 1981) monkeys took apart puzzles that hid no food (1950), rats worked for non-nutritive saccharin, and students paid to lie in sensory-deprivation cubicles (1954) quit within days. **Donald Olding Hebb**'s (1904 to 1985) **optimal arousal theory** (1955) says organisms seek a moderate level of arousal. In 1908 **Robert Mearns Yerkes** and **John Dillingham Dodson** had found that mice learned a brightness discrimination fastest under intermediate shock, and the harder the discrimination, the weaker the best shock. The **Yerkes-Dodson law**, as later taught (Figure 1.12), makes performance an **inverted-U** function of arousal or anxiety, with a lower optimum for complex tasks than for simple ones.

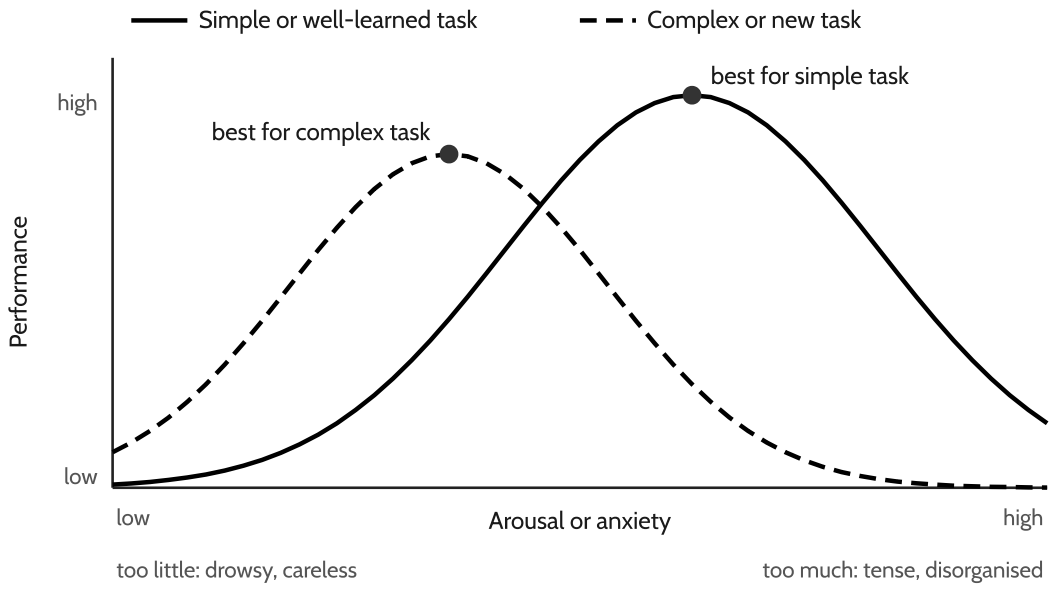


Figure 1.12 The Yerkes-Dodson law. Performance is an inverted-U function of arousal, and the peak for a complex or new task (dashed) lies at a lower arousal level than the peak for a simple or well-learned task (solid).

Robert Zajonc (1965) explained social facilitation the same way: an audience raises arousal, which strengthens the dominant response, helping easy, well-learned tasks and impairing novel or difficult ones. People differ too: **Marvin Zuckerman**'s (1928 to 2018) **sensation seeking** is the urge for varied, novel, intense experience despite risk.

> **! EXAM TRAP**
>
> Yerkes-Dodson is an **inverted-U** relationship between performance and anxiety: more precise than "curvilinear", and never linear or M-shaped. A complex task is done best at **moderate arousal, with a lower optimum**, not at very high arousal or at a level independent of difficulty.

Drives push; incentives pull. The **incentive theory** holds that behaviour is driven mainly by external rewards: **Otto Tinklepaugh**'s monkeys (1928) refused lettuce after seeing a banana hidden. The **expectancy-value theory** multiplies the expectancy of success by the value of the outcome, so either at zero kills motivation. Its roots are the learned expectancies of **Edward Chace Tolman** (1886 to 1959) and the **social learning** theory of **Julian Bernard Rotter** (1954), in which behaviour depends on expectancy and reinforcement value; organisational texts give the social learning name to the framework built on expectancy, demand and incentive, which combines the **cognitive** framework (Tolman's expectancies) with the **behaviouristic** one (reinforcement); "supportive" is not a framework but a model of organisational behaviour (autocratic, custodial, supportive, collegial). **Richard Lester Solomon** and John Corbit's **opponent-process theory** (1974) explains acquired motives such as addiction: an opposite after-reaction grows with repetition, bringing tolerance and withdrawal.

Maslow's hierarchy and other lists of needs ●●●

Abraham Harold Maslow (1908 to 1970), with Carl Rogers a founder of humanistic psychology (Section 1.2), set out in 1943 ("A Theory of Human Motivation") and 1954 (*Motivation and Personality*) a **hierarchy of needs** (Figure 1.13): physiological, safety, belongingness and love (social), esteem, and self-actualisation, becoming all that one can be. By **prepotency**, unmet lower needs are strongest and must be largely satisfied before higher ones take charge: a starving man dreams of food, not fame. Physiological and safety needs are thus the **lower needs**. The four lower levels are **deficiency needs** (D-needs), which stop motivating once met; self-actualisation is a **growth** or being (B) need, which strengthens as it is satisfied.

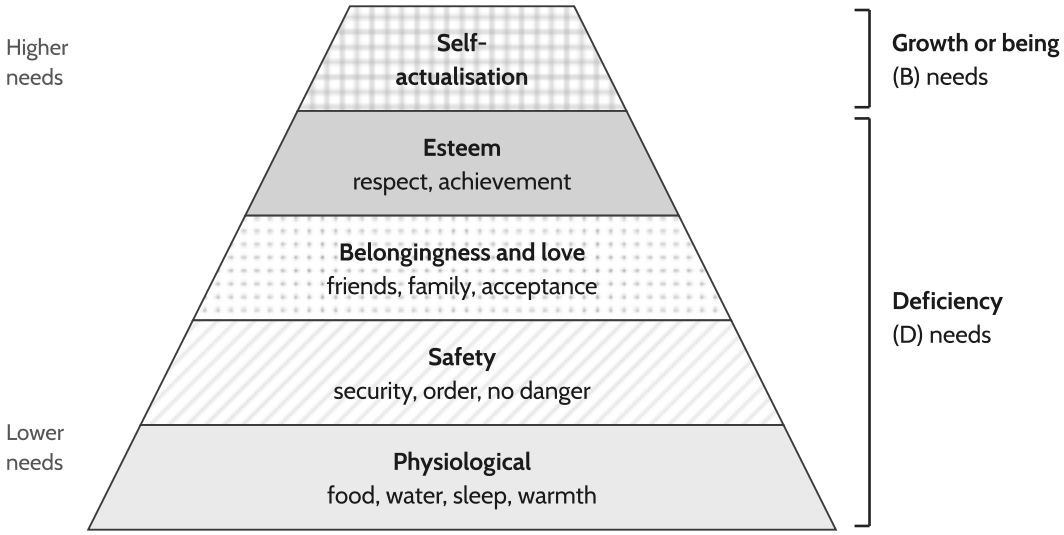


Figure 1.13 Maslow's hierarchy of needs (1943, 1954). The four lower levels are deficiency (D) needs; self-actualisation is a growth or being (B) need. Lower needs are prepotent: when unmet they dominate behaviour, and they must be largely met before higher needs take charge.

> **! EXAM TRAP**
>
> Bottom to top: **physiological, safety, belongingness, esteem, self-actualisation**. "Which is NOT a level?" **Ego** (Freud). Physiological and safety needs are **lower needs**: "greatest potency" is half-true (they are prepotent), and "transformational" is a leadership term. The hierarchical theory is **A. H. Maslow's**, not McClelland's, Atkinson's or Mowrer's. RML 2025 keyed **physiological** as the need mostly met before esteem; the level immediately below esteem is **belongingness (social belonging)**, so choose that when offered.

Self-actualisers, by Maslow's estimate 1 per cent of people or fewer, show **metamotivation** towards metaneeds or B-values such as truth, beauty and justice; frustrated, these breed **metapathologies** such as cynicism and apathy. They report **peak experiences**, moments of ecstatic awe. Maslow later added cognitive and aesthetic needs and, beyond self-actualisation, **self-transcendence**. The term self-actualisation was coined by **Kurt Goldstein** (1878 to 1965) in 1939. Critics fault Maslow's hand-picked sample, weak evidence for the ranking and Western bias.

> **! EXAM TRAP**
>
> "Striving for fulfilment of one's potential" is **self-actualisation**, not self-concept, ideal self (Rogers) or self-efficacy (Bandura). Self-actualisers show **metamotivation**, not D-motivation; "E-" and "F-motivation" are invented.

Clayton Paul Alderfer's **ERG theory** (1969) shrinks the ladder to existence, relatedness and growth, lets several needs act at once and adds **frustration-regression:** blocked at a higher need, a person falls back on a lower one (Table 1.21).

Table 1.21 Who proposed which list of needs.

Theorist	Year	The needs
Erich Fromm	1955	Relatedness, transcendence, rootedness, identity, frame of orientation
David McClelland	1953, 1961	Achievement, affiliation, power (learned)
Clayton Alderfer	1969	Existence, relatedness, growth
Deci and Ryan	1985	Autonomy, competence, relatedness (innate)
William Glasser	1965, 1998	Survival, love and belonging, power (achievement), freedom, fun

> **! EXAM TRAP**
>
> One of **William Glasser**'s five basic genetic needs in reality therapy is **power or achievement**; self-actualisation (Maslow), the collective unconscious (Jung) and unconditional positive regard (Rogers) are baits. Viktor Frankl's **will to meaning** belongs to **logotherapy** (Section 1.2), not REBT, existential-humanistic therapy or ACT.

Murray, McClelland and achievement ●●

Henry Alexander Murray (1893 to 1988), in *Explorations in Personality* (1938), listed about twenty psychogenic needs, among them n Achievement, n Affiliation (companionship), n Dominance (control of others) and n Exhibition (being seen and heard). A **press** is an environmental force that helps or hinders a need. **Christiana Morgan** and Murray's **Thematic Apperception Test** (TAT, 1935) reads needs and presses from stories told about ambiguous pictures (Assessment chapter).

> **! EXAM TRAP**
>
> Murray's **need-press theory** underlies the **TAT**, a projective test whose stories reveal unconscious needs, conflicts and motives; not the Rorschach, Jung's word association test or sentence completion, and not a test of IQ or memory.

David Clarence McClelland (1917 to 1998) scored TAT stories for the **need for achievement** (nAch, 1953), a learned desire to meet standards of excellence, and later for **power** (nPow) and **affiliation** (nAff). High scorers prefer moderate risk, personal responsibility and quick feedback; businessmen he trained in Kakinada, Andhra Pradesh, in the 1960s went on to start more businesses. **John William Atkinson** (1923 to 2003) explained why in 1957: the tendency to approach success = motive to succeed × probability of success (Ps) × incentive (1 − Ps), and Ps × (1 − Ps) is 0.25 at Ps = 0.5 but only 0.09 at 0.1 or 0.9, so success-oriented people choose in-

termediate tasks, while those ruled by the **motive to avoid failure** pick very easy or very hard ones. **Matina Souretis Horner** (1972) proposed a **fear of success** in women, later read as cultural stereotyping.

Explanations matter too. Rotter's **locus of control** (1966) is internal when people credit their own effort, external when they credit luck or others (not low self-efficacy or Adler's inferiority complex). **Bernard Weiner** (born 1935) classified the causes of success and failure by **locus**, **stability** and **controllability**: ability is internal, stable and uncontrollable; effort internal, unstable and controllable. "I am useless at maths" breeds hopelessness; "I did not practise enough" breeds effort.

Intrinsic motivation and the self ●●

Daniel Ellis Berlyne (1960) traced curiosity to novelty, complexity and incongruity, and **Robert White** (1959) united Harlow's manipulation drive, exploration and play as **effectance motivation**, the urge to deal effectively with the environment. That competence motive has two modern heirs.

The first is **self-efficacy**, introduced by **Albert Bandura** (1925 to 2021) in 1977: the belief that one can carry out the actions a particular task needs. It governs which tasks people attempt, how hard they try and how long they persist, and it grows mainly from mastery experiences, then from watching others, persuasion and bodily states. It is the core of his social cognitive theory (1986), built on **triadic reciprocal causation**: person, behaviour and environment influence one another. (Observational learning belongs to Section 1.6.)

> **! EXAM TRAP**
>
> The reciprocal causation model belongs to **social cognitive theory**, not drive reduction or constructivism (learners building their own knowledge, after Piaget and Vygotsky), and that theory highlights **self-efficacy**, not external rewards only, biological drives or goal difficulty. "Two-factor theories" are Herzberg's (work), Mowrer's (avoidance), Schachter and Singer's (emotion) and Spearman's (intelligence).

KEY STUDY: LEPPER, GREENE AND NISBETT'S MAGIC MARKERS (1973)

Mark Lepper, **David Greene** and **Richard Nisbett** let nursery-school children who liked drawing with magic markers draw for an expected certificate, an unexpected one, or nothing. One to two weeks later, the expected-award children spent about half as much free time drawing as the others; surprise awards did no harm. **Criticism.** Verbal or informative rewards do not undermine interest either, so the effect is narrower than often claimed.

This **overjustification effect** occurs because the reward moves the reason for an enjoyed activity outside the person. **Edward L. Deci** and **Richard M. Ryan** built such findings into the second heir, **self-determination theory** (SDT, 1985), with three innate needs: **autonomy**, **competence** and **relatedness**. Its **cognitive evaluation theory** explains the magic markers: rewards felt as controlling lower intrinsic motivation, while informational ones, such as unexpected praise, raise it. Motivation runs from amotivation through increasingly internalised extrinsic regulation to intrinsic motivation, driven by internal reward and growth.

> ! EXAM TRAP
>
> SDT's needs are **autonomy, competence and relatedness**, not **affiliation** or achievement (McClelland), independence, superiority or excellence. Its motivation driven by internal reward and growth is **intrinsic**; amotivation is its absence, external regulation the least autonomous form.

Mihaly Csikszentmihalyi (1934 to 2021) called intrinsic motivation at its most absorbing **flow** (1975, 1990): action and awareness merge and time vanishes when **challenge and skill are balanced** (too hard brings anxiety, too easy boredom). It is not Maslow's peak experience, Rogers's fully functioning person or Frankl's will to meaning. Finally, **Leon Festinger**'s (1957) **cognitive dissonance**, the tension of acting against one's beliefs (Arvind, who finds smoking disgusting, smokes at a party), motivates its reduction by a change of attitude or behaviour; effort justification is one form of that reduction, not the state (Social Psychology chapter).

Conflict and frustration ○

The blocking of goal-directed behaviour by obstacles, personal limits or conflicting motives is **frustration**. **Pressure** is the demand to perform or conform: Ramesh, made to work hard, fast and a lot in a call centre, is under pressure, not blocked (frustration), torn (conflict) or facing unpredictability.

Kurt Lewin (1890 to 1947) described the conflicts in 1935. **Approach-approach** (two attractive goals) is easiest to resolve; **avoidance-avoidance** (two evils) produces vacillation and attempts to leave the field; in an **approach-avoidance conflict** one goal both attracts and repels; and **double approach-avoidance**, several such goals, is the commonest in real life. **Neal Elgar Miller**'s (1909 to 2002) gradients (1944) explain the vacillation: both tendencies strengthen near the goal, but avoidance is steeper, so the person hovers where they cross.

Frustration may lead to **aggression**, direct or displaced; **regression**, as in Roger Barker, Tamara Dembo and Lewin's children (1941), who played like toddlers once attractive toys were screened off; **fixation**, as in **Norman Maier**'s (1949) rats, which kept jumping one way after the problem became soluble; withdrawal; or defence mechanisms (Personality chapter). The **frustration-aggression hypothesis** of **John Dollard**, Leonard Doob, Miller, **Orval Hobart Mowrer** and Robert Sears (1939) held that frustration always leads to aggression; **Leonard Berkowitz** (1989) recast it in terms of negative affect.

> ! EXAM TRAP
>
> An after-the-fact, socially acceptable reason for behaviour really driven by other motives is **rationalisation**; re-channelling sexual or aggressive impulses into valued outlets is **sublimation**; a safer target is **displacement**; attributing one's own impulse to others is **projection**; turning it into its exaggerated opposite is **reaction formation**; pushing a threatening thought out of awareness is **repression**; refusing to acknowledge an unpleasant reality is **denial**; compensation pushed to excess is **overcompensation** (the full list: Personality chapter).

Hunger, thirst and sex ●●

In 1912 Walter Cannon and **A. L. Washburn** showed that stomach contractions coincide with hunger pangs, yet people without stomachs still feel hunger. **Glucostatic** (blood sugar) and **lipostatic** (body fat) set-point theories followed; John Pinel and Steven Barnes argue instead that we eat mainly for anticipated pleasure, with weight drifting to a **settling point**.

In 1940 Albert Hetherington and Stephen Ranson found that lesions of the **ventromedial hypothalamus** (VMH) made rats overeat and grow obese; in 1951 the Indian physiologist **Bal Krishan Anand** and John Brobeck found that lesions of the **lateral hypothalamus** (LH) caused aphagia (no eating) and adipsia (no drinking). Eliot Stellar's **dual-centre theory** (1954) made the LH the feeding centre and the VMH the satiety centre. Since revised (VMH lesions raise insulin; other nuclei matter), it now shares the stage with leptin from fat, insulin from the pancreas and ghrelin from the empty stomach.

> **! EXAM TRAP**
>
> Satiety centre: **VMH**, lesion gives hyperphagia and obesity. Feeding centre: **LH**, lesion gives aphagia and weight loss (**L**ateral, **L**ose). "Ventromedial hippocampus" and "lateral thalamus" are invented; the cerebrum and basal ganglia are not hunger centres. The hypothalamus as a whole is the master homeostatic regulator (hunger, thirst, temperature; its suprachiasmatic nucleus is the circadian clock), not the cerebellum (movement) or pineal gland (melatonin).

Stanley Schachter's externality hypothesis (1968 to 1971) held that obese people respond more to external food cues, such as the clock, than to internal ones (eating disorders: Psychiatric Disorders chapter). Thirst is **osmotic** (salty body fluid draws water from cells) or **hypovolaemic** (blood volume falls, signalled through angiotensin II). **Alfred Charles Kinsey** surveyed the sexual behaviour of men (1948) and women (1953); **William Howell Masters** and **Virginia Eshelman Johnson** (1966) observed a four-phase **sexual response cycle** of excitement, plateau, orgasm and resolution; **Helen Singer Kaplan** (1979) put **desire** first (desire, excitement, orgasm). Hormones such as testosterone matter less in human sexuality than in other mammals. In Carl Warden's obstruction box (1931) the maternal drive outran thirst and hunger; pain avoidance (aroused by tissue damage) and sleep (Section 1.5) complete the primary motives.

> **! EXAM TRAP**
>
> Masters and Johnson: **excitement, plateau, orgasm, resolution**. Sequences that insert desire (Kaplan), start with plateau or end in "refractory" (the male recovery period) are scrambles. The reports on sexual behaviour are **Kinsey's**, not Freud's, Anna Freud's or Horney's.

Motivation at work, school and clinic ●●

Frederick Herzberg (1923 to 2000) asked about two hundred engineers and accountants when they had felt exceptionally good or bad at work. His **two-factor theory** (1959) separates **motivators** (Table 1.22), in the work itself, which create satisfaction, from **hygiene factors**, in its context, which cause dissatisfaction when poor but, when improved, merely remove it.

Table 1.22 Herzberg's two factors.

Motivators (job content)	Hygiene factors (job context)
Achievement, recognition	Company policy, supervision
The work itself, responsibility	Relationships, working conditions
Advancement, growth	Salary, status, job security

Victor Harold Vroom's **expectancy theory** (1964) makes motivational force = expectancy (effort leads to performance) × instrumentality (performance leads to outcomes) × valence (value of the outcomes). If Priya, a sales executive, has E = 0.8, I = 0.9 and V = 0.7, her force is about 0.50; if her firm is known not to pay bonuses, I drops to 0.2 and her force to about 0.11. **John Stacy Adams**'s **equity theory** (1963) says people compare their ratio of outcomes to inputs with others' and act to remove inequity. In the **goal-setting theory** of **Edwin Locke** and **Gary Latham** (from 1968), specific, difficult but attainable goals, with commitment and feedback, beat vague "do your best" goals.

> **! EXAM TRAP**
>
> Hygiene factors **prevent dissatisfaction but do not by themselves motivate** (they are not self-actualisation needs, nor for high achievers only); job satisfaction depends on **work motivation and job environment**, not "or" or "only". Vroom's outcomes must be **valuable**; guaranteed rewards, effort unrelated to performance and unimportant rewards kill motivation. Goals should be **specific and challenging**, with feedback, not vague, easy or value-free.

In school, informational feedback, praise for effort and attribution retraining (Carol Dweck, 1975) sustain effort. In the clinic, **motivational interviewing** (William Richard Miller and Stephen Rollnick, 1991) treats resistance to change as ambivalence, an approach-avoidance conflict, and draws out the client's own reasons to change.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A physiological deficiency is a _____; the tense state it produces is a _____.
2. For a complex task the optimal arousal is _____ than for a simple one.
3. The level immediately below esteem in Maslow's hierarchy is _____.
4. Murray's environmental force that helps or hinders a need is a _____.
5. Self-determination theory's three needs are autonomy, competence and _____.
6. In Herzberg's theory, salary is a _____ factor.

Answers: (1) need, drive, (2) lower, (3) belongingness and love, (4) press, (5) relatedness, (6) hygiene.

1.11 Emotion and Stress MEDIUM YIELD · 74 QS

Your heart pounds as the question paper is handed out. Is that pounding the fear itself, or something your mind has still to interpret? This section takes emotion apart, then turns to stress and coping.

What an emotion is ○

An **emotion** is a brief, whole-person response to an event that matters to you, with three components: physiological arousal (a pounding heart, sweating palms), expressive behaviour (face, voice, posture, action) and conscious experience ("I am afraid"). The theories that follow quarrel mainly about the order of these three. A feeling is the subjective part of an emotion. An emotion lasts seconds to minutes and has an object (you are angry *at* someone); a **mood** lasts hours or days, is weaker and often has no object. **Affect** is the umbrella term for your present good or bad feeling state and, clinically, its observable expression, rated for quality, range and reactivity; persistence is what defines mood. Temperament is the inborn style of emotional reactivity (Alexander Thomas and Stella Chess's easy, difficult and slow-to-warm-up types are in the Developmental chapter).

Are there basic emotions, innate and universal, each with its own face? **Paul Ekman** (born 1934) and **Wallace V. Friesen** found six with universally recognised expressions: happiness, sadness, fear, anger, surprise and disgust, with contempt added in 1986. **Silvan Tomkins** (1962) listed nine innate affects; **Carroll Izard**'s differential emotions theory (1977) listed ten, including shame and guilt; and **Robert Plutchik**'s wheel (1980) set eight primaries opposite one another (joy and sadness, trust and disgust, fear and anger, surprise and anticipation), with neighbours blending into dyads such as love (joy plus trust). Dimensional models place emotions in a space instead: Wundt's three-dimensional theory of feeling (Section 1.1) and the circumplex of **James Russell** (1980), a circle of valence (pleasant to unpleasant) by arousal.

> ! EXAM TRAP
>
> **Jealousy, shame, guilt, pride and love are not among Ekman's six.** Shame and guilt are Izard's, trust and anticipation Plutchik's. "Our current mood and feelings" is **affect**, not mood.

Body first or label first ●●

Common sense says you see a bear, feel afraid, and therefore tremble. **William James** (1842 to 1910) in 1884 and the Danish physician **Carl Georg Lange** (1834 to 1900) in 1885 reversed the order. The **James-Lange theory** holds that the stimulus first triggers bodily changes and that the emotion *is* our perception of them: "we feel sorry because we cry, angry because we strike, afraid because we tremble." The textbook sequence is stimulus, arousal, awareness ("interpretation") of the arousal, then emotion (Figure 1.14, top row), where the "interpretation" is just the brain noticing what the body is doing.

Walter Bradford Cannon (1871 to 1945) objected in 1927 that emotion survives when the viscera are cut off from the brain; that the same visceral changes accompany fear, anger, fever and exercise; that the viscera are too insensitive and slow for quick, fine distinctions; and that injected adrenaline produces symptoms without real emotion. In the **Cannon-Bard theory**, developed with **Philip Bard** (1898 to 1977), the thalamus signals the cortex and the body at once, so feeling and arousal occur simultaneously and independently (second row). If arousal is so similar across emotions, what tells fear from joy? In 1962 **Stanley Schachter** and **Jerome Singer** answered with the **two-factor theory**: emotion is physiological arousal, which sets its intensity, plus a cognitive label drawn from the situation, which names it.

KEY STUDY: SCHACHTER AND SINGER'S ADRENALINE EXPERIMENT (1962)

Students given adrenaline (epinephrine), described as a vitamin, were told its true effects, told nothing or misinformed; a placebo group got saline. Each then waited with a stooge acting euphoric or angry. The aroused but uninformed caught the stooge's mood most; the informed, who could blame the injection, least. **Criticism.** The effects were small, and in replications unexplained arousal simply felt anxious.

A label can be wrong. In **misattribution of arousal** people mistake the source of their arousal: in 1974 **Donald Dutton** and **Arthur Aron** found that men approached by an attractive woman on the swaying Capilano suspension bridge telephoned her far more often than men on a low, solid bridge. Fear had been read as attraction. **Dolf Zillmann**'s excitation transfer theory (1970s) adds that leftover arousal from one event intensifies the next emotion.

! EXAM TRAP

Match theory to sequence. Common sense: emotion, then body (the view James rejected). James-Lange: body first, then emotion. Cannon-Bard: emotion and visceral reaction *simultaneous*, via the thalamus. Schachter-Singer: *cognition plus arousal*, and the only one that explains misattribution. No classic theory makes the somatic nervous system the seat of emotion, and "emotions are learned through reinforcement" is behaviourism.

Appraisal, faces and opposing processes

Richard S. Lazarus (1922 to 2002) put thought first. In his cognitive appraisal (cognitive-mediational) theory, **primary appraisal** asks "Does this matter to me?" (harm-loss, threat or challenge), **secondary appraisal** asks "What can I do about it?", and reappraisal revises both (Figure 1.14, bottom row). **Robert Zajonc** (1923 to 2008) argued for affective primacy ("preferences need no inferences", 1980; debated with Lazarus in 1984): by the mere exposure effect, repeated exposure raises liking even for shapes seen too briefly to be recognised, so affect can come first. The **facial feedback hypothesis**, rooted in Tomkins's work, says that our own expressions modulate our feelings; in the pen study of **Fritz Strack**, **Leonard Martin** and **Sabine Stepper** (1988), holding a pen in the teeth, which forces a smile, made cartoons seem funnier, an effect now judged real but small. **Charles Darwin**'s *The Expression of the Emotions in Man and Animals* (1872) founded the evolutionary view of expressions as inherited adaptations. **Richard Solomon**'s opponent-process theory (1974) holds that every strong emotion triggers a slow, opposite after-reaction that grows with repetition: a drug's high fades while withdrawal deepens.

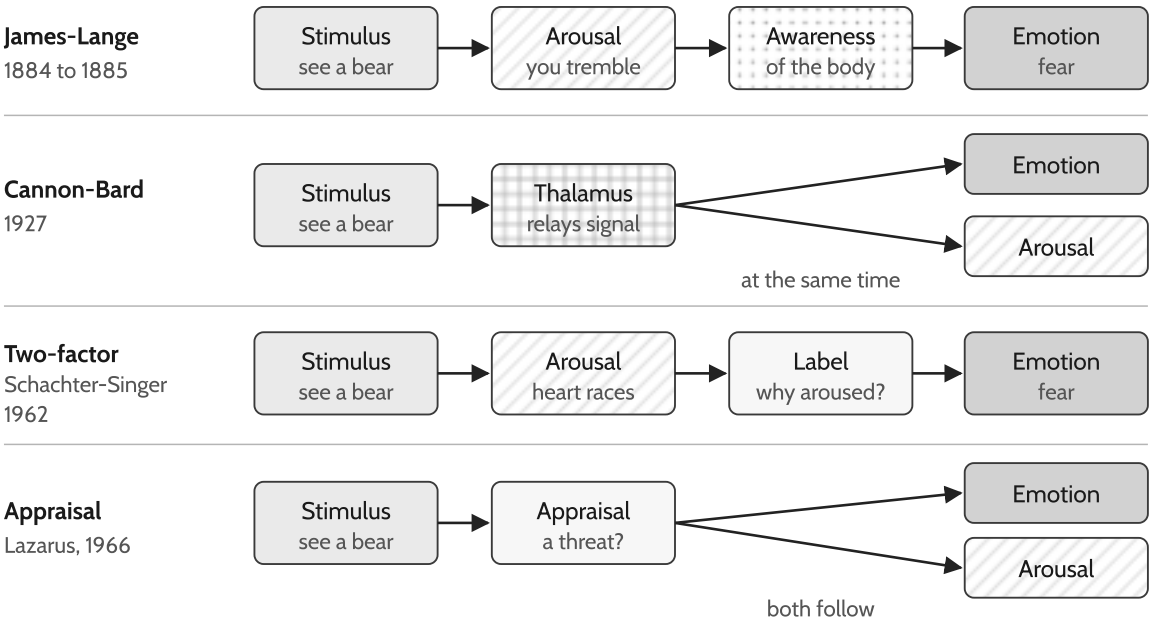


Figure 1.14 Four classic theories of emotion compared. Follow each row from the stimulus to the emotion. James-Lange: the body reacts first and the felt emotion is awareness of that reaction. Cannon-Bard: the thalamus sends signals that produce feeling and arousal at the same time. Two-factor: arousal plus a cognitive label. Appraisal (cognitive-mediational): an evaluation of the stimulus comes first, and both emotion and arousal follow from it.

The emotional brain and body ●●●

The body's side of emotion is run by the autonomic nervous system. Its sympathetic division prepares for fight or flight: it dilates the pupils, speeds the heart and breathing, dries the mouth, halts digestion and releases adrenaline from the adrenal medulla, and its sweating lowers skin resistance (the galvanic skin response). The parasympathetic division ("rest and digest") reverses each effect, constricting the pupils. Heartbeat and breathing themselves are paced by the medulla of the hindbrain; the hypothalamus only orchestrates them.

Bard's cats, with the cortex removed, showed **sham rage**, an undirected hissing fury that vanished only when the hypothalamus was removed too: the hypothalamus organises rage and the cortex restrains it. In 1937 **James Papez** (1883 to 1958) proposed an emotional loop, the **Papez circuit**: hippocampus, fornix, mammillary bodies, anterior thalamus, cingulate gyrus, entorhinal cortex and back to the hippocampus. **Paul MacLean** (1913 to 2007) added the amygdala and other structures in 1952 and named the whole the **limbic system**, the classic "seat of emotion", which regulates emotion, motivation and memory (Table 1.23).

Its key member is the **amygdala**, deep in each temporal lobe. When **Heinrich Klüver** and **Paul Bucy** removed both anterior temporal lobes of rhesus monkeys (1937, 1939), wild animals became tame. **Klüver-Bucy syndrome** is placidity (loss of fear, even of snakes), psychic blindness or visual agnosia, hyperorality, hyperphagia and hypersexuality; human cases often add aphasia and amnesia, because nearby temporal cortex is damaged too. The amygdala detects threat, drives fear conditioning and gives emotional events their extra memorability, an advantage that disappears when noradrenaline is blocked with propranolol (Cahill and McGaugh, 1994). **Joseph LeDoux** (born 1949) found a fast, crude low road from the thalamus straight to the amygdala and a slower, detailed high road through the cortex. Such subcorti-

cal routes probably explain affective blindsight, the recognition of emotional faces shown in a cortically blind field.

The prefrontal cortex regulates emotion (**Phineas Gage**'s rod injury of 1848; **Antonio Damasio**'s somatic markers, 1994). Damage to its control of the brainstem, after stroke, multiple sclerosis, ALS or brain injury, can cause **pseudobulbar affect**: sudden, involuntary laughing or crying out of proportion to the actual mood, which is not depression. The polygraph records arousal, not lies, so frightened innocent suspects can fail it; **David T. Lykken**'s guilty knowledge test (1959), which asks about details only the culprit knows, is more accurate.

Table 1.23 Brain structures in emotion.

Structure	Role	Classic evidence
Hypothalamus	Rage, stress hormones, hunger, thirst	Sham rage (Bard)
Limbic system	Emotion, motivation, memory	Papez (1937); MacLean (1952)
Amygdala	Fear, aggression, emotional memory	Klüver and Bucy; LeDoux
Hippocampus	Memory for facts	Amnesia, as in H.M.
Distractors	Reticular formation (arousal), pituitary, corpus callosum, cerebellum (coordination), caudate (movement), suprachiasmatic nucleus (body clock)	Not emotion centres

> **! EXAM TRAP**
>
> **Limbic system**, not one of its parts or the broad forebrain, for "seat of emotion"; its function is emotion, while hunger and thirst are the hypothalamus's. **Amygdala**, not hippocampus, for fear, aggression and the emotional enhancement of memory. Amygdala removal gives tameness, not attack (sham rage), obesity (ventromedial hypothalamus) or amnesia (hippocampus); options with "hyperalertness" or "avolition, alogia" are wrong. Pupil constriction is the odd one out among sympathetic effects.

Expression and four emotions ○

The Fore of Papua New Guinea, who had seen no films or magazines, matched Ekman and Friesen's faces to stories far above chance (1971), confusing only fear with surprise. Cultures differ instead in **display rules**, learned norms for when an emotion may be shown. The Facial Action Coding System (1978) codes faces as muscle action units, and a genuine Duchenne smile engages the muscles around the eyes. Posture, gesture and voice carry emotion too. Women are, on average, more facially expressive, largely through socialisation. Fear answers a present danger; anxiety is diffuse and future-oriented; a phobia is a persistent, irrational fear of a specific object or situation. The catharsis hypothesis, that venting anger drains it, fails: hitting a punching bag while thinking of an insulter increased aggression (**Brad Bushman**, 2002). Disgust guards against contamination (**Paul Rozin** and **April Fallon**, 1987).

Ed Diener (1946 to 2021) defined **subjective well-being** (1984) by three components: life satisfaction (cognitive), frequent positive affect and infrequent negative affect. Good fortune fades: by **Harry Helson**'s adaptation-level theory (1964) and the hedonic treadmill of **Philip**

Brickman and **Donald Campbell** (1971), lottery winners are no happier than others a year or so later, and relative deprivation (**Samuel Stouffer**, 1949) makes us judge ourselves against others. **Barbara Fredrickson**'s **broaden-and-build theory** (1998) says positive emotions widen thought and build lasting resources. **Savoring** (**Fred Bryant** and **Joseph Veroff**, 2007) is enhancing and prolonging the enjoyment of positive experiences.

> ! EXAM TRAP
>
> **Diener:** life satisfaction, not physical, spiritual or social wellness. **Fredrickson:** broaden-and-build; flow is Csikszentmihalyi, cognitive distortions Beck, psychodynamic theory Freud. **Savoring** is not avoiding negatives, relaxing or reflecting on trauma.

How emotions develop and are regulated ●●

Katharine Banham Bridges (1932) saw the newborn's general excitement split first into *distress* and *delight*, and only then into anger, fear, elation and the rest. Primary emotions are present by about six months. The **self-conscious emotions** (embarrassment, pride, shame, guilt) need a sense of self, so they appear after 12 months, from about 18 months. **Edward Tronick**'s still-face procedure (1978), in which a mother suddenly freezes her face, shows that even young infants expect emotional give-and-take.

In the process model of **James Gross** (1998), emotion regulation has five families of strategies, in the order an emotion unfolds: situation selection, situation modification, attentional deployment (distraction, rumination), cognitive change, chiefly **cognitive reappraisal** ("my racing heart is my body getting ready"), and response modulation, which includes expressive suppression. **Alexithymia** (**Peter Sifneos**, 1973) is difficulty identifying and describing one's own feelings, not anhedonia (loss of pleasure). **Daniel Goleman**'s emotional intelligence (1995; Section 1.9) has five domains: knowing one's emotions, managing them, motivating oneself, recognising others' emotions and handling relationships (later relabelled self-awareness, self-regulation, motivation, empathy and social skills); empathy has emotional, cognitive and empathic-concern components.

> ! EXAM TRAP
>
> **Reappraisal is cognitive change**; suppression is response modulation. **First infant emotions:** distress and delight (Bridges); fear, rage and love are Watson's. **Goleman:** if the 1995 list (knowing and managing one's emotions, motivating oneself, recognising others' emotions, handling relationships) and the later self-awareness relabelling both appear, choose the 1995 wording.

Stress and the body's emergency systems ●●

Stress has been defined as a stimulus (the stressor), as a response (Selye's "nonspecific response of the body to any demand") and as a transaction in which demands are appraised as exceeding one's resources. Cannon's **fight-or-flight response** (1915) is the fast route: within seconds the sympathetic nerves act and the adrenal medulla releases adrenaline.

What if the emergency never ends? **Hans Selye** (1907 to 1982) found that every noxious agent he gave rats, from tissue extracts to cold, produced the same triad: enlarged adrenal glands, shrunken immune tissue and stomach ulcers. In 1936 he proposed the **general adaptation**

syndrome (*The Stress of Life*, 1956), with three stages in fixed order. In the *alarm reaction* the body mobilises, as in fight or flight. In the *stage of resistance* cortisol stays high and the body adapts, at a cost. In the *stage of exhaustion* adaptive capacity runs out and diseases of adaptation such as ulcers and hypertension appear. He later (1974) separated **eustress**, beneficial stress, from distress.

Resistance runs on the slower **hypothalamic-pituitary-adrenal axis** (HPA axis): the hypothalamus releases corticotropin-releasing hormone (CRH), the anterior pituitary adrenocorticotropic hormone (ACTH), and the adrenal cortex cortisol, which feeds back on the hypothalamus and anterior pituitary to switch the system off. **Shelley E. Taylor** (2000) added tend-and-befriend: females especially meet stress by protecting offspring and seeking allies, a pattern linked to oxytocin.

> ! EXAM TRAP
>
> **Alarm, resistance, exhaustion.** Illness is not a stage, and the stages are Selye's, not life events theory, the health belief model or protection motivation theory. **The adrenal glands** are the stress glands; the pituitary only releases ACTH, and the posterior pituitary is not in the HPA axis. Insulin (pancreas), thyroxine (thyroid) and melatonin (pineal) are distractors.

Sources of stress and the transactional model ●●

Thomas Holmes and **Richard Rahe**'s **Social Readjustment Rating Scale** (1967) scores 43 life events in life change units, from death of a spouse (100) through marriage (50) to minor violations of the law (11); because change itself is the stressor, even holidays count. **Allen Kanner**, Lazarus and colleagues (1981) showed that **daily hassles**, the irritations of everyday life, predict symptoms better than major events, while uplifts, the small pleasures that offset them, matter less. An acute stressor is brief (a pop quiz); a chronic one lasts (HIV disease, cancer, marital discord); a cataclysmic one, such as a flood or war, strikes many people at once. Frustration and conflict are covered under motivation (Section 1.10). At work, organisational stressors are task, role and interpersonal demands, structure and leadership, as distinct from environmental (political uncertainty) and individual (family, money) factors. Role conflict (incompatible expectations) and **Robert Karasek**'s high-demand, low-control jobs (1979) can end in **burnout** (**Herbert Freudenberger**, 1974), which **Christina Maslach**'s inventory (1981) measures as exhaustion, depersonalisation and reduced accomplishment.

In Lazarus and **Susan Folkman**'s **transactional model of stress** (1984), stress is a person-environment relationship appraised as taxing or exceeding one's resources: primary appraisal judges harm-loss, threat or challenge, secondary appraisal judges the resources for coping, and coping and reappraisal follow. In one exam hall, Ananya appraises the paper as a threat and her preparation as thin, and shakes; Kiran sees a challenge, trusts his practice, and feels keyed up but focused. The **diathesis-stress model** (**Paul Meehl**, 1962; **Joseph Zubin** and **Bonnie Spring**, 1977) explains why stress makes one person ill and spares another: a disorder appears when a pre-existing vulnerability, often biological, is triggered by stress; usually neither is enough alone.

! EXAM TRAP

The SRRS measures life events, not hassles, physiological reactions or cognitive change. **Secondary appraisal** is the resources question; "coping resource assessment" is not the model's term. **Diathesis-stress** is not "genes only", "environment only", "unconscious conflicts only", "learned experiences alone" or supernatural; humanistic, sociocultural and cognitive models blame blocked growth, social forces and faulty thinking.

Coping, personality and health ●●●

Problem-focused coping acts on the stressor (a revision plan); **emotion-focused coping** acts on the feelings (reappraisal, support, acceptance, but also denial). The best choice fits controllability: plan while the exam can still be prepared for, reappraise once the result is out. Approach coping beats avoidance in the long run. Defence mechanisms (Personality chapter) differ in being unconscious: repression involuntarily blocks painful feelings (suppression is deliberate); displacement redirects anger to a safer target (anger at a husband shouted at a child); sublimation channels it into a valued outlet (kickboxing); projection attributes one's own feelings to others; isolation of affect strips the feeling from an idea (a gruesome fantasy told calmly), whereas denial rejects the reality; rationalisation offers a plausible excuse; and reaction formation turns a feeling into its opposite.

Social support protects both directly and as a buffer when stress is high (**Sheldon Cohen** and **Thomas Wills**, 1985). **Robert S. Weiss** (1973) separated **emotional isolation**, lacking a deep attachment to one specific person, from social isolation, lacking a circle of friends. **Suzanne Kobasa**'s **hardiness** (1979) is commitment, control and challenge; **Aaron Antonovsky**'s sense of coherence (1979) is comprehensibility, manageability and meaningfulness; resilience is bouncing back from adversity. Dispositional optimism (**Michael Scheier** and **Charles Carver**, 1985), a general expectancy of good outcomes, aids coping and recovery. **Martin Seligman** and **Steven Maier** (1967) found that dogs given inescapable shock stopped trying to escape, learned helplessness (Section 1.6), which Seligman later reversed into learned optimism.

In 1959 the cardiologists **Meyer Friedman** and **Ray Rosenman** described the **Type A behaviour pattern**: competitive striving, time urgency and hostility. In their Western Collaborative Group Study, Type A men developed coronary heart disease about twice as often as relaxed Type B men; later work found hostility to be the toxic component. A cancer-prone Type C, unassertive and suppressing negative emotion, has weak support. **Johan Denollet**'s **Type D personality** (mid-1990s) is negative affectivity plus social inhibition, and predicts a poor outlook in heart disease.

George Engel's **biopsychosocial model** (1977) explains illness through biological, psychological and social factors together. **Psychoneuroimmunology** began when **Robert Ader** and **Nicholas Cohen** (1975) conditioned immune suppression in rats by pairing saccharin with an immunosuppressive drug; chronic stress suppresses immunity and slows healing. Three models predict health behaviour: the health belief model (**Irwin Rosenstock** and colleagues, 1950s: perceived susceptibility, severity, benefits, barriers and cues to action), **Ronald Rogers**'s protection motivation theory (1975: threat appraisal and coping appraisal) and **Icek Ajzen**'s theory of planned behaviour (1985: attitude, subjective norm and perceived behavioural control, close to Albert Bandura's self-efficacy, belief in one's capability to act).

Stress management (Clinical chapter) includes **Edmund Jacobson**'s progressive muscle relaxation (1929), tensing and releasing muscle groups in turn; **Herbert Benson**'s relaxation response (1975); biofeedback, learning to control a bodily signal displayed on a monitor; meditation, as in **Jon Kabat-Zinn**'s mindfulness-based stress reduction (1979); regular aerobic exercise; and cognitive restructuring, replacing distorted thoughts with realistic ones, which **Donald Meichenbaum** combined with coping skills in stress inoculation training (1970s).

> **! EXAM TRAP**
>
> **Type A** is competitiveness, impatience and hostility; options with "non-competitiveness" are wrong, and introversion or extraversion is no risk factor. Offered both Type A and hostility, RML 2024 keyed Type A; for the toxic *component*, answer hostility. **Type D** is not hostility, optimism or perfectionism. **Learned helplessness** is Seligman and Maier, not Beck, Sullivan or Kahneman. **Biopsychosocial** means social, not cultural. Emotional isolation is not incompetence or low self-esteem.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. "We feel afraid because we tremble" summarises the _____ theory.
2. MacLean's name for the classic "seat of emotion" is the _____ system.
3. Tameness, hyperorality and hypersexuality after bilateral amygdala damage make up _____ syndrome.
4. In Gross's process model, reappraisal belongs to _____ change.
5. Selye's general adaptation syndrome runs alarm, _____, exhaustion.
6. Denollet's Type D combines negative affectivity with social _____.

Answers: (1) James-Lange, (2) limbic, (3) Klüver-Bucy, (4) cognitive, (5) resistance, (6) inhibition.

The Case of Max Friedrich's Key, Revisited

Go back to the winter of 1879 and the student pressing his key. You can now take his act apart with the whole chapter. The stimulus had to cross a threshold (Section 1.3) and be organised into an object (Section 1.4); apperception was attention bringing it into focus (Section 1.5); practice shortened the times, which is learning (Section 1.6); the instruction was held in memory (Section 1.7); choosing when to press was a small decision (Section 1.8); observers differed in speed, one root of the study of individual differences and intelligence (Section 1.9); and the readiness and strain of waiting were motivation and emotion (Sections 1.10 and 1.11). Friedrich's numbers are forgotten, but his question, how long a mental act takes and what it is made of, runs through every section of this chapter.

Chapter summary

1.1 Psychology as a Science: Roots, Founders and the First Schools. Psychology, once the study of the soul, is now the scientific study of behaviour and covert mental processes. Its long past lies in philosophy (Plato's nativism, Aristotle's laws of association, Descartes' dualism, Locke's *tabula*

rasa, the associationists, Leibniz and Herbart on apperception, Kant) and in physiology (Muller, Helmholtz, localisation), with Darwin and Galton adding function and individual differences. Fechner's psychophysics (1860) and Donders' chronometry measured the mind. **Wundt** founded the first laboratory at Leipzig in 1879 and taught voluntarism through controlled introspection. **Titchener**'s structuralism sought the elements of consciousness; **James**'s functionalism asked what the mind is for, and Witmer opened the first clinic (1896).

1.2 Schools and Systems, the Cognitive Revolution and Indian Psychology. From about 1900 to 1960 psychology was a *preparadigmatic* field of rival schools. **Watson** founded behaviourism in 1913; Tolman, Hull, Guthrie and Skinner refined it. **Gestalt psychology** (Wertheimer's phi phenomenon, 1912; Kohler; Koffka) held that the whole differs from the sum of its parts, and Lewin defined the life space. **Psychoanalysis** was the first force, **humanism** (Maslow, Rogers) the third and transpersonal psychology the fourth, beside existential psychology and Seligman's positive psychology. The **cognitive revolution** (1956 onwards) restored the mind. Sengupta and Bose pioneered modern psychology at Calcutta; **Indian psychology** rests on Samkhya's purusha, prakriti and three gunas, Patanjali's eight limbs, the four states and the five koshas.

1.3 Sensation and Psychophysics. Sensation is the detection and transduction of physical energy; perception is its interpretation, so illusions distort perception. Muller's specific nerve energies and the all-or-none law explain how quality and intensity are coded, and every sense but smell relays through the thalamus. Fechner's psychophysics (1860) defines the absolute threshold as 50 percent detection and the JND as the smallest detectable change; Weber's law, Fechner's logarithmic law and Stevens' power law link stimulus to sensation, and signal detection theory separates sensitivity (d′) from response bias. Rods serve scotopic vision and foveal cones colour and acuity; colour is trichromatic then opponent; pitch is coded by place and timing; gate control explains pain.

1.4 Perception: Organising and Interpreting the World. Perception is the selection, organisation and interpretation of sensory signals. **Constructivist** theories (Helmholtz's unconscious inference, Gregory's hypotheses, the Bayesian union of priors and evidence) treat it as top-down inference; **Gibson's** direct, ecological theory, with its affordances, is the classic bottom-up account, and Neisser's perceptual cycle joins the two. The Gestalt laws, crowned by Prägnanz, organise the field into figure, ground and groups. Depth comes from two binocular cues (retinal disparity, convergence) and many monocular ones; the visual cliff (Gibson and Walk, 1960) showed depth perception by crawling age. Constancies keep objects stable, while illusions such as the Müller-Lyer reveal misapplied size scaling and differ from hallucinations and delusions.

1.5 Attention and Consciousness. Attention selects the little that reaches awareness; its span, studied by Jevons (1871) and with the tachistoscope, is about six to eight items. Cherry's dichotic listening (1953) led to Broadbent's early filter (1958), Treisman's attenuation model (1964) and late selection (Deutsch and Deutsch, 1963). In the **Stroop effect** (1935) automatic reading defeats selective attention, and vigilance declines over a watch (Mackworth). James described a stream of consciousness, Freud conscious, preconscious and unconscious levels. The suprachiasmatic nucleus times sleep, whose NREM and REM stages cycle every 90 minutes, four to six times a night. Dreams are explained by Freud, Jung and activation-synthesis; hypnosis rests on suggestibility.

1.6 Learning: Conditioning, Cognition and Observation. Learning is a relatively permanent change in behaviour, or its potential, resulting from experience (Kimble). In **classical conditioning** (Pavlov)

a neutral stimulus paired with a UCS becomes a CS; extinction, spontaneous recovery, generalisation and discrimination follow. Watson's Little Albert showed conditioned fear, and Jones and Wolpe reversed it through counterconditioning and systematic desensitisation (reciprocal inhibition). Rescorla, Kamin and Garcia showed that conditioning depends on prediction and preparedness. In **operant conditioning** (Thorndike's law of effect, Skinner) consequences change behaviour: reinforcement increases it, punishment decreases it, and schedules shape its pattern, variable ratio most strongly. Tolman's latent learning, Köhler's insight and Bandura's observational learning showed much learning without direct reinforcement.

1.7 Memory: Encoding, Storage, Retrieval and Forgetting. Memory is studied as **encoding**, **storage** and **retrieval**; Ebbinghaus (1885) measured forgetting with nonsense syllables and savings. Atkinson and Shiffrin's **multi-store model** (1968) separates sensory, short-term (seven plus or minus two chunks, mainly acoustic) and long-term memory, and Baddeley and Hitch (1974) recast short-term memory as **working memory**: central executive, phonological loop, visuospatial sketchpad and, from 2000, the episodic buffer. Depth of processing, Tulving's episodic and semantic memory, implicit memory, schemas and Bartlett's reconstruction, encoding specificity, decay, interference and repression explain remembering and forgetting, while H.M., Korsakoff's syndrome and Lashley's search for the engram reveal its biology.

1.8 Thinking, Problem Solving, Reasoning, Decision Making and Language. Thinking manipulates internal representations: images (Shepard and Metzler's mental rotation; Paivio's dual coding), concepts (Bruner's hypothesis testing for artificial concepts, Rosch's prototypes for natural ones) and propositions. Problems yield to algorithms, heuristics or Gestalt insight, and are blocked by **functional fixedness** (Duncker, 1945) and **mental set** (Luchins, 1942). Deduction runs from theory to data, induction from data to theory. Simon's bounded rationality and satisficing, and Tversky and Kahneman's heuristics and prospect theory, describe real decisions. Creativity is divergent production passing through Wallas's four stages. Language is built from phonemes and morphemes; Chomsky's LAD, the critical period, Whorf's linguistic relativity and Vygotsky's inner speech complete the picture.

1.9 Intelligence: Theories, Measurement, Heredity and Environment. Intelligence has been defined as judgement (Binet), *what the tests test* (Boring) and Wechsler's aggregate or global capacity (1944). Binet and Simon's 1905 scale introduced mental age; Stern coined the IQ (1912), Terman's Stanford-Binet (1916) made the ratio IQ famous, and Wechsler's deviation IQ (mean 100, SD 15) replaced it. The WAIS-IV has four indices; Raven's matrices test fluid intelligence, and Bhatia's Battery suits illiterate Indian children. Spearman's g, Thurstone's primary abilities, Guilford's cube, Cattell's fluid and crystallised intelligence, Gardner's multiple intelligences and Sternberg's triarchic theory offer rival structures. Heritability is a population statistic, and intellectual disability requires adaptive deficits as well as low IQ.

1.10 Motivation. Motivation energises and directs behaviour: a **need** produces a **drive** that pushes, while incentives pull. McDougall's instincts gave way to Hull's **drive reduction** (1943), which could not explain curiosity; Hebb proposed optimal arousal, and the **Yerkes-Dodson law** (1908) makes performance an inverted U of arousal with a lower optimum for complex tasks. Maslow's **hierarchy of needs** runs physiological, safety, belongingness, esteem, self-actualisation; Alderfer reduced it to ERG. Murray's needs and press underlie the TAT; McClelland studied achievement, power and affiliation. Self-efficacy, **self-determination theory** and flow explain intrinsic motivation. Lewin classified conflicts; the VMH is the satiety centre, the LH the feeding centre; Herzberg, Vroom and Locke explain work.

1.11 Emotion and Stress. An **emotion** joins arousal, expression and conscious experience; a mood is longer and diffuse, and affect is the current feeling state. Ekman named six universal basic emotions. **James-Lange** puts bodily change before feeling, **Cannon-Bard** makes them simultaneous via the thalamus, **Schachter and Singer** add a cognitive label (explaining misattribution), and Lazarus puts appraisal first. The **limbic system**, above all the **amygdala**, organises fear. Stress runs through Selye's alarm, resistance and exhaustion and the HPA axis; life events, hassles and appraisal shape it, and coping, support, hardiness and Type A or D personality link it to illness.

Who's who

Table 1.24 The people in this chapter.

Name	When	Examined for
Hermann Ebbinghaus	1850 to 1909	Memory measured with nonsense syllables and savings (On Memory, 1885); "Psychology has a long past, but only a short history" (1908)
Wilhelm Maximilian Wundt	1832 to 1920	First psychology laboratory, Leipzig (1879); Principles of Physiological Psychology (1873 to 1874); voluntarism; experimental introspection; Volkerpsychologie (folk psychology)
Aristotle	384 to 322 BCE	De Anima, the first treatise on the soul; empiricist; laws of association (similarity, contrast, contiguity)
William James	1842 to 1910	Father of American psychology; The Principles of Psychology (1890); stream of consciousness; habit; the self; inspiration of functionalism
William McDougall	1871 to 1938	Defined psychology as the science of conduct or behaviour (1905, 1908)
John Broadus Watson	1878 to 1958	Founder of behaviourism (1913); Angell's student at Chicago
Robert Sessions Woodworth	1869 to 1962	S-O-R formula; Dynamic Psychology (1918); psychology as the activities of the individual in relation to the environment
Norman L. Munn	Twentieth century	"The scientific study of behaviour", the definition credited to him in Indian textbooks
Plato	About 427 to 347 BCE	Nativist and rationalist; learning as recollection of inborn knowledge
Rene Descartes	1596 to 1650	Dualism; interaction through the pineal gland; innate ideas; the reflex
John Locke	1632 to 1704	Tabula rasa (1690); sensation and reflection; simple and complex ideas; primary and secondary qualities
David Hartley	1705 to 1757	Founder of associationism as a doctrine (1749)
James Mill	1773 to 1836	Mental mechanics: a complex idea is the sum of its parts
John Stuart Mill	1806 to 1873	Mental chemistry: ideas fuse into a new whole
Thomas Hobbes	1588 to 1679	Materialism; forerunner of British empiricism
George Berkeley	1685 to 1753	To be is to be perceived (esse est percipi)
David Hume	1711 to 1776	Laws of association: resemblance, contiguity, cause and effect
Alexander Bain	1818 to 1903	Associationism tied to the nervous system; founded the journal Mind (1876)
Gottfried Wilhelm Leibniz	1646 to 1716	Monads; petites perceptions; coined apperception; psychophysical parallelism
Johann Friedrich Herbart	1776 to 1841	Threshold of consciousness; apperceptive mass
Immanuel Kant	1724 to 1804	Innate categories; held that psychology could not be a science
Thomas Henry Huxley	1825 to 1895	Epiphenomenalism: mind as a by-product of the brain
Baruch Spinoza	1632 to 1677	Double-aspect theory of mind and body
Charles Bell	1774 to 1842	Bell-Magendie law (1811)
Francois Magendie	1783 to 1855	Bell-Magendie law (1822): dorsal roots sensory, ventral roots motor
Johannes Muller	1801 to 1858	Doctrine of specific nerve energies; Handbook of Human Physiology (1833 to 1840)

Name	When	Examined for
Hermann von Helmholtz	1821 to 1894	Speed of the nerve impulse (1850); unconscious inference; trichromatic theory of colour vision
Thomas Young	1773 to 1829	Suggested three colour receptors (1802): the Young-Helmholtz theory
Franz Joseph Gall	1758 to 1828	Founder of phrenology
Johann Gaspar Spurzheim	1776 to 1832	Popularised phrenology under that name
Pierre Flourens	1794 to 1867	Ablation studies of the brain (1820s)
Paul Broca	1824 to 1880	Speech production area in the left frontal lobe; patient Tan (1861)
Leborgne ("Tan")	1861	Broca's patient who understood speech but could say little more than "tan"
Carl Wernicke	1848 to 1905	Area for understanding speech in the left temporal lobe (1874)
Gustav Fritsch	1838 to 1927	Motor cortex mapped by electrical stimulation, with Hitzig (1870)
Eduard Hitzig	1838 to 1907	Motor cortex mapped by electrical stimulation, with Fritsch (1870)
Charles Darwin	1809 to 1882	Natural selection (1859); The Expression of the Emotions (1872); continuity, function and variation
Francis Galton	1822 to 1911	Individual differences; Hereditary Genius (1869); anthropometric laboratory (1884); correlation; questionnaire; word association; twin studies; eugenics
George John Romanes	1848 to 1894	Animal Intelligence (1882); anecdotal comparative psychology
Conwy Lloyd Morgan	1852 to 1936	Morgan's canon of parsimony (1894)
Ernst Heinrich Weber	1795 to 1878	Just noticeable difference as a constant fraction of the standard (Weber's law)
Gustav Theodor Fechner	1801 to 1887	Founder of psychophysics; insight of 22 October 1850; Elements of Psychophysics (1860); S = k log I
Franciscus Cornelis Donders	1818 to 1889	Subtraction method (1868); mental chronometry
Emil Kraepelin	1856 to 1926	Worked in Wundt's laboratory; classification of psychiatric disorders
Charles Spearman	1863 to 1945	Leipzig doctorate; factor analysis
Vladimir Mikhailovich Bekhterev	1857 to 1927	Studied with Wundt; reflexology
Edward Bradford Titchener	1867 to 1927	Structuralism (1898) at Cornell; elements and attributes; stimulus error; context theory of meaning; the Experimentalists (1904)
Oswald Kulpe	1862 to 1915	Wurzburg school; imageless thought; Aufgabe and set
Narziss Ach	1871 to 1946	Wurzburg: the determining tendency and set (1905)
Franz Brentano	1838 to 1917	Act psychology; intentionality (1874)
Carl Stumpf	1848 to 1936	Tone psychology; teacher of the Gestalt founders
Georg Elias Muller	1850 to 1934	Gottingen laboratory; memory research
Carl Lange	1834 to 1900	Theory of emotion (1885), independently of James: the James-Lange theory
John Dewey	1859 to 1952	"The Reflex Arc Concept in Psychology" (1896); Chicago functionalism
James Rowland Angell	1869 to 1949	"The Province of Functional Psychology" (1906); taught Watson
Harvey A. Carr	1873 to 1954	The adaptive act; mature Chicago functionalism
James McKeen Cattell	1860 to 1944	Coined "mental test" (1890); reaction time and individual differences; Columbia
Edward Lee Thorndike	1874 to 1949	Animal learning at Columbia; law of effect
Granville Stanley Hall	1844 to 1924	First American PhD in psychology (1878); first American research laboratory (1883); American Journal of Psychology (1887); first APA president (1892); adolescence as storm and stress; recapitulation
Lightner Witmer	1867 to 1956	First psychological clinic (1896); first used "clinical psychology" in print (1907); father of clinical psychology
Hugo Munsterberg	1863 to 1916	Founder of forensic and industrial psychology; Wundt's student
Mary Whiton Calkins	1863 to 1930	First woman APA president (1905); refused a Harvard doctorate
Margaret Floy Washburn	1871 to 1939	First woman PhD in psychology (1894); The Animal Mind (1908)
Leta Stetter Hollingworth	1886 to 1939	Rejected claims of female inferiority with experimental evidence
Thomas Carlyle	1795 to 1881	Great-man (great-person) theory of history (1841)
Edwin Garrigues Boring	1886 to 1968	Historian of experimental psychology; champion of the Zeitgeist view
Alfred Russel Wallace	1823 to 1913	Natural selection, independently of Darwin (1858): a multiple discovery

Name	When	Examined for
Alfred Binet	1857 to 1911	First practical intelligence test (1905); father of intelligence testing
Narendra Nath Sengupta	1889 to 1944	First Indian laboratory (1915) and department (1916) at Calcutta; father of modern psychology in India
Girindrasekhar Bose	1887 to 1953	Indian Psychoanalytic Society (1922); father of psychoanalysis in India
M. V. Gopalaswamy	1924	Founded the psychology department at the University of Mysore
Sigmund Freud	1856 to 1939	Founder of psychoanalysis; free association; the unconscious mind
Carl Jung	1875 to 1961	Used word association as a clinical technique
Max Wertheimer	1880 to 1943	Founder of Gestalt psychology (1912)
Ivan Pavlov	1849 to 1936	Classical conditioning
Wilhelm Roentgen	1895	Physicist who discovered X-rays (1895); a distractor for folk psychology
Abraham Maslow	1908 to 1970	Humanistic psychology
Carl Rogers	1902 to 1987	Humanistic psychology
Burrhus Frederic Skinner	1904 to 1990	Operant conditioning; radical behaviourism
Jerome Bruner	1915 to 2016	"New Look" research on perception
Roy Schafer	1940s	"New Look" research on perception; a distractor for Titchener's definition of perception
Gregory Kimble	1961	Learning theorist (the standard definition of learning, 1961); a distractor for Titchener's definition of perception
Erik Erikson	1902 to 1994	Psychosocial stages; a distractor for "clinical psychology" in print
Karen Horney	1885 to 1952	Psychoanalyst; never APA president, a distractor for Calkins
Thomas Samuel Kuhn	1922 to 1996	Paradigm, paradigm shift and preparadigmatic science (The Structure of Scientific Revolutions, 1962)
Robert Irving Watson	1909 to 1980	Historian who named the prescriptions, the enduring issues of psychology (1967)
Ivan Petrovich Pavlov	1849 to 1936	Conditioned reflexes, a forerunner of behaviourism
Rosalie Rayner	1898 to 1935	Little Albert study with Watson (1920)
Percy Williams Bridgman	1882 to 1961	Operationism (1927), the basis of the operational definition
Edward Chace Tolman	1886 to 1959	Purposive behaviourism; molar behaviour; intervening variables; cognitive map; latent learning
Charles Honzik	1930 (study)	Latent learning experiment with Tolman
Clark Leonard Hull	1884 to 1952	Hypothetico-deductive system; drive reduction; reaction potential = habit strength × drive
John Dollard	1950 (with Miller)	Early social learning theory: Freud recast in Hull's terms
Neal Elgar Miller	1950 (with Dollard)	Early social learning theory with Dollard
Edwin Ray Guthrie	1886 to 1959	Contiguity; one-trial learning
Zing-Yang Kuo	1898 to 1970	Behaviourist who rejected the concept of instinct
Harry Harlow	1905 to 1981	Learning sets (a wrong option for the cognitive map)
James L. Gould	1986 (honeybee study)	Tested Tolman's cognitive map idea in honeybees; did not propose it
Joseph Wolpe	1915 to 1997	Systematic desensitisation (1958)
Norman Leslie Munn	Modern	Textbook author who popularised "the scientific study of behaviour"; keyed in one Panwar item as first to establish it (prefer Watson if offered)
Albert Bandura	1925 to 2021	Self-efficacy (1977); later social learning theory with Walters (1963)
Richard Walters	1963 (with Bandura)	Later social learning theory
Wolfgang Kohler	1887 to 1967	Insight in Sultan the chimpanzee on Tenerife (1913 to 1920); transposition; isomorphism
Kurt Koffka	1886 to 1941	Gestalt co-founder; geographical versus behavioural environment; Principles of Gestalt Psychology (1935)
Kurt Lewin	1890 to 1947	Field theory; life space; B = f(P, E); group dynamics; action research; father of modern social psychology
Bluma Zeigarnik	1900 to 1988	Better recall of interrupted tasks (1927)
Maria Ovsiankina	1898 to 1993	Spontaneous resumption of interrupted tasks (1928)

Name	When	Examined for
Frederick Salomon Perls	1893 to 1970	Fritz Perls: Gestalt therapy, here-and-now awareness, the empty chair
Laura Perls	1940s and 1950s	Co-developer of Gestalt therapy
Paul Goodman	1951 (Gestalt Therapy)	Co-developer of Gestalt therapy
Jean-Martin Charcot	1825 to 1893	Showed Freud hysterical symptoms produced and removed under hypnosis (Paris, 1885)
Pierre Janet	1859 to 1947	Explained hysteria as dissociation
Hippolyte Bernheim	1840 to 1919	Nancy school: hypnosis as suggestion
Josef Breuer	1842 to 1925	Anna O. and the cathartic method; Studies on Hysteria with Freud (1895)
Anna O. (Bertha Pappenheim)	Treated 1880 to 1882	Breuer's patient; the "talking cure"
Alfred Adler	1870 to 1937	Individual psychology
Carl Gustav Jung	1875 to 1961	Analytical psychology; the collective unconscious
Erich Fromm	1900 to 1980	Humanistic psychoanalysis
Harry Stack Sullivan	1892 to 1949	Interpersonal psychiatry
Erik Homburger Erikson	1902 to 1994	Psychosocial stages
Anna Freud	1895 to 1982	Ego psychology; defence mechanisms
Melanie Klein	1882 to 1960	Object relations
Abraham Harold Maslow	1908 to 1970	Named the third force; self-actualisation; foresaw the fourth force (1968)
Carl Ransom Rogers	1902 to 1987	Actualising tendency; unconditional positive regard
Charlotte Buhler	1893 to 1974	Humanistic founder; lifespan view
Rollo May	1909 to 1994	Brought existentialism to American psychology; normal versus neurotic anxiety (1950)
Edmund Husserl	1859 to 1938	Phenomenology; bracketing
Irvin David Yalom	Born 1931	Four ultimate concerns: death, freedom, isolation, meaninglessness (1980)
Soren Kierkegaard	1813 to 1855	Anxiety as the dizziness of freedom
Martin Heidegger	1889 to 1976	Dasein, being-in-the-world (1927)
Jean-Paul Sartre	1905 to 1980	Existence precedes essence
Ludwig Binswanger	1881 to 1966	Daseinsanalysis
Medard Boss	1903 to 1990	Daseinsanalysis
Viktor Emil Frankl	1905 to 1997	Logotherapy; will to meaning; existential vacuum; noogenic neurosis
Martin Elias Peter Seligman	Born 1942	Founded positive psychology as President of the American Psychological Association (1998)
Mihaly Csikszentmihalyi	1934 to 2021	Co-founder of positive psychology; flow
C. R. Snyder	Modern	Hope theory: the will and the ways to reach a goal
Howard Gardner	Born 1943	Multiple intelligences (a wrong option for the founder of positive psychology)
Steven Pinker	Born 1954	Language and evolutionary psychology (The Language Instinct, 1994); a wrong option for the founder of positive psychology
Karl Spencer Lashley	1890 to 1958	Serial order in behaviour needs central plans (1948)
Allen Newell	1927 to 1992	Logic Theorist with Simon (1956)
Herbert Alexander Simon	1916 to 2001	Logic Theorist with Newell (1956)
Noam Chomsky	Born 1928	Formal models of language (1956); review of Skinner's Verbal Behavior (1959)
George Armitage Miller	1920 to 2012	"The magical number seven, plus or minus two" (1956)
Donald Eric Broadbent	1926 to 1993	First major information-processing theory (1958)
Ulric Neisser	1928 to 2012	Father of cognitive psychology; Cognitive Psychology (1967)
Frederic Charles Bartlett	1886 to 1969	Schemas (1932); forerunner of cognitive psychology
Jean Piaget	1896 to 1980	Studied children's thinking from the 1920s; pioneer of cognitive psychology
David Rumelhart	1942 to 2011	Parallel distributed processing with McClelland (1986)
James McClelland	Born 1948	Parallel distributed processing with Rumelhart (1986)

Name	When	Examined for
Albert Ellis	1913 to 2007	Rational emotive behaviour therapy
Aaron Temkin Beck	1921 to 2021	Cognitive therapy; cognitive distortions
Anne Treisman	1935 to 2018	Attenuation work from about 1960 (a wrong option for the first information-processing theory)
Lawrence Kohlberg	1981 (The Philosophy of Moral Development)	Moral development (a wrong option for the pioneer of cognitive psychology)
George Libman Engel	1913 to 1999	Biopsychosocial model (1977)
Asutosh Mookerjee	1864 to 1924	Calcutta vice-chancellor (garbled as "Amitosh Mukherjee" in one item); not a founder of psychology
Durganand Sinha	1922 to 1998	Indigenisation of psychology in India
Uma Chowdhury	About 1960	Indian adaptation of the Thematic Apperception Test (also spelled Chaudhuri; probably the "Kamala Chaudhuri" of one item)
Jai B. P. Sinha	1980	Nurturant-task leader
Udai Pareek	1925 to 2010	Role stress
Sudhir Kakar	Born 1938	Psychoanalysis of Indian childhood and society
S. K. Mangal	Modern	Indian textbook author; did not found a laboratory
A. K. Singh	Modern	Indian textbook author; did not found a laboratory
G. S. Rathi	Modern	A wrong option for India's first laboratory
Matthijs Cornelissen	Early 2000s	Led the Indian Psychology movement; founded the Indian Psychology Institute
Koneru Ramakrishna Rao	Born 1932	Co-editor of the Handbook of Indian Psychology (2008); probably the "G. R. S. Rao" of one item
Anand C. Paranjpe	2008 (Handbook)	Co-editor of the Handbook of Indian Psychology
Ajit K. Dalal	2008 (Handbook)	Co-editor of the Handbook of Indian Psychology
Girishwar Misra	2011	Co-editor of Foundations of Indian Psychology
Suneet Varma	2011	Co-editor of Foundations of Indian Psychology
Kapila	Ancient (traditional)	Traditional founder of Samkhya
Ishvarakrishna	About fourth or fifth century CE	Samkhya Karika
Patanjali	Between about the second century BCE and the fourth century CE	Yoga Sutras; chitta-vritti-nirodha; the eight limbs
Adi Shankaracharya	Eighth century CE	Advaita Vedanta: atman is Brahman; the world is maya
Siddhartha Gautama (the Buddha)	Traditionally 563 to 483 BCE	Four Noble Truths; Eightfold Path; anatta
Sri Aurobindo	1872 to 1950	Integral yoga; the psychic being
Marsha Linehan	Born 1943	Dialectical behaviour therapy, with Zen mindfulness
Jerome Seymour Bruner	1915 to 2016	New Look in perception (Section 1.4); distractor for Titchener's definition
Johannes Peter Muller	1801 to 1858	Doctrine of specific nerve energies
Eugene Galanter	1962	Classic absolute threshold examples (a candle flame 48 km away, a watch tick at 6 m)
Stanley Smith Stevens	1906 to 1973	Power law of sensation (1957); magnitude estimation; cross-modality matching
Wilhelm Wundt	1832 to 1920	Father of experimental psychology; distractor for the founder of psychophysics
David M. Green	1966	Signal detection theory, with Swets (1966)
John A. Swets	1966	Signal detection theory, with Green (1966)
Jan Evangelista Purkyne (Purkinje)	1787 to 1869	Purkinje shift: reds darken before blues at dusk (1825)
Ernst Mach	1838 to 1916	Mach bands (1865), explained by lateral inhibition
Ewald Hering	1834 to 1918	Opponent-process theory of colour vision (1878)

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Shinobu Ishihara	1879 to 1963	Ishihara colour vision plates (1917)
Georg von Bekesy	1899 to 1972	Travelling wave on the basilar membrane, place theory of pitch; Nobel Prize 1961
William Rutherford	1839 to 1899	Frequency (telephone) theory of pitch (1886)
Ernest Rutherford	1871 to 1937	Physicist wrongly credited with the frequency theory of pitch in some answer keys
Ernest Glen Wever	1902 to 1991	Volley principle of pitch (1930, with Bray)
Charles William Bray	born 1904	Volley principle of pitch (1930, with Wever)
Kikunae Ikeda	1864 to 1936	Named umami, the fifth basic taste (1908)
Hans Henning	1916	Smell prism (six primary odours) and taste tetrahedron (four tastes)
Ronald Melzack	1929 to 2019	Gate control theory of pain (1965, with Wall)
Patrick David Wall	1925 to 2001	Gate control theory of pain (1965, with Melzack)
Silas Weir Mitchell	1829 to 1914	Coined the term phantom limb (1871)
Vilayanur Subramanian Ramachandran	born 1951	Phantom limbs explained by cortical reorganisation (1990s)
Harry McGurk	1976	McGurk effect, with MacDonald (1976)
John MacDonald	1976	McGurk effect, with McGurk (1976)
Donald Olding Hebb	1904 to 1985	McGill sensory deprivation studies in his laboratory
William Bexton	1954	McGill sensory deprivation study (Bexton, Heron and Scott, 1954)
Woodburn Heron	1954	McGill sensory deprivation study (Bexton, Heron and Scott, 1954)
Thomas Scott	1954	McGill sensory deprivation study (Bexton, Heron and Scott, 1954)
Charles Bonnet	1720 to 1793	Charles Bonnet syndrome: visual hallucinations with insight in people with sight loss (1760)
Richard Langton Gregory	1923 to 2010	Constructivist approach and perceptual hypotheses (Eye and Brain, 1966); inappropriate constancy scaling (1963); studied S.B. (1963)
James Jerome Gibson	1904 to 1979	Direct (ecological) perception, the classic bottom-up theory; affordances; optic flow
Wolfgang Köhler	1887 to 1967	Gestalt co-founder
Edgar Rubin	1886 to 1951	Figure and ground (1915); Rubin's vase
Eleanor Jack Gibson	1910 to 2002	Visual cliff (with Walk, 1960)
Richard Walk	1920 to 1999	Visual cliff (with Eleanor Gibson, 1960)
Emil Emmert	1844 to 1911	Emmert's law of afterimage size (1881)
Franz Carl Müller-Lyer	1857 to 1916	Müller-Lyer illusion (1889)
Marshall Segall	1963, 1966	Carpentered-world hypothesis (with Campbell and Herskovits)
Donald Thomas Campbell	1916 to 1996	Carpentered-world hypothesis (with Segall and Herskovits)
Melville Jean Herskovits	1895 to 1963	Carpentered-world hypothesis (with Segall and Campbell)
Mario Ponzo	1882 to 1960	Ponzo illusion (1911)
Adelbert Ames Jr.	1880 to 1955	Ames room
Johann Karl Friedrich Zöllner	1834 to 1882	Zöllner illusion (1860)
Johann Christian Poggendorff	1796 to 1877	Poggendorff illusion (1860)
Louis Albert Necker	1786 to 1861	Necker cube, an ambiguous figure (1832)
Lionel Penrose	1898 to 1972	Penrose triangle, an impossible figure (1958, with his son Roger)
Roger Penrose	born 1931	Penrose triangle (1958, with his father Lionel)
Muzafer Sherif	1906 to 1988	Used the autokinetic effect to study norm formation (1935)
Oliver Selfridge	1926 to 2008	Pandemonium model of feature analysis (1959)
David Hubel	1926 to 2013	Feature detectors in cat visual cortex (with Wiesel, from 1959)
Torsten Wiesel	born 1924	Feature detectors in cat visual cortex (with Hubel, from 1959)
Eleanor Rosch	born 1938	Prototypes (1970s)
Irving Biederman	1987	Recognition-by-components theory; about 36 geons
David Marr	1945 to 1980	Computational theory: primal sketch, 2.5-D sketch, 3-D model
Gerald Reicher	1969	Word superiority effect
Leigh Minturn	1955	B or 13 context effect (with Bruner)

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Cecile Goodman	1947	Coin study: economic deprivation affects perception (with Bruner)
Elliott McGinnies	1949	Perceptual defence for taboo words
George Malcolm Stratton	1865 to 1957	Inverting lenses (1896, 1897): perceptual adaptation
Jean Wallace	1963	Studied the restored-sight patient S.B. with Gregory
Herman Witkin	1916 to 1979	Field dependence-independence; rod-and-frame and embedded figures tests
James Vicary	1915 to 1977	Subliminal advertising hoax (1957, admitted 1962)
Anthony Marcel	1983	Masked semantic priming without awareness
Lawrence Weiskrantz	1926 to 2018	Blindsight (1974)
Joseph Banks Rhine	1895 to 1980	Parapsychology at Duke; Zener-card studies
Karl Zener	1903 to 1964	Zener cards
K. D. Broota	1970s	Indian work on cultural differences in perceptual selectivity (with Ganguli)
H. C. Ganguli	1970s	Indian work on cultural differences in perceptual selectivity (with Broota)
William Stanley Jevons	1835 to 1882; 1871	Bean experiment on the span of attention
George Miller	1956	Magical number seven, plus or minus two: memory span, not span of attention (Section 1.7)
Colin Cherry	1914 to 1979; 1953	Cocktail party problem; dichotic listening and shadowing
Neville Moray	1959	Own name heard in the unattended ear; no memory for words repeated 35 times
Donald Broadbent	1926 to 1993; 1958	Filter model (early selection); first major information-processing theory
J. Anthony Deutsch and Diana Deutsch	1963	Late selection model
Donald Norman	born 1935; 1968	Pertinence model of late selection
William Johnston and Steven Heinz	1978	Multimode (flexible) theory of selection
Nilli Lavie	1994 to 1995	Perceptual load theory
Michael Posner	born 1936	Spotlight; cueing task (1980); endogenous and exogenous attention; automaticity criteria (with Charles Snyder, 1975)
Charles Snyder	1975	Criteria for automatic processing (with Posner)
Garry Gelade	1980	Feature integration theory (with Treisman)
Daniel Simons and Christopher Chabris	1999	The invisible gorilla (inattentional blindness)
Daniel Kahneman	1934 to 2024; 1973	Capacity model (Attention and Effort)
Christopher Wickens	1984	Multiple resource theory
Richard Shiffrin and Walter Schneider	1977	Automatic versus controlled processing
Patricia Devine	1989	Stereotypes activated automatically; low-prejudiced people refute them by controlled processing
John Ridley Stroop	1897 to 1973; 1935	The Stroop effect
Norman Mackworth	1948	Clock test; vigilance decrement
Jerome L. Singer	1924 to 2010	Research on daydreaming
B. F. Skinner	1904 to 1990	Operant conditioning (Section 1.6); not the coiner of stream of consciousness
Anthony Greenwald and colleagues	1998	Implicit Association Test (unconscious stereotyping)
David Chalmers	born 1966; 1995	The hard problem of consciousness
Charles Tart	born 1937; 1969	Altered states of consciousness
Hans Berger	1873 to 1941; 1929	First human EEG
Eugene Aserinsky	1921 to 1998; 1953	Discovered REM sleep with Kleitman
Nathaniel Kleitman	1895 to 1999	REM discovery (1953); sleep cycle with Dement (1957)

Name	When	Examined for
William Charles Dement	1928 to 2020	Sleep cycle (1957, with Kleitman); REM deprivation and REM rebound (1960)
Howard Roffwarg and colleagues	1966	Sleep and REM across the lifespan
Viktor Frankl	1905 to 1997	Paradoxical intention (logotherapy), used for insomnia (Chapter 7)
J. Allan Hobson and Robert McCarley	1977	Activation-synthesis hypothesis of dreams
Antti Revonsuo	2000	Threat-simulation theory of dreams
Stephen LaBerge	early 1980s	Verified lucid dreaming with eye-movement signals
Franz Anton Mesmer	1734 to 1815	Animal magnetism (1770s)
James Braid	1795 to 1860	Coined hypnotism; Neurypnology (1843)
Ambroise-Auguste Liébeault	1823 to 1904	Founder of the Nancy school
André Weitzenhoffer	1959	Stanford Hypnotic Susceptibility Scales (with Hilgard)
Ernest Ropiequet Hilgard	1904 to 2001	Stanford scales (1959); neodissociation theory and the hidden observer
Theodore Xenophon Barber	1927 to 2005	Social-cognitive view of hypnosis
Nicholas Spanos	1942 to 1994	Social-cognitive theory of hypnosis
Maharishi Mahesh Yogi	1918 to 2008	Transcendental meditation (1950s)
Herbert Benson	1935 to 2022; 1975	The relaxation response
Nikolaas Tinbergen	1907 to 1988	Fixed action patterns triggered by sign stimuli (stickleback)
Eric Kandel	Born 1929	Habituation and sensitisation traced to synapses in Aplysia
Robert Fantz	1964	Habituation method for studying infant discrimination
Konrad Lorenz	1903 to 1989	Imprinting in greylag goslings (1935); critical period
Shenger-Krestovnikova	1921	Experimental neurosis (circle and ellipse), in Pavlov's laboratory
W. J. Brogden	1939	Sensory preconditioning
Mary Cover Jones	1897 to 1987	Peter and the rabbit (1924); counterconditioning
Edmund Jacobson	1888 to 1983	Progressive muscle relaxation
Thomas Stampfl	1960s	Implosive therapy
Joseph Cautela	1967	Covert sensitisation
Victor Meyer	1966	Exposure and response prevention for obsessive-compulsive disorder
Robert Rescorla	1940 to 2020	Contingency (1968); Rescorla-Wagner model (1972)
Leon Kamin	1927 to 2017	Blocking (1969)
Robert Lubow	1959	Latent inhibition (with Moore)
A. U. Moore	1959	Latent inhibition (with Lubow)
Allan Wagner	1934 to 2018	Rescorla-Wagner model (1972)
Wolfram Schultz	1990s	Dopamine neurons of the ventral tegmental area signal reward prediction error
John Garcia	1917 to 2012	Taste aversion; Garcia and Koelling (1966); equipotentiality refuted
Robert Koelling	1966	Bright, noisy, tasty water (with Garcia)
Martin Seligman	Born 1942	Preparedness (1970); learned helplessness (1967, with Maier)
Teodoro Ayllon	1960s; book 1968	Token economy at Anna State Hospital (with Azrin)
Nathan Azrin	1960s; book 1968	Token economy (with Ayllon)
Orval Hobart Mowrer	1907 to 1982	Two-factor theory of avoidance (1947); bell-and-pad for bed-wetting (1938)
William Kaye Estes	1919 to 2011	Punishment suppresses rather than erases a response (1944); stimulus sampling theory (1950)
Charles Ferster	1957	Schedules of reinforcement (with Skinner)
David Premack	1925 to 2015	Premack principle (1959)
Steven Maier	1967	Learned helplessness (with Seligman)
Keller Breland	1961	Instinctive drift (with Marian Breland)
Marian Breland	1961	Instinctive drift (with Keller Breland)
Richard Herrnstein	1930 to 1994	Matching law (1961)
Leo Crespi	1942	Contrast effects (the Crespi effect)

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Neal Miller	1909 to 2002	Operant conditioning of autonomic responses and biofeedback; imitation (with Dollard, 1941)
Arnold Kegel	1948	Perineometer, an early form of biofeedback
Dorothea Ross	1961	Bobo doll studies (with Bandura and Sheila Ross)
Sheila Ross	1961	Bobo doll studies (with Bandura and Dorothea Ross)
Julian Rotter	1916 to 2014	Social learning theory (1954): behaviour potential, expectancy, reinforcement value
Robert Mills Gagné	1916 to 2002	Eight types of learning; The Conditions of Learning (1965)
William Lowe Bryan	1860 to 1955	Telegraphy learning curves and the plateau (with Harter, 1897, 1899)
Noble Harter	1897, 1899	The plateau in telegraphy learning (with Bryan)
William C. F. Krueger	1929	Overlearning improves retention
Edwin Locke	Born 1938	Goal setting: specific goals improve performance
Gary Latham	1970s onwards	Goal setting (with Locke)
Sidney Leavitt Pressey	1888 to 1979	The first teaching machine (1920s)
Clive Wearing	1985	British musician whose encephalitis (1985) left dense amnesia with spared musical skill
Raymond Nickerson	1979	Coin-drawing study showing encoding failure (with Marilyn Adams)
Marilyn Adams	1979	Coin-drawing study showing encoding failure (with Raymond Nickerson)
Richard C. Atkinson	1968	Multi-store (modal) model with Shiffrin; keyword method (1975)
Richard M. Shiffrin	1968	Multi-store (modal) model with Atkinson
George Sperling	born 1934	Partial-report experiments on iconic memory (1960)
Christopher Darwin	1972	Three-eared man study of echoic memory (with Turvey and Crowder)
Michael Turvey	1972	Three-eared man study of echoic memory (with Darwin and Crowder)
Robert Crowder	1972	Three-eared man study of echoic memory (with Darwin and Turvey)
Nelson Cowan	2001	Pure short-term memory capacity of about four chunks
John Brown	1958	Brown-Peterson technique for short-term memory duration
Lloyd Peterson	1959	Brown-Peterson technique: unrehearsed short-term memory lasts about 18 to 30 seconds
Margaret Peterson	1959	Brown-Peterson technique: unrehearsed short-term memory lasts about 18 to 30 seconds
Geoffrey Keppel	1962	Showed Brown-Peterson forgetting was interference, not decay (with Underwood)
Benton Underwood	1962	Showed Brown-Peterson forgetting was interference, not decay (with Keppel)
Reuben Conrad	1964	Acoustic confusions show acoustic coding in short-term memory
Alan Baddeley	born 1934	Acoustic versus semantic coding (1966); working memory model with Hitch (1974); episodic buffer (2000); divers study with Godden (1975)
Bennet Murdock	1962	Reliability of the serial position curve in free recall
Murray Glanzer	1966	A filled delay abolishes recency but spares primacy (with Cunitz)
Anita Cunitz	1966	A filled delay abolishes recency but spares primacy (with Glanzer)
Tim Shallice	1970	Patient K.F.: impaired short-term memory with normal long-term learning (with Warrington)
Elizabeth Warrington	1970	Patient K.F.: impaired short-term memory with normal long-term learning (with Shallice)
J. W. McCrary	1953	Keyed in RML 2024 (as McCray) for first observing the primacy-recency effect, with Hunter
W. S. Hunter	1953	Keyed in RML 2024 for first observing the primacy-recency effect, with McCrary
Graham Hitch	1974	Working memory model with Baddeley
Robert Logie	1995	Visual cache and inner scribe of the visuospatial sketchpad
Fergus Craik	1972	Levels of processing with Lockhart; Craik and Tulving (1975)
Robert Lockhart	1972	Levels of processing with Craik
T. B. Rogers	1977	Self-reference effect (with Kuiper and Kirker)
Nicholas Kuiper	1977	Self-reference effect (with Rogers and Kirker)

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W. S. Kirker	1977	Self-reference effect (with Rogers and Kuiper)
Donald Morris	1977	Transfer-appropriate processing (with Bransford and Franks)
John Bransford	1977	Transfer-appropriate processing (with Morris and Franks)
Jeffery Franks	1977	Transfer-appropriate processing (with Morris and Bransford)
Allan Paivio	1925 to 2016	Dual coding theory (1971): verbal and imagery codes
Endel Tulving	1927 to 2023	Episodic versus semantic memory (1972); encoding specificity with Thomson (1973); cue-dependent forgetting (1974); patient K.C.
Peter Graf	1985	Named explicit and implicit memory (with Schacter)
Daniel Schacter	1985	Named explicit and implicit memory (with Graf)
Larry Squire	1980	Declarative versus nondeclarative memory and their taxonomy
David Meyer	1971	Semantic priming in lexical decision, BREAD and BUTTER (with Schvaneveldt)
Roger Schvaneveldt	1971	Semantic priming in lexical decision, BREAD and BUTTER (with Meyer)
David Rubin	1986	Reminiscence bump (with colleagues); flashbulb versus everyday memories with Talarico (2003)
Roger Brown	1925 to 1997	Flashbulb memory with Kulik (1977); tip-of-the-tongue state with McNeill (1966)
James Kulik	1977	Flashbulb memory and the Now Print! mechanism (with Brown)
Nicole Harsch	1992	Challenger flashbulb memory study (with Neisser)
Jennifer Talarico	2003	9/11 flashbulb memories: confidence, not consistency (with Rubin)
John Flavell	1971	Coined metamemory (1971); metacognition
David McNeill	1966	Tip-of-the-tongue state (with Brown)
Allan Collins	1969	Hierarchical network model with Quillian (1969); spreading activation with Loftus (1975)
Ross Quillian	1969	Hierarchical network model and cognitive economy (with Collins)
Elizabeth Loftus	born 1944	Misinformation effect (Loftus and Palmer, 1974); spreading activation with Collins (1975); false memory implantation; critic in the recovered memory controversy
Roger Schank	1977	Scripts, such as the restaurant script (with Abelson)
Robert Abelson	1977	Scripts, such as the restaurant script (with Schank)
Gordon Allport	1947	Levelling and sharpening, from the study of rumour (with Postman)
Leo Postman	1947	Levelling and sharpening, from the study of rumour (with Allport)
John Palmer	1974	Smashed versus hit study of the misinformation effect (with Loftus)
Marcia Johnson	1993	Source monitoring framework (with colleagues)
Zena Pearlstone	1966	Category cues show available but inaccessible memories (with Tulving)
Donald Thomson	1973	Encoding specificity principle (with Tulving)
Duncan Godden	1975	Context-dependent memory in divers (with Baddeley)
Donald Goodwin	1969	State-dependent memory with alcohol
Gordon Bower	1932 to 2020	Mood-dependent and mood-congruent memory (1981)
John Jenkins	1924	Less forgetting after sleep than after waking (with Dallenbach)
Karl Dallenbach	1924	Less forgetting after sleep than after waking (with Jenkins)
Arthur Melton	1940	Two-factor theory of interference: response competition and unlearning (with Irwin)
J. M. Irwin	1940	Two-factor theory of interference (with Melton)
Michael Anderson	1994	Retrieval-induced forgetting (with Robert and Elizabeth Bjork)
Robert Bjork	1994	Retrieval-induced forgetting (with Anderson and Elizabeth Bjork)
Elizabeth Bjork	1994	Retrieval-induced forgetting (with Anderson and Robert Bjork)
Georg Elias Müller	1900	Consolidation (with Pilzecker)
Alfons Pilzecker	1900	Consolidation (with Müller)
Théodule Ribot	1881	Ribot's law: the new perishes before the old
Philip Ballard	1913	Reminiscence: recall rises after a delay without study
Hedwig von Restorff	1933	Von Restorff (isolation) effect: a distinctive item is remembered better
Arthur Gates	1917	Testing effect: recitation beats rereading

Name	When	Examined for
William Beecher Scoville	1953	Removed H.M.'s medial temporal lobes; reported the case with Milner (1957)
Henry Gustav Molaison	1926 to 2008	Patient H.M.: profound anterograde amnesia after bilateral medial temporal lobe removal, with spared procedural learning
Brenda Milner	1957	Studied H.M. (Scoville and Milner, 1957); mirror-tracing learning without awareness
Sergei Korsakoff	late 1880s	Described Korsakoff's syndrome
Richard Semon	1904	Coined the term engram
Timothy Bliss	1973	Discovered long-term potentiation (with Lømo)
Terje Lømo	1973	Discovered long-term potentiation (with Bliss)
Simonides of Ceos	about 556 to 468 BCE	Credited with the method of loci
Francis P. Robinson	1946	SQ3R study method
Roger Newland Shepard	1929 to 2022	Mental rotation (1971, with Jacqueline Metzler)
Jacqueline Metzler	1971	Mental rotation, with Shepard
Stephen Michael Kosslyn	born 1948	Image scanning (1978): images as depictive representations
Zenon Walter Pylyshyn	1937 to 2022	Propositional theory and the imagery debate (1973)
Eugen Bleuler	1857 to 1939	Autistic (dereistic) thinking, 1911
Robert Ennis	20th century	A widely used definition of critical thinking
Jacqueline Goodnow	1956	A Study of Thinking, with Bruner and Austin
George Austin	1956	A Study of Thinking, with Bruner and Goodnow
Ludwig Wittgenstein	1889 to 1951	Family resemblance (1953)
Mary Gick	1980	Analogical problem solving: the radiation problem, with Holyoak
Keith Holyoak	1980	Analogical problem solving: the radiation problem, with Gick
Robert Jeffrey Sternberg	born 1949	Problem-solving cycle; investment theory of creativity (1991, with Lubart); analytical, creative and practical intelligence
Norman Maier	1900 to 1977	Nine-dot problem (1930)
Janet Metcalfe	1987	Warmth ratings show insight solutions arrive suddenly (with Wiebe)
David Wiebe	1987	Warmth ratings, with Metcalfe
Karl Duncker	1903 to 1940	Candle problem and functional fixedness (1945)
Abraham Luchins	1914 to 2005	Water jars; mental set (Einstellung), 1942
Charles Sanders Peirce	1839 to 1914	Abductive reasoning, inference to the best explanation
Karl Raimund Popper	1902 to 1994	Hypothetico-deductive method
Robert Woodworth	1935	Atmosphere effect in syllogisms, with Sells
Saul Sells	1935	Atmosphere effect in syllogisms, with Woodworth
Peter Cathcart Wason	1924 to 2003	2-4-6 task (1960), selection task (1966), confirmation bias
Philip Nicholas Johnson-Laird	born 1936	Mental models theory of reasoning (1983)
Gerd Gigerenzer	born 1947	Fast and frugal heuristics; recognition heuristic; ecological rationality
Amos Tversky	1937 to 1996	Heuristics and biases (1972 to 1974), prospect theory (1979), framing (1981), with Kahneman
Baruch Fischhoff	born 1946	Hindsight bias, the knew-it-all-along effect (1975)
Joy Paul Guilford	1897 to 1987	Convergent and divergent thinking; 1950 APA address; fluency, flexibility, originality, elaboration
Graham Wallas	1858 to 1932	The Art of Thought (1926): preparation, incubation, illumination, verification
Ellis Paul Torrance	1915 to 2003	Torrance Tests of Creative Thinking (1966)
Sarnoff Mednick	1928 to 2015	Remote Associates Test (1962)
Edward de Bono	1933 to 2021	Lateral thinking (1967)
Alex Faickney Osborn	1888 to 1966	Brainstorming; Applied Imagination (1953)
Todd Lubart	1991	Investment theory of creativity, with Sternberg
Charles Francis Hockett	1916 to 2000	Design features of language (1960)
Dan Isaac Slobin	born 1939	Language-making capacity (1985)
Eric Heinz Lenneberg	1921 to 1975	Critical period for language (1967)

Name	When	Examined for
Genie	found 1970	Child isolated until 13: learned words but not grammar
Katherine Nelson	1973	Referential style of early word learning
Benjamin Lee Whorf	1897 to 1941	Linguistic relativity (the Sapir-Whorf hypothesis)
Edward Sapir	1884 to 1939	Whorf's teacher; the Sapir-Whorf hypothesis
Brent Berlin	1969	Basic colour terms in a near-universal order, with Kay
Paul Kay	1969	Basic colour terms, with Berlin
Lev Semyonovich Vygotsky	1896 to 1934	Thinking and Speech (1934): thought depends on language; private speech becomes inner speech; heaps, complexes, true concepts
David Wechsler	1896 to 1981	Aggregate or global capacity definition (1944); Wechsler-Bellevue (1939); deviation IQ; WAIS, WISC and WPPSI
Douglas Detterman	1986	With Sternberg, the 1986 expert symposium on defining intelligence
Carol Susan Dweck	Born 1946	Entity versus incremental theories of intelligence; fixed and growth mind-sets (Mindset, 2006); praise effort, not ability
Clark Wissler	1901	Showed that Cattell's mental tests did not predict college grades
Théodore Simon	1873 to 1961	Binet's collaborator on the 1905, 1908 and 1911 scales
Henry Herbert Goddard	1866 to 1957	Translated the Binet-Simon scale into English (1908); coined moron (1910)
William Stern	1871 to 1938	Coined the intelligence quotient (1912): mental age divided by chronological age
Lewis Madison Terman	1877 to 1956	Stanford-Binet (1916) and ratio IQ = MA ÷ CA × 100; gifted Termites study (from 1921)
Robert Mearns Yerkes	1876 to 1956	Army Alpha (verbal) and Army Beta (non-verbal) group tests (1917)
Gale Roid	2003	Stanford-Binet Fifth Edition (SB5): five factors, verbal and non-verbal, mean 100 and SD 15
Frank Schmidt and John Hunter	1998	General mental ability predicts job performance with a validity of about .5
John Carlyle Raven	1902 to 1970	Standard Progressive Matrices (1938): culture-fair test of fluid intelligence and the best single marker of g
Raymond Bernard Cattell	1905 to 1998	Fluid versus crystallised intelligence (1963); investment theory; Culture Fair Intelligence Test; the 16PF (personality)
V. V. Kamat	1930s	Binet-Kamat Test, a heavily verbal Indian adaptation of the Stanford-Binet
Arthur J. Malin	1969	Malin's Intelligence Scale for Indian Children (MISIC), a half-verbal adaptation of the WISC
C. M. Bhatia	1955	Bhatia's Battery of Performance Tests of Intelligence: five performance subtests, no reading, rural and urban norms
J. B. P. Sinha	1980	Nurturant-task leader (organisational psychology); a distractor, not the author of Bhatia's Battery
Charles Edward Spearman	1863 to 1945	Two-factor theory (1904): general factor g and specific factors s; positive manifold; first factor analysis
Louis Leon Thurstone	1887 to 1955	Seven primary mental abilities (1938)
Philip Ewart Vernon	1905 to 1987	Hierarchical model (1950): g, then the major group factors v:ed and k:m
John Bissell Carroll	1916 to 2003	Three-stratum theory (1993): g, eight broad and about 70 narrow abilities
Kevin McGrew	Late 1990s	Merged the Cattell-Horn and Carroll models into the CHC theory
John Leonard Horn	1928 to 2006	Developed the Gf-Gc theory with Cattell; extended it to about nine broad abilities
Godfrey Hilton Thomson	1881 to 1955	Sampling theory (1916): test correlations arise from overlapping samples of many small bonds
Arthur Robert Jensen	1923 to 2012	Level I (associative) and Level II (conceptual) abilities; 1969 claim that group IQ gaps are partly genetic
Jagannath Prasad Das	Born 1931	PASS model with Naglieri and Kirby (1994); Cognitive Assessment System
Jack A. Naglieri	1994	Co-author of the PASS model and the Cognitive Assessment System
John R. Kirby	1994	Co-author of the PASS model
Alexander Romanovich Luria	1902 to 1977	Functional units of the brain, the basis of the PASS model

Name	When	Examined for
Stephen J. Ceci	1990	Bioecological view: cognitive potentials develop only in supporting contexts
Peter Salovey	1990	With Mayer, emotional intelligence (1990); four-branch ability model; MSCEIT
John D. Mayer	1990	With Salovey, emotional intelligence (1990); four-branch ability model; MSCEIT
Daniel Goleman	Born 1946	Popularised emotional intelligence (1995); five-part mixed model
Reuven Bar-On	1997	Emotional Quotient Inventory (EQ-i), a self-report mixed model; the term EQ
Danah Zohar	2000	Spiritual intelligence
Richard Haier	1988	PET study: high scorers used less brain glucose (neural efficiency)
Thomas J. Bouchard Jr.	1981; from 1979	Review of IQ correlations among relatives with McGue (1981); Minnesota Study of Twins Reared Apart
Matt McGue	1981	With Bouchard, the 1981 review of IQ correlations among relatives
Eric Turkheimer	2003	Heritability of IQ near .10 in the poorest families: gene-environment interaction
Richard Lewontin	1929 to 2021	Seed analogy (1970): heritability within groups says nothing about differences between groups
Irving Gottesman	1930 to 2016	Reaction range (1963): genes set limits, environment decides where within them a person lands
James Robert Flynn	1934 to 2020	The Flynn effect: average IQ scores rose about 3 points per decade (1984)
Richard J. Herrnstein	1930 to 1994	The Bell Curve (1994) with Murray
Charles Murray	Born 1943	The Bell Curve (1994) with Herrnstein
Claude Steele	1995	Stereotype threat, with Aronson (1995)
Joshua Aronson	1995	Stereotype threat, with Steele (1995)
Edgar Arnold Doll	1889 to 1968	Vineland Social Maturity Scale (1935): social age and social quotient
Joseph Renzulli	Born 1936	Three-ring conception of giftedness (1978): above-average ability, task commitment, creativity
Henry Alexander Murray	1893 to 1988	Needs and press (Explorations in Personality, 1938); viscerogenic and psychogenic needs; TAT with Morgan (1935)
Walter Bradford Cannon	1871 to 1945	Coined homeostasis (1926; The Wisdom of the Body, 1932); stomach contractions and hunger with Washburn (1912)
Claude Bernard	1813 to 1878	The milieu intérieur, the constant internal environment
Leon Festinger	1919 to 1989	Cognitive dissonance (1957)
Harry Frederick Harlow	1905 to 1981	Puzzle-solving monkeys and the manipulation drive (1950); Maslow's doctoral adviser
John Dillingham Dodson	1879 to 1955	Yerkes-Dodson law (1908)
Robert Zajonc	1923 to 2008	Drive theory of social facilitation (1965): arousal strengthens the dominant response
Marvin Zuckerman	1928 to 2018	Sensation seeking
Otto Tinklepaugh	1928	Monkeys expecting banana refused lettuce: incentive theory
Julian Bernard Rotter	1916 to 2014	Social learning theory: expectancy and reinforcement value (1954); locus of control (1966)
Richard Lester Solomon	1918 to 1995	Opponent-process theory of acquired motivation (with Corbit, 1974)
John Corbit	1974	Opponent-process theory with Solomon
Kurt Goldstein	1878 to 1965	Coined the term self-actualisation (1939)
Clayton Paul Alderfer	1940 to 2015	ERG theory (1969): existence, relatedness, growth; frustration-regression
David Clarence McClelland	1917 to 1998	Need for achievement (1953), power and affiliation; the Kakinada training studies
Edward L. Deci	born 1942	Self-determination theory with Ryan (1985); cognitive evaluation theory
Richard M. Ryan	born 1953	Self-determination theory with Deci (1985): autonomy, competence, relatedness
William Glasser	1925 to 2013	Reality therapy: five basic needs, including power or achievement
Christiana Morgan	1897 to 1967	Thematic Apperception Test with Murray (1935)

Name	When	Examined for
John William Atkinson	1923 to 2003	Achievement motivation theory (1957): motive to succeed versus motive to avoid failure; preference for intermediate risk
Matina Souretis Horner	born 1939	Fear of success (1972)
Bernard Weiner	born 1935	Attribution theory of achievement: locus, stability, controllability
Daniel Ellis Berlyne	1924 to 1976	Curiosity: novelty, complexity, incongruity (1960)
Robert White	1904 to 2001	Effectance or competence motivation (1959)
Lev Vygotsky	1896 to 1934	Constructivism (a wrong option for reciprocal causation)
Frederick Herzberg	1923 to 2000	Two-factor (motivator-hygiene) theory (1959)
Stanley Schachter	1922 to 1997	Externality hypothesis of obesity (1968 to 1971); two-factor theory of emotion with Singer
Jerome Singer	1962	Two-factor theory of emotion with Schachter
Mark Lepper	1973	Overjustification effect: the magic-marker study with Greene and Nisbett
David Greene	1973	Magic-marker study with Lepper and Nisbett
Richard Nisbett	born 1941	Magic-marker study with Lepper and Greene (1973)
Roger Barker	1903 to 1990	Regression after frustration: toys behind a screen (with Dembo and Lewin, 1941)
Tamara Dembo	1941	Regression study with Barker and Lewin
Leonard Doob	1909 to 2000	Co-author of Frustration and Aggression (1939)
Robert Sears	1908 to 1989	Co-author of Frustration and Aggression (1939)
Leonard Berkowitz	1926 to 2016	Reformulated frustration-aggression hypothesis (1989): negative affect
A. L. Washburn	1912	Swallowed the balloon in Cannon's hunger study
John P. J. Pinel	2021 (Biopsychology, with Barnes)	Positive-incentive view of eating; settling point
Steven J. Barnes	2021 (Biopsychology, with Pinel)	Positive-incentive view of eating; settling point
Albert Hetherington	1940 (with Ranson)	VMH lesions cause hyperphagia and obesity
Stephen Walter Ranson	1880 to 1942	VMH lesions cause hyperphagia and obesity (with Hetherington, 1940)
Bal Krishan Anand	1951 (with Brobeck)	Lateral hypothalamic lesions cause aphagia: the feeding centre
John Brobeck	1951 (with Anand)	Lateral hypothalamic lesions cause aphagia
Eliot Stellar	1919 to 1993	Dual-centre theory of hunger (1954)
Alfred Charles Kinsey	1894 to 1956	Reports on male (1948) and female (1953) sexual behaviour
William Howell Masters	1915 to 2001	Sexual response cycle: excitement, plateau, orgasm, resolution (1966)
Virginia Eshelman Johnson	1925 to 2013	Sexual response cycle with Masters (1966)
Helen Singer Kaplan	1929 to 1995	Triphasic model: desire, excitement, orgasm (1979)
Carl Warden	1931	Columbia obstruction box: maternal drive strongest
Victor Harold Vroom	born 1932	Expectancy (VIE) theory of work motivation (1964)
John Stacy Adams	1963	Equity theory
Carol Dweck	born 1946	Attribution retraining of helpless children (1975)
William Richard Miller	born 1947	Motivational interviewing (with Rollnick, 1991)
Stephen Rollnick	1991 (with Miller)	Motivational interviewing
Paul Ekman	born 1934	Six universal basic emotions (contempt added 1986); Fore study (1971); Facial Action Coding System (1978)
Wallace V. Friesen	1971; 1978	With Ekman: Fore universality study and the Facial Action Coding System
Silvan Tomkins	1962	Affect theory: nine innate affects; the face as the seat of affect, root of the facial feedback hypothesis
Carroll Izard	1977	Differential emotions theory: ten basic emotions, including shame and guilt
Robert Plutchik	1980	Wheel of eight primary emotions and their dyads
James Russell	1980	Circumplex model: emotions on valence and arousal
Alexander Thomas		With Chess: easy, difficult and slow-to-warm-up temperaments (pointer)

Name	When	Examined for
Stella Chess		With Thomas: easy, difficult and slow-to-warm-up temperaments (pointer)
Carl Georg Lange	1834 to 1900	James-Lange theory of emotion (1885)
Philip Bard	1898 to 1977	Cannon-Bard theory; sham rage in decorticate cats
Donald Dutton	1974	Capilano bridge study: misattribution of arousal
Arthur Aron	1974	Capilano bridge study: misattribution of arousal
Dolf Zillmann	1970s	Excitation transfer theory
Richard S. Lazarus	1922 to 2002	Cognitive appraisal theory (primary and secondary appraisal); daily hassles (1981); transactional model of stress (1984)
Fritz Strack	1988	Pen study supporting the facial feedback hypothesis
Leonard Martin	1988	Pen study of facial feedback, with Strack and Stepper
Sabine Stepper	1988	Pen study of facial feedback, with Strack and Martin
Richard Solomon	1918 to 1995	Opponent-process theory of emotion (1974)
James Papez	1883 to 1958	Papez circuit of emotion (1937)
Paul MacLean	1913 to 2007	Named the limbic system (1952), the classic seat of emotion
Heinrich Klüver	1897 to 1979	Klüver-Bucy syndrome after bilateral temporal lobe removal in monkeys (1937, 1939)
Paul Bucy	1904 to 1992	Klüver-Bucy syndrome (1937, 1939)
Joseph LeDoux	born 1949	Low road and high road to the amygdala in fear
Larry Cahill	1994	With McGaugh: propranolol abolishes the emotional enhancement of memory
James McGaugh	1994	With Cahill: the amygdala and noradrenaline in emotional memory
H.M. (Henry Molaison)	operated on 1953; died 2008	Amnesic patient: the hippocampus, not the amygdala, stores facts
Phineas Gage	1823 to 1860	Prefrontal injury (1848) that changed his emotional and social conduct
Antonio Damasio	born 1944	Somatic marker hypothesis (1994)
David T. Lykken	1959	Guilty knowledge test, more accurate than the polygraph
Brad Bushman	2002	Venting experiment against the catharsis hypothesis
Paul Rozin	1987	With Fallon: disgust as protection against contamination
April Fallon	1987	With Rozin: disgust
Ed Diener	1946 to 2021	Subjective well-being, tripartite model (1984): life satisfaction, positive affect, negative affect
Harry Helson	1898 to 1977	Adaptation-level theory (1964)
Philip Brickman	1971	With Campbell: the hedonic treadmill
Donald Campbell	1971	With Brickman: the hedonic treadmill
Samuel Stouffer	1949	Relative deprivation (The American Soldier)
Barbara Fredrickson	1998	Broaden-and-build theory of positive emotions
Fred Bryant	2007	With Veroff: savoring
Joseph Veroff	2007	With Bryant: savoring
Aaron Beck	1967	Cognitive model of depression (1967) and cognitive distortions (a distractor in questions on Fredrickson and learned helplessness)
John B. Watson	1878 to 1958	Three innate emotions: fear, rage and love
Katharine Banham Bridges	1932	Differentiation of infant emotions: distress and delight first
Edward Tronick	1978	Still-face procedure: infants expect emotional give-and-take
James Gross	1998	Process model of emotion regulation: reappraisal is cognitive change, suppression is response modulation
Peter Sifneos	1920 to 2008	Coined alexithymia (1973)
Hans Selye	1907 to 1982	General adaptation syndrome (1936; The Stress of Life, 1956): alarm, resistance, exhaustion; eustress (1974)
Shelley E. Taylor	born 1946	Tend-and-befriend (2000)
Thomas Holmes	1967	With Rahe: Social Readjustment Rating Scale (life events)
Richard Rahe	1967	With Holmes: Social Readjustment Rating Scale (life events)
Allen Kanner	1981	With Lazarus and colleagues: daily hassles and uplifts

Name	When	Examined for
Robert Karasek	1979	Job demand-control model
Herbert Freudenberger	1926 to 1999	Introduced the term burnout (1974)
Christina Maslach	born 1946	Maslach Burnout Inventory (1981): exhaustion, depersonalisation, reduced accomplishment
Susan Folkman	1984	With Lazarus: transactional model of stress
Paul Meehl	1920 to 2003	Diathesis-stress model of schizophrenia (1962)
Joseph Zubin	1977	With Spring: vulnerability-stress model
Bonnie Spring	1977	With Zubin: vulnerability-stress model
Sheldon Cohen	1985	With Wills: direct and stress-buffering effects of social support
Thomas Wills	1985	With Cohen: stress-buffering effect of social support
Robert S. Weiss	1973	Emotional isolation versus social isolation
Suzanne Kobasa	1979	Hardiness: commitment, control, challenge
Aaron Antonovsky	1923 to 1994	Sense of coherence (1979): comprehensibility, manageability, meaningfulness
Michael Scheier	1985	With Carver: dispositional optimism
Charles Carver	1985	With Scheier: dispositional optimism
Meyer Friedman	1910 to 2001	With Rosenman: Type A behaviour pattern (1959); Western Collaborative Group Study
Ray Rosenman	1920 to 2013	With Friedman: Type A behaviour pattern (1959)
Johan Denollet	mid-1990s	Type D personality: negative affectivity plus social inhibition
George Engel	1913 to 1999	Biopsychosocial model (1977)
Robert Ader	1932 to 2011	With Nicholas Cohen: conditioned immunosuppression (1975), the founding study of psychoneuroimmunology
Nicholas Cohen	1975	With Ader: conditioned immunosuppression
Irwin Rosenstock	1950s	Health belief model
Ronald Rogers	1975	Protection motivation theory
Icek Ajzen	1985	Theory of planned behaviour: attitude, subjective norm, perceived behavioural control
Jon Kabat-Zinn	born 1944	Mindfulness-based stress reduction (1979)
Donald Meichenbaum	born 1940	Stress inoculation training (1970s)

Key terms

mental processes, clinical psychology, comparative psychology, evolutionary psychology, nativism, empiricism, rationalism, laws of association, dualism, innate ideas, tabula rasa, associationism, mental chemistry, apperceptive mass, psychophysical parallelism, Bell-Magendie law, psychophysics, mental chronometry, immediate experience, voluntarism, creative synthesis, tridimensional theory of feeling, introspection, Volkerpsychologie, structuralism, stimulus error, context theory of meaning, imageless thought, functionalism, stream of consciousness, reflex arc, mental test, psychological clinic, great-person theory, Zeitgeist, preparadigmatic science, prescriptions, behaviourism, peripheralism, operationism, cognitive map, latent learning, phi phenomenon, insight, transposition, life space, action research, Gestalt therapy, structural model, basic anxiety, phenomenology, Dasein, logotherapy, noogenic neurosis, transpersonal psychology, savouring, cognitive revolution, information-processing approach, connectionism, eclecticism, Indian psychology, theory of opposite wishes, indigenisation, nurturant-task leader, astika, purusha, prakriti, gunas, antahkarana, chitta-vritti-nirodha, nishkama karma, atman, turiya, panchakosha, psychic being, sensation, perception, transduction, doctrine of specific nerve energies, all-or-none law, sensory adaptation, absolute threshold, just noticeable difference, Weber's law, Fechner's law, magnitude estimation, Stevens' power law, method of limits, method of constant

stimuli, signal detection theory, d prime, response bias, receiver operating characteristic, accommodation, fovea, blind spot, scotopic vision, dark adaptation, Purkinje shift, saccades, Mach bands, lateral inhibition, trichromatic theory, opponent-process theory, place theory, frequency theory, volley principle, papillae, umami, gate control theory, phantom limb, semicircular canals, synaesthesia, Charles Bonnet syndrome, Bottom-up processing, top-down processing, unconscious inference, constructivist approach, Bayesian approach, direct perception, affordances, law of proximity, law of closure, Prägnanz, binocular cues, retinal disparity, convergence, monocular cues, texture gradient, Motion parallax, visual cliff, Perceptual constancy, size constancy, size-distance invariance, Emmert's law, illusion, Müller-Lyer illusion, inappropriate constancy scaling, carpentered-world hypothesis, hallucination, autokinetic effect, Pandemonium, prototype, recognition-by-components, perceptual set, field dependence-independence, Subliminal perception, Extrasensory perception, span of attention, cocktail party phenomenon, dichotic listening, filter model, early selection, attenuation model, late selection, feature integration theory, inattentional blindness, change blindness, automatic process, controlled processing, Stroop effect, sustained attention, Consciousness, topographic model, preconscious, unconscious, psychodynamic perspective, hard problem of consciousness, altered state of consciousness, circadian rhythm, suprachiasmatic nucleus, REM sleep, slow-wave sleep, sleep cycle, REM rebound, cataplexy, manifest content, latent content, dream work, collective unconscious, activation-synthesis hypothesis, Hypnosis, suggestibility, neodissociation theory, Concentrative meditation, mindfulness meditation, tolerance, learning, habituation, sensitisation, imprinting, classical conditioning, unconditioned stimulus, conditioned stimulus, extinction, spontaneous recovery, stimulus generalisation, stimulus discrimination, counterconditioning, reciprocal inhibition, systematic desensitisation, flooding, conditioned taste aversion, preparedness, law of effect, operant conditioning, negative reinforcement, secondary reinforcers, token economy, two-factor theory, response cost, shaping, schedule of reinforcement, Premack principle, learned helplessness, biofeedback, insight learning, observational learning, reciprocal determinism, self-efficacy, savings method, multi-store model, sensory memory, chunking, serial position effect, working memory, phonological loop, visuospatial sketchpad, central executive, episodic buffer, levels of processing, self-reference effect, episodic memory, semantic memory, procedural memory, implicit memory, flashbulb memory, tip-of-the-tongue state, schema, misinformation effect, source monitoring, encoding specificity principle, context-dependent memory, state-dependent memory, decay theory, proactive interference, retroactive interference, repression, anterograde amnesia, retrograde amnesia, Korsakoff's syndrome, confabulation, engram, long-term potentiation, Thinking, mental rotation, dual-coding theory, concept, family resemblance, problem space, algorithm, heuristic, Means-end analysis, hill climbing, functional fixedness, mental set, Deductive reasoning, Inductive reasoning, abductive reasoning, belief bias, confirmation bias, bounded rationality, satisficing, conjunction fallacy, Prospect theory, loss aversion, Hindsight bias, Creativity, phoneme, morpheme, pragmatics, prosody, deep structure, language acquisition device, poverty of the stimulus, critical period, holophrase, telegraphic speech, overregularisation, linguistic relativity, inner speech, Aphasia, operational definition, Achievement, aptitude, growth mindset, mental age, intelligence quotient (IQ), Stanford-Binet, ratio IQ, deviation IQ, concurrent validity, culture-fair test, Raven's Progressive Matrices, Factor analysis, positive manifold, general factor (g), primary mental abilities, structure of intellect, Fluid intelligence (Gf), Crystallised intelligence (Gc), multiple intelligences, triarchic theory, successful intelligence, PASS model, emotional intelligence, twin studies, Heritability, reaction range, Flynn effect, stereotype threat, Intellectual disability, adaptive functioning, Global developmental delay, Vineland Social Maturity Scale, Savant syndrome, motivation, need, drive, primary motives, secondary motives, intrinsic motivation, extrinsic motivation, instinct, hormic psychology, homeostasis, drive-reduction theory, optimal arousal theory, Yerkes-

Dodson law, incentive theory, expectancy-value theory, hierarchy of needs, prepotency, deficiency needs, metamotivation, ERG theory, press, need for achievement, triadic reciprocal causation, overjustification effect, self-determination theory, cognitive evaluation theory, flow, frustration, ventromedial hypothalamus, lateral hypothalamus, sexual response cycle, expectancy theory, goal-setting theory, emotion, mood, Affect, James-Lange theory, Cannon-Bard theory, misattribution of arousal, primary appraisal, secondary appraisal, facial feedback hypothesis, sham rage, Papez circuit, limbic system, amygdala, Klüver-Bucy syndrome, pseudobulbar affect, display rules, subjective well-being, broaden-and-build theory, Savoring, self-conscious emotions, cognitive reappraisal, Alexithymia, fight-or-flight response, general adaptation syndrome, eustress, hypothalamic-pituitary-adrenal axis, Social Readjustment Rating Scale, daily hassles, burnout, transactional model of stress, diathesis-stress model, Problem-focused coping, emotion-focused coping, emotional isolation, hardiness, Type A behaviour pattern, Type D personality, biopsychosocial model, Psychoneuroimmunology

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From the past papers

1. Functionalism, an early school of thought in psychology, was most closely associated with which psychologist? (IHBAS 2024)

   (a) Edward Titchener (b) John Watson (c) William James (d) G. Stanley Hall

   Answer: (c) William James. James's *Principles of Psychology* (1890) inspired the functionalist question of what the mind is for, and Dewey and Angell made it a formal school at Chicago. Titchener founded the rival structuralism, Watson founded behaviourism, and Hall, though functionalist in spirit, is examined for institutions such as the first American research laboratory (1883) and the APA (1892).

2. The first person to use the term clinical psychology in print was (RML 2020)

   (a) Erik Erikson (b) Sigmund Freud (c) Lightner Witmer (d) Cattell

   Answer: (c) Lightner Witmer. Witmer opened the first psychological clinic at the University of Pennsylvania in 1896 and used the term in print in his journal *The Psychological Clinic* (1907). Erikson is known for the psychosocial stages, Freud for psychoanalysis, and James McKeen Cattell for coining "mental test".

3. Match List I with List II LIST I (Psychologist) LIST 2 (Theory) A. Dollard and Miller 1. Radical Behaviorism B. Skinner 2. Self-Actualization C. Bandura and Walters 3. Early Social Learning D. Maslow 4. Later Social Learning (IHBAS 2024)

   (a) A-1, B-2, C-3, D-4 (b) A-3, B-1, C-4, D-2 (c) A-2, B-1, C-3, D-4 (d) A-3, B-2, C-4, D-1

   Answer: (b) A-3, B-1, C-4, D-2. Dollard and Miller recast Freud in Hull's terms in 1950 (early social learning); Skinner's system is radical behaviourism; Bandura and Walters gave the later social learning theory (1963); Maslow is self-actualisation. Option (d) tempts only if you forget that Skinner, not Maslow, is the radical behaviourist.

4. Which is the correct order of the last four of Patanjali's eight limbs? (NIMHANS 2026 recall)

   (a) Dharana, pratyahara, dhyana, samadhi (b) Pratyahara, dharana, dhyana, samadhi (c) Pratyahara, dhyana, dharana, samadhi (d) Dhyana, dharana, pratyahara, samadhi

   Answer: (b) Pratyahara, dharana, dhyana, samadhi. The senses are first withdrawn inward (pratyahara), the mind is then fixed on one point (dharana), the fixing becomes an unbroken flow (dhyana) and finally absorption (samadhi). Each wrong option swaps two of these steps.

5. Which statement is INCORRECT? (Amity 2023)

   (a) Cones are most prevalent in the central retina (b) Cones are sensitive to moderate to high levels of light (c) Cones provide only monochromatic information (d) Cones provide excellent acuity

   Answer: (c) Cones provide only monochromatic information. Cones, with three pigments, give colour vision; it is the rods, with one pigment (rhodopsin), that give colourless scotopic vision. The other three statements are true: cones are packed in the fovea, work in moderate to bright light and give excellent acuity.

6. In signal detection theory, the ability to distinguish a stimulus from background noise is: (IHBAS 2025)

   (a) Sensitivity (d′) (b) Criterion (β) (c) ROC curve (d) JND

   Answer: (a) Sensitivity (d'). d′ is the separation between the noise and signal-plus-noise distributions and does not depend on the observer's willingness to say "yes". The criterion (β) is that willingness (response bias), the ROC curve plots hit rate against false alarm rate, and the JND belongs to classical psychophysics.

7. Among monocular cues for depth perception is (JEMAS 2025)

   (a) Convergence (b) Retinal disparity (c) Assimilation (d) Interposition

   Answer: (d) Interposition. An object that partly hides another is seen as nearer, and one eye is enough to see it. Convergence and retinal disparity are the only two binocular cues, and assimilation is not a depth cue at all (it is Piaget's term).

8. Illusions are (JEMAS 2025)

   (a) misinterpretation of stimuli arising from an external object (b) real perceptual object perceived in a distorted way (c) vivid mental images (d) perception without an adequate external stimulus

   Answer: (a) Misinterpretation of stimuli arising from an external object. In an illusion a real stimulus is misread. Option (b) is a sensory distortion such as micropsia, (c) is imagery, and (d) is the definition of a hallucination.

9. The Stroop effect demonstrates interference in (JEMAS 2025)

   (a) Long-term memory retrieval (b) Working memory (c) Selective attention (d) Procedural memory

   Answer: (c) Selective attention. Automatic word reading cannot be filtered out while naming the ink colour, so the task shows a failure of selective attention. The memory options are distractors; no stored memory is being retrieved or rehearsed.

10. According to Freud, the ___ mind includes accessible memories, which can be recalled at will (RML 2020)

    (a) Conscious (b) Preconscious (c) Unconscious (d) Psychic determinism

    Answer: (b) Preconscious. The conscious holds only what is in awareness now, and the unconscious cannot be recalled at will; psychic determinism is a principle (no mental event is accidental), not a level of mind.

11. Systematic desensitization technique is based upon the idea of: (IHBAS 2025)

    (a) Reciprocal inhibition (b) Law of effect (c) Habituation (d) Adaptation to stress

    Answer: (a) Reciprocal inhibition. Wolpe paired each step of an anxiety hierarchy with relaxation, a response incompatible with anxiety, so the anxiety bond weakens. The law of effect is Thorndike's operant principle, habituation underlies exposure and response prevention, and adaptation to stress belongs to stress theory.

12. Among the following schedules of reinforcement, which typically produces the highest rate of response and is most resistant to extinction? (Real PYQ 2024)

    (a) Fixed Ratio (b) Variable Ratio (c) Fixed Interval (d) Variable Interval

    Answer: (b) Variable Ratio. Ratio schedules reward fast responding, and unpredictability removes any reason to pause and makes it hard to tell that reinforcement has stopped, as with slot machines. Fixed ratio shows a pause after each reinforcer, fixed interval a scallop and variable interval only a moderate, steady rate.

13. The famous patient H.M. (Henry Molaison) underwent surgical removal of the medial temporal lobes, including the hippocampus, to treat severe epilepsy. While he retained most of his intellectual abilities and working memory, his primary and most profound deficit was in: (Real PYQ 2024)

    (a) Procedural memory (b) Semantic memory (c) Anterograde amnesia (d) Retrograde amnesia

    Answer: (c) Anterograde amnesia. After the 1953 operation H.M. could not form new declarative memories, although his intelligence, digit span and childhood memories survived. Procedural memory was spared (he learned mirror tracing), old semantic knowledge was intact, and his retrograde amnesia covered only the years before surgery.

14. Which of the following terms refers to difficulty recalling information due to interference from new information? (IHBAS 2024)

    (a) Retroactive interference (b) Proactive interference (c) Decay theory (d) Retrieval failure

    Answer: (a) Retroactive interference. New learning reaching back to disrupt older memories is retroactive (RETRO: new disrupts old); proactive interference is the reverse, old disrupting new. Decay blames the passage of time and retrieval failure a mismatch of cues, not competition from new learning.

15. In which method does the researcher gather data, look for patterns, and then generate a theory or hypothesis? (RML 2023)

    (a) Deductive reasoning (b) Hypothetico-deductive method (c) Inductive reasoning (d) Abductive reasoning

    Answer: (c) Inductive reasoning. Induction runs from specific observations to a probable generalisation, so data come first and theory last, as in most qualitative research. Deduction and the hypothetico-deductive method run the other way, from a theory to a tested prediction; abduction infers the best explanation of a surprising fact.

16. What is the primary difference between Piaget and Vygotsky's theories of language development? (RML 2021)

    (a) Piaget argued thought depends on language while Vygotsky argued language depends on thought (b) Vygotsky argued thought depends on language while Piaget argued language depends on thought (c) Vygotsky was concerned with early language development while Piaget was concerned with later language development. (d) None of the above

    Answer: (b). For Piaget, language expresses concepts already built through action; for Vygotsky, thought and speech merge at about two, after which thought becomes verbal and depends on language. Option (a) swaps the names, and (c) is false because both studied the same preschool years.

17. The term IQ was invented by (JEMAS 2025)

    (a) Lewis Termon (sic) (b) Gordon Allport (c) William Stern (d) John Watson

    Answer: (c) William Stern. Stern proposed the intelligence quotient, mental age divided by chronological age, in 1912. Terman is the tempting wrong answer: he multiplied the quotient by 100 and popularised it in the 1916 Stanford-Binet, but did not coin it; Allport (traits) and Watson (behaviourism) never worked on the IQ.

18. A 10-year-old child from a rural area, with no formal schooling, is brought for an intellectual assessment due to academic struggles. The child is illiterate and has had minimal exposure to formal testing materials. Which of the following tests would be the MOST appropriate to use to minimize cultural and educational bias? (Real PYQ 2024)

    (a) Bhatia's Battery of Performance Tests of Intelligence (b) Wechsler Intelligence Scale for Children (WISC-IV) (c) Stanford-Binet Intelligence Scale (5th ed.) (d) Malin's Intelligence Scale for Indian Children (MISIC)

    Answer: (a) Bhatia's Battery of Performance Tests of Intelligence. Its five performance subtests need no reading, and it has Indian rural norms. The WISC-IV and SB5 rest on school-built verbal knowledge and foreign norms; MISIC is the tempting Indian option, but half of it is verbal, adapted from the WISC.

19. The 'satiety centre' of the brain is located in the: (RML 2024)

    (a) Lateral hypothalamus (b) Ventromedial hippocampus (c) Lateral thalamus (d) Ventromedial hypothalamus

    Answer: (d) Ventromedial hypothalamus. Bilateral VMH lesions make rats overeat and grow obese (Hetherington and Ranson, 1940), so Stellar's dual-centre theory (1954) called the VMH the satiety centre. The lateral hypothalamus is the opposite, the feeding centre, whose destruction causes aphagia (Anand and Brobeck, 1951); "ventromedial hippocampus" and "lateral thalamus" are invented by changing one word.

20. According to Herzberg's Two-Factor Theory, which of the following factors are considered motivators that lead to job satisfaction? (RML 2023)

    (a) Salary and benefits (b) Job security and work conditions (c) Recognition and opportunities for advancement (d) Company policies and interpersonal relationships

    Answer: (c) Recognition and opportunities for advancement. Herzberg's motivators lie in the content of the work: achievement, recognition, the work itself, responsibility, advancement and growth. Salary, benefits, job security, working conditions, company policies and relationships are hygiene factors, which prevent dissatisfaction when good but do not by themselves motivate.

21. The seat of emotion is to be found in _____. (RML 2024)

    (a) Reticular formation (b) Hind brain (c) Limbic system (d) Forebrain

    Answer: (c) Limbic system. MacLean's limbic system (1952), built on Papez's circuit and including the amygdala, is the classic seat of emotion. The reticular formation governs arousal and wakefulness, the hindbrain paces heartbeat and breathing, and "forebrain" is too broad, although the limbic system lies within it.

22. 'Psychological disorders develop when a biological predisposition is set off by a stressful situation', is proposed by (JEMAS 2025)

    (a) Humanistics existential models. (b) Socio-cultural model. (c) Diathesis-stress model. (d) Cognitive model.

    Answer: (c) Diathesis-stress model. A pre-existing vulnerability (diathesis) set off by a stressful trigger is the model's definition (Meehl, 1962; Zubin and Spring, 1977). Humanistic-existential models blame blocked growth, sociocultural models social forces and cognitive models faulty thinking; none is defined by predisposition plus stress.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 **Systematic desensitisation** (Joseph Wolpe, *Psychotherapy by Reciprocal Inhibition*, 1958) rests on **reciprocal inhibition**: relaxation, a response incompatible with anxiety, is paired step by step with a graded anxiety hierarchy. ASKED 17 TIMES

2 **Working memory** (Alan Baddeley and Graham Hitch, 1974) has four components: the central executive (an attentional controller that stores nothing, loaded by random number generation), the phonological loop (auditory-verbal information), the visuospatial sketchpad (visual and spatial information, including mental rotation) and the episodic buffer, added in 2000, which binds information into coherent episodes. ASKED 11 TIMES

3 The **token economy** is operant conditioning with tokens as secondary reinforcers, developed by Teodoro Ayllon and Nathan Azrin (*The Token Economy*, 1968); it is the listed therapy not derived from classical conditioning. ASKED 10 TIMES

4 **Abraham Maslow**'s hierarchy of needs (1943, 1954), bottom to top: physiological, safety, belongingness and love (social), esteem, self-actualisation. Physiological and safety needs are the lower needs; ego is not a level. ASKED 10 TIMES

5 **Korsakoff's syndrome** follows thiamine (vitamin B1) deficiency in chronic alcoholism, often after Wernicke's encephalopathy (confusion, ataxia, nystagmus); it damages the mammillary bodies and mediodorsal thalamus and causes dense anterograde amnesia, extensive retrograde amnesia and confabulation, sincere false stories that fill memory gaps. ASKED 8 TIMES

6 The **empty chair** technique belongs to **Gestalt therapy** (Fritz Perls, with Laura Perls and Paul Goodman), an experiential here-and-now therapy that is not the Gestalt psychology of Wertheimer, Kohler and Koffka. ASKED 7 TIMES

7 **Variable ratio** reinforcement gives the highest, steadiest rate and the greatest resistance to extinction; fixed ratio gives high rates with a post-reinforcement pause, fixed interval the scallop, variable interval moderate steady rates, and continuous reinforcement the fastest learning and fastest extinction. ASKED 7 TIMES

8 **Extinction** is the fading of a conditioned response when the CS is repeatedly presented without the UCS, or of an operant response that is no longer reinforced; a parent who ignores tantrums is using extinction, not operant escape. ASKED 7 TIMES

9 Reinforcement always increases behaviour and punishment decreases it; positive adds and negative removes a stimulus, so switching off a shock is **negative reinforcement**, while time-out and response cost are **negative punishment**. ASKED 7 TIMES

10 **H.M.** (Henry Molaison) had both medial temporal lobes, including the hippocampus and amygdala, removed by William Beecher Scoville in 1953; Scoville and Brenda Milner (1957) reported a profound anterograde amnesia for declarative memory, with normal short-term memory, limited retrograde amnesia and spared procedural learning (mirror tracing). ASKED 7 TIMES

11 **Raymond Cattell** (with John Horn) split g into **fluid intelligence** (Gf), reasoning with novel problems independently of acquired knowledge and measured best by Raven's Progressive Matrices, and **crystallised intelligence** (Gc), the repertoire of information, cognitive skills and strategies, measured by vocabulary and information. ASKED 7 TIMES

12 **The limbic system** is the classic "seat of emotion": Paul MacLean named it in 1952, enlarging James Papez's 1937 circuit, and it regulates emotion, motivation and memory; hunger and thirst belong to the hypothalamus. ASKED 7 TIMES

13 **The amygdala** is the structure for fear and aggression: it detects threat, drives fear conditioning and gives emotional events their extra memorability, while the hippocampus stores the facts. ASKED 7 TIMES

14 The **hippocampus** is the structure damaged in anterograde amnesia and is critical for consolidating new declarative memories; it is not the store of remote memories. ASKED 6 TIMES

15 **Narendra Nath Sengupta** set up India's first psychology laboratory (1915) and first Department of Psychology (1916) at the University of Calcutta and is the father of modern psychology in India; keys naming Bose or Uma Chaudhuri for the first laboratory are wrong. ASKED 5 TIMES

16 **Humanistic psychology** (Maslow, Rogers) is the **third force**, after psychoanalysis (first) and behaviourism (second); **transpersonal psychology** is the fourth force. ASKED 5 TIMES

17 The **Stroop effect** (John Ridley Stroop, 1935): naming the ink colour of a conflicting colour word (RED printed in green) is slowed because automatic word reading interferes; it shows a limit of selective attention (interference in reaction time), not a memory effect. ASKED 5 TIMES

18 **Proactive interference**: earlier learning disrupts newer learning (dialling your old phone number); **retroactive interference**: newer learning disrupts older memories. RML 2024 keys English grammar intruding on newly learned Spanish as retroactive, but it is proactive. ASKED 5 TIMES

19 **Inductive reasoning** moves from specific observations to probable generalisations (gather data, find patterns, build theory: the logic of theory development and qualitative research); **deductive reasoning** starts from a theory and tests its predictions against observations. ASKED 5 TIMES

20 **Graham Wallas** (*The Art of Thought*, 1926): preparation, incubation, illumination, verification; incubation (setting the problem aside) is second and illumination (the insight) third. ASKED 5 TIMES

21 **Functional fixedness** (Karl Duncker's candle problem, 1945) is seeing an object only in its usual use; **mental set** (Abraham Luchins's water jars, 1942) is reusing a strategy that worked before, and was keyed as the most basic and common obstacle to problem solving. ASKED 5 TIMES

22 **Henry Murray**'s need-press theory (1938) underlies the **Thematic Apperception Test** (Morgan and Murray, 1935), a projective test whose stories reveal the teller's needs, conflicts and motives. ASKED 5 TIMES

23 **Wilhelm Wundt** opened the first psychology laboratory at **Leipzig in 1879** (not Vienna, Paris or Cambridge); Galton's London laboratory (1884) was anthropometric. ASKED 4 TIMES

24 Patanjali's **eight limbs**, in order: yama, niyama, asana, pranayama, pratyahara (withdrawal of the senses), dharana (concentration on one point), dhyana (unbroken meditation), samadhi (absorption); karma and jnana are not limbs. ASKED 4 TIMES

25 **Tamas** is the guna of inertia, lethargy, darkness, dullness and ignorance; a person who has stopped eating and cannot get herself to act shows tamas alone. ASKED 4 TIMES

26 **Every sense except smell relays through the thalamus** on its way to the cortex (vision via the lateral geniculate nucleus, hearing via the medial geniculate nucleus); the olfactory nerve (cranial nerve I) bypasses it. ASKED 4 TIMES

27 The **suprachiasmatic nucleus** (SCN) of the hypothalamus is the master circadian pacemaker, entrained by light through the retinohypothalamic tract; SCN lesions abolish circadian rhythms, not total sleep. ASKED 4 TIMES

28 **REM (paradoxical) sleep**: a desynchronised, fast, low-voltage EEG like waking, rapid eye movements, atonia of the core muscles, genital arousal and vivid dreams; EEG synchrony and delta waves belong to N3, spindles and K-complexes to N2. ASKED 4 TIMES

29 **Carl Jung** split the unconscious into a personal (individual) unconscious and the **collective unconscious**, an inherited layer of archetypes shared by all humanity, which he explained by the similar brain structure of all human races. ASKED 4 TIMES

30 **William Stern** coined the intelligence quotient in 1912; **Lewis Terman**'s Stanford-Binet (1916) multiplied it by 100, giving the ratio IQ = (MA ÷ CA) × 100, so a mental age of 8 at age 10 gives an IQ of 80. ASKED 4 TIMES

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